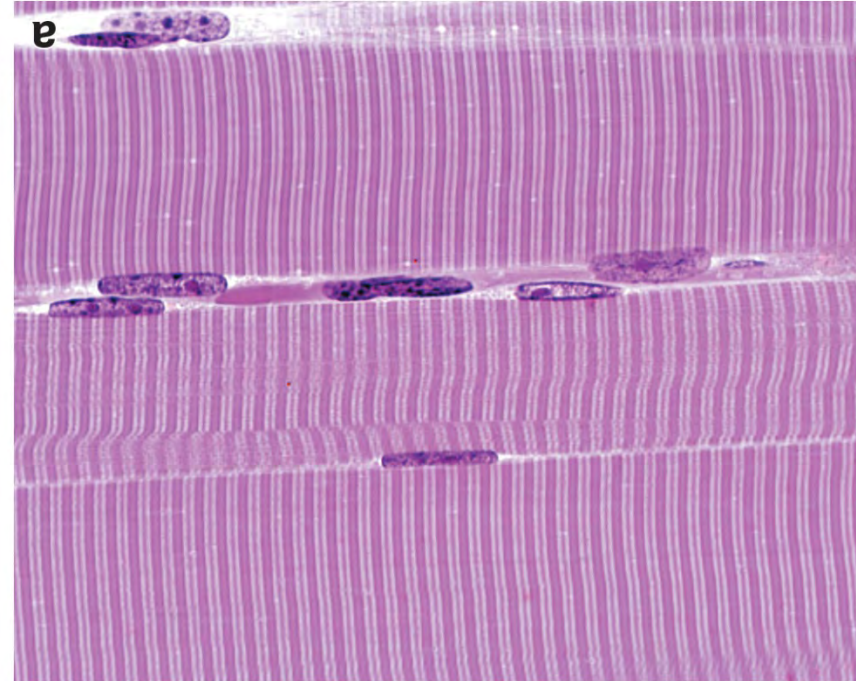
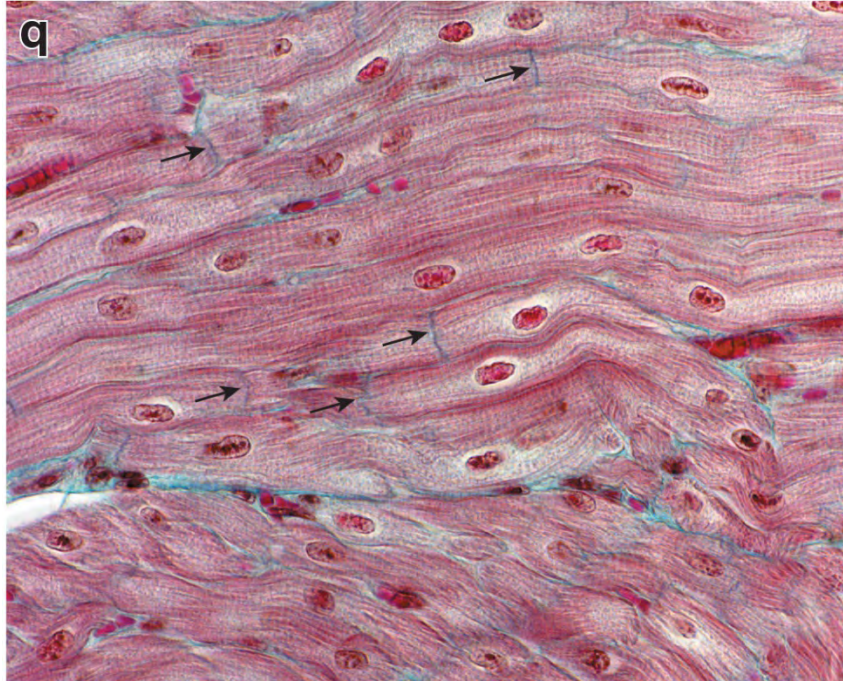
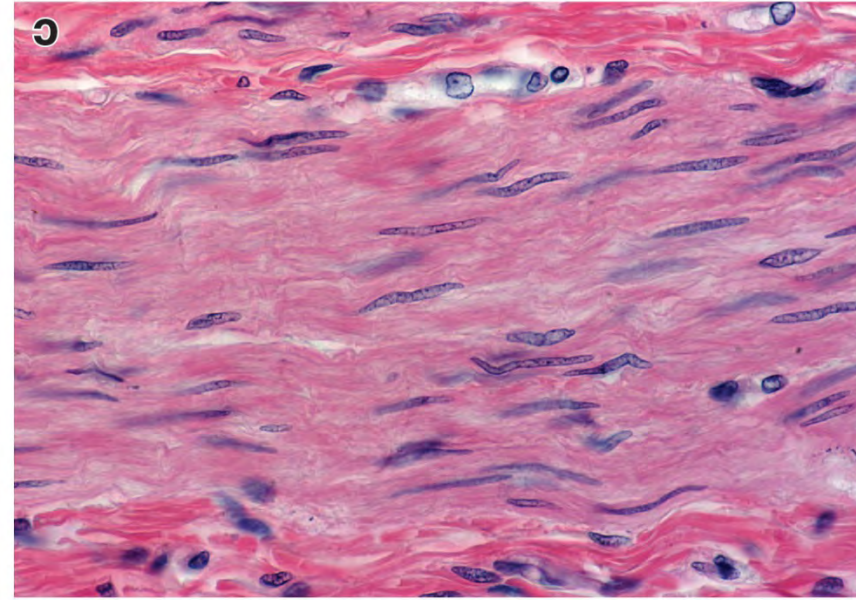


주차	학습내용	주차	학습내용
1 (9/4)	세포학 상피조직	9 (10/31)	호흡기
2 (9/11)	휴강	10 (11/6)	소화관, 소화샘
3 (9/18)	결합조직	11 (11/13)	내분비계통
4 (9/25)	추석연휴	12 (11/20)	비뇨기
5 (10/2)	근육조직, 연골&뼈	13 (11/27)	특수감각기
6 (10/9)	한글날 (온라인: 림프, 순환기)	14 (12/4)	피부
7 (10/14)	신경조직, 혈구 (수요일 강의)	15 (12/11)	기말고사
8 (10/24)	중간고사		

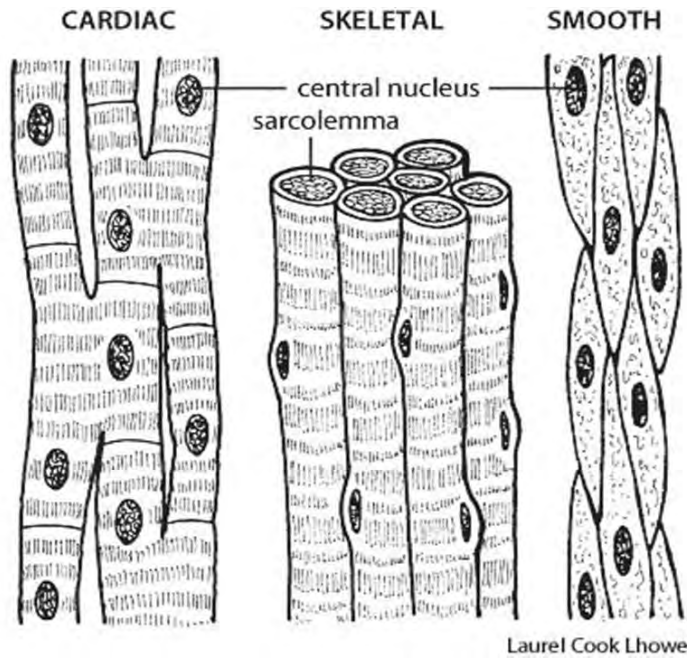
주차	학습내용	주차	학습내용
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6 (10/9)	한글날 (온라인: 림프, 순환기)	14 (12/4)	피부
7 (10/16)	신경조직, 혈구 (실습: 림프, 순환기, 신경조직)	15 (12/11)	기말고사
8 (10/24)	중간고사		

Muscle



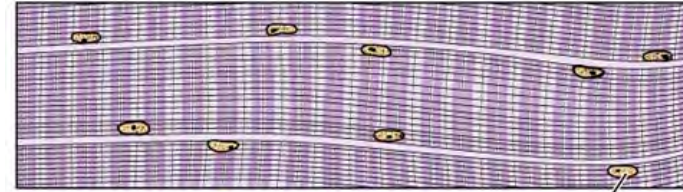
- Types of muscles

1. Skeletal muscle: striation, multiple nuclei, peripheral nuclei, voluntary
2. Cardiac muscle: striation, one or two nuclei, central nucleus, involuntary
3. Smooth muscle: no striation, one nucleus, central nucleus, involuntary



Muscle types

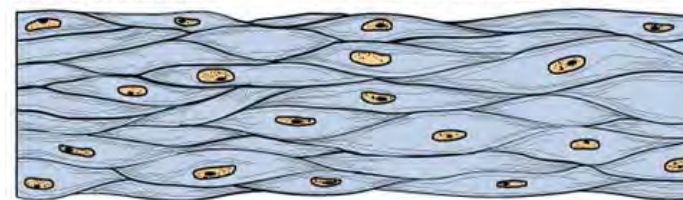
Skeletal muscle



Cardiac muscle



Smooth muscle



Activity

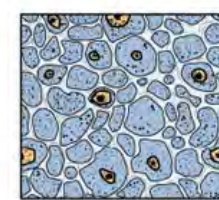
Cross sections



Strong, quick
discontinuous
voluntary
contraction



Strong, quick
continuous
involuntary
contraction



Weak, slow
involuntary
contraction

Muscle

- Muscle cell – muscle fiber
 - Sarcolemma: cell membrane
 - Sarcoplasm: cytoplasm
 - Sarcoplasmic Reticulum: Smooth ER
 - Sarcosome: mitochondria
 - Myofibril ← myofilament
 - Thin myofilament : actin
 - Thick myofilament : myosin



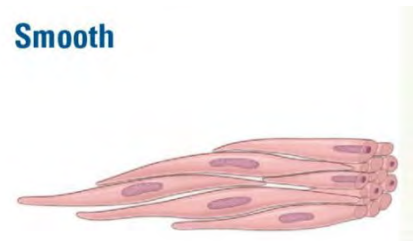
Skeletal



Cardiac

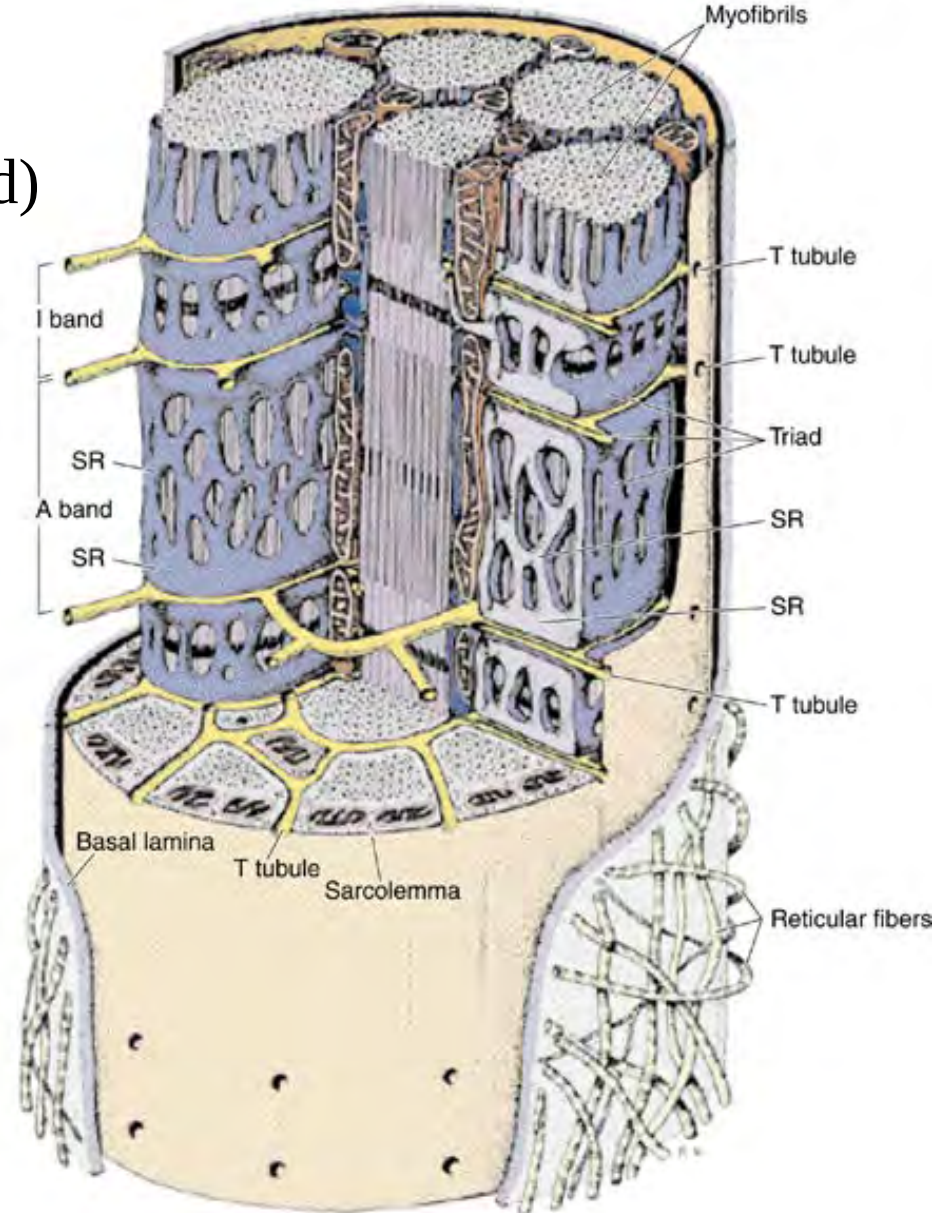


Smooth



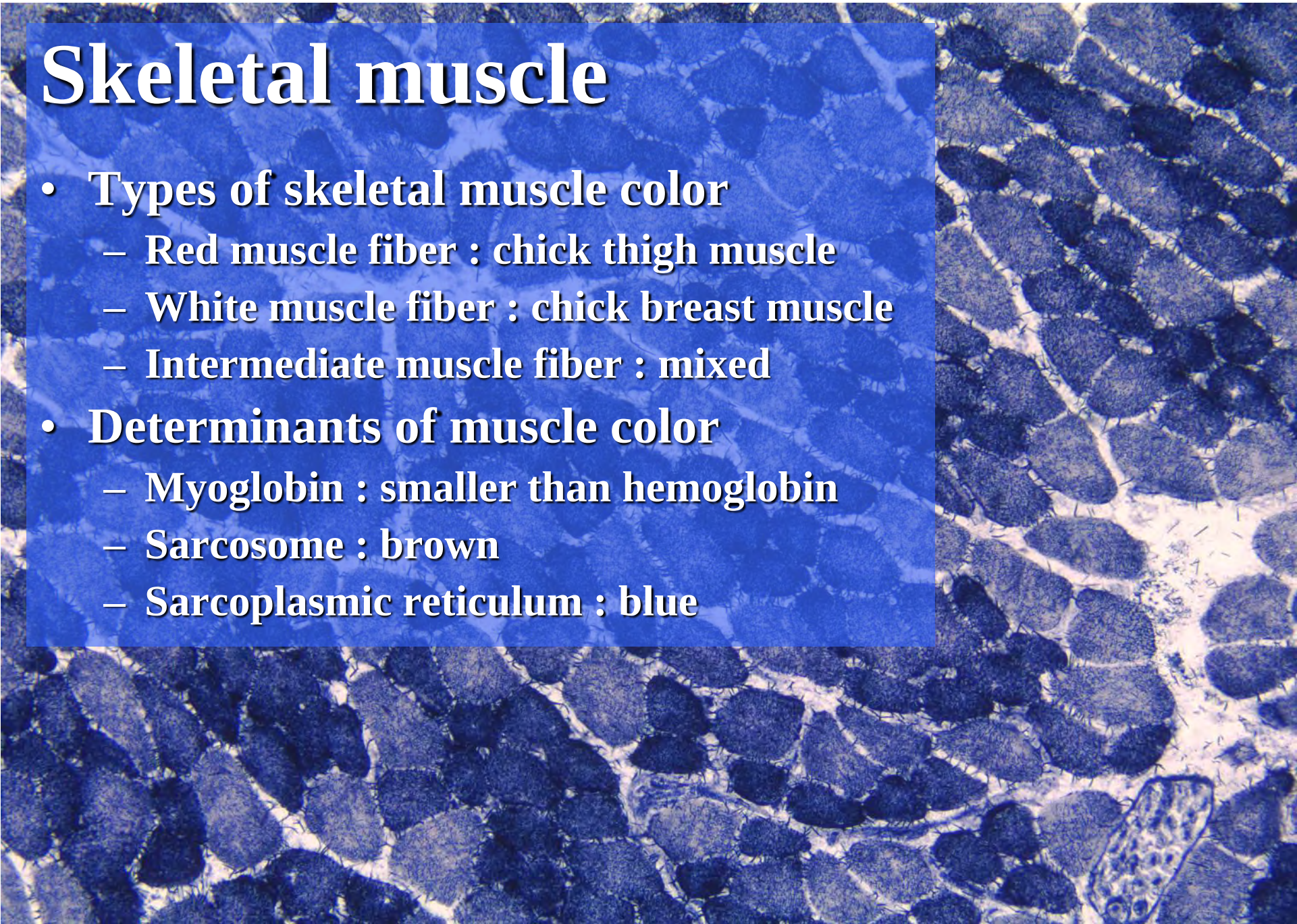
Muscle

- Formation of muscle cell
 - Skeletal muscle (multinucleated)
 - Fusion of hundreds myoblasts during development → myotubes
 - Cardiac muscle (1 or 2 nuclei)
 - No fusion
 - Smooth muscle (one nucleus)
 - No fusion



Skeletal muscle

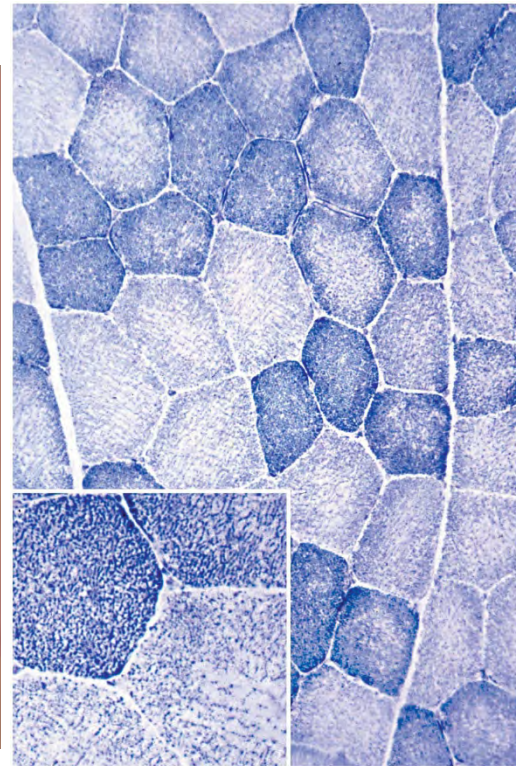
- Types of skeletal muscle color
 - Red muscle fiber : chick thigh muscle
 - White muscle fiber : chick breast muscle
 - Intermediate muscle fiber : mixed
- Determinants of muscle color
 - Myoglobin : smaller than hemoglobin
 - Sarcosome : brown
 - Sarcoplasmic reticulum : blue



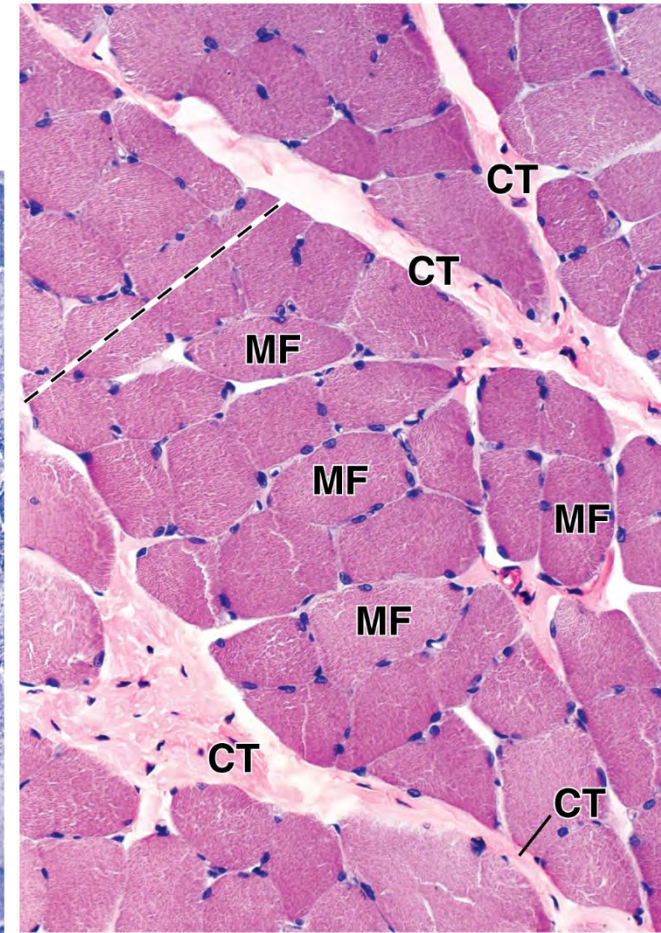
Red fiber vs white fiber

Characteristics	Red	White
Contraction	Slow, repetitive, not easily fatigued, weaker	Fast, “burst”, easily fatigued, stronger contraction
Fiber diameter	Thin	Thick → big muscle
Energy	Aerobic activity using fats or carbohydrates as fuel	Anaerobic Glycolysis using glucose as fuel
Vascularization	Rich	Poorer
Mitochondria	Numerous	Few
Myoglobin	Rich	Poor
Sarcoplasmic Reticulum	Not extensive	Extensive
Innervation	Smaller nerve fibers	Larger nerve fibers

Red fiber vs white fiber



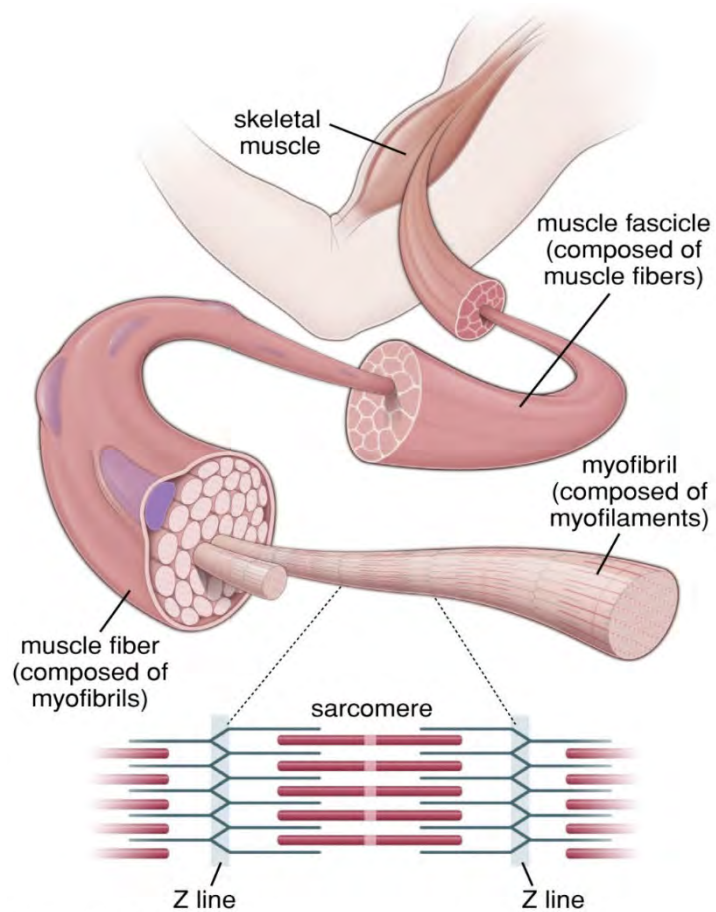
NADH-TR stain



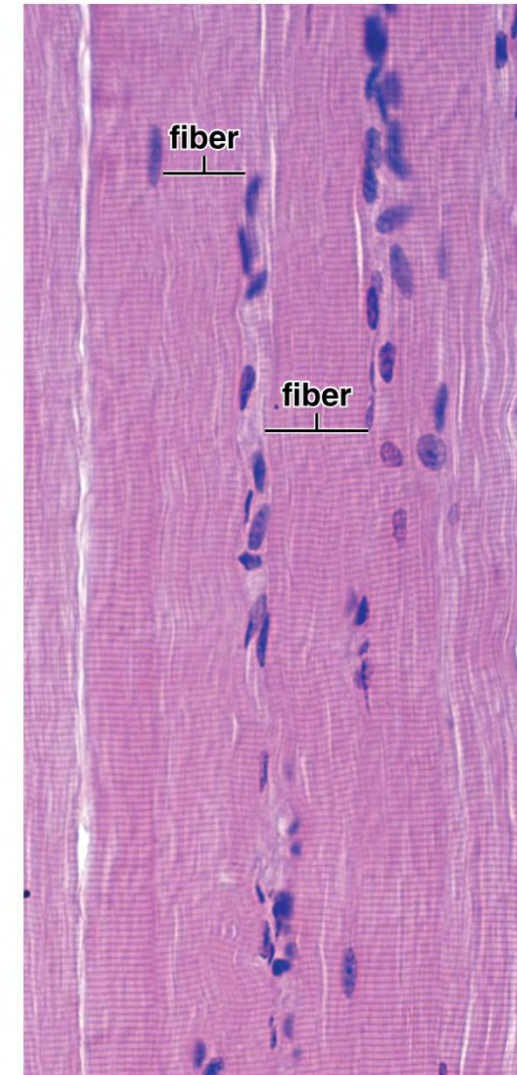
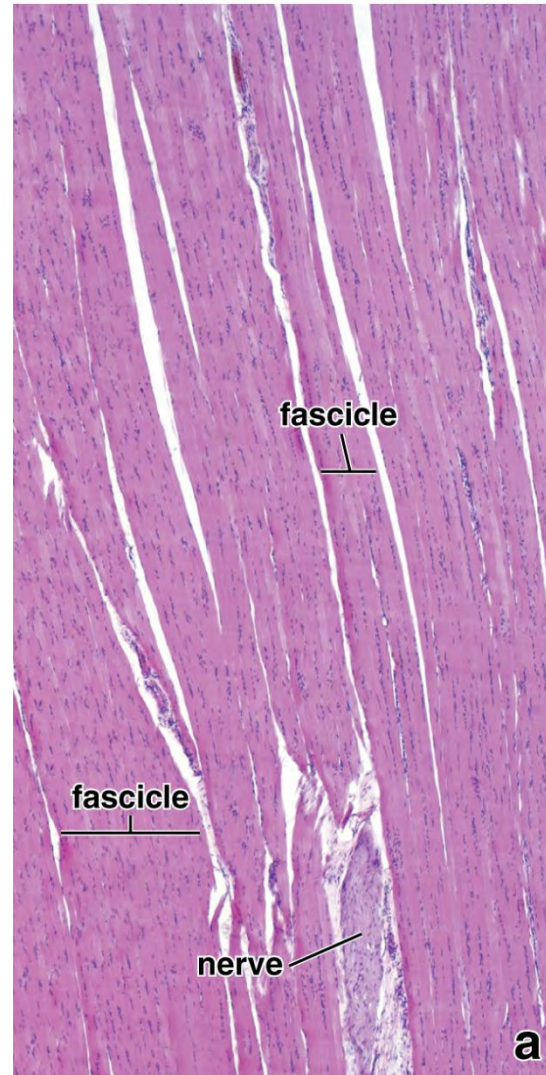
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Structures in Skeletal muscle

Myofilament → myofibril → muscle fiber → muscle fascicle → biceps brachii muscle



fascicle = bundle



Connective tissue in Skeletal muscle

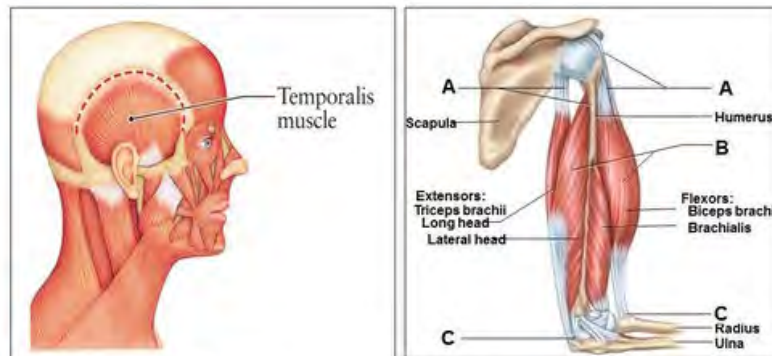
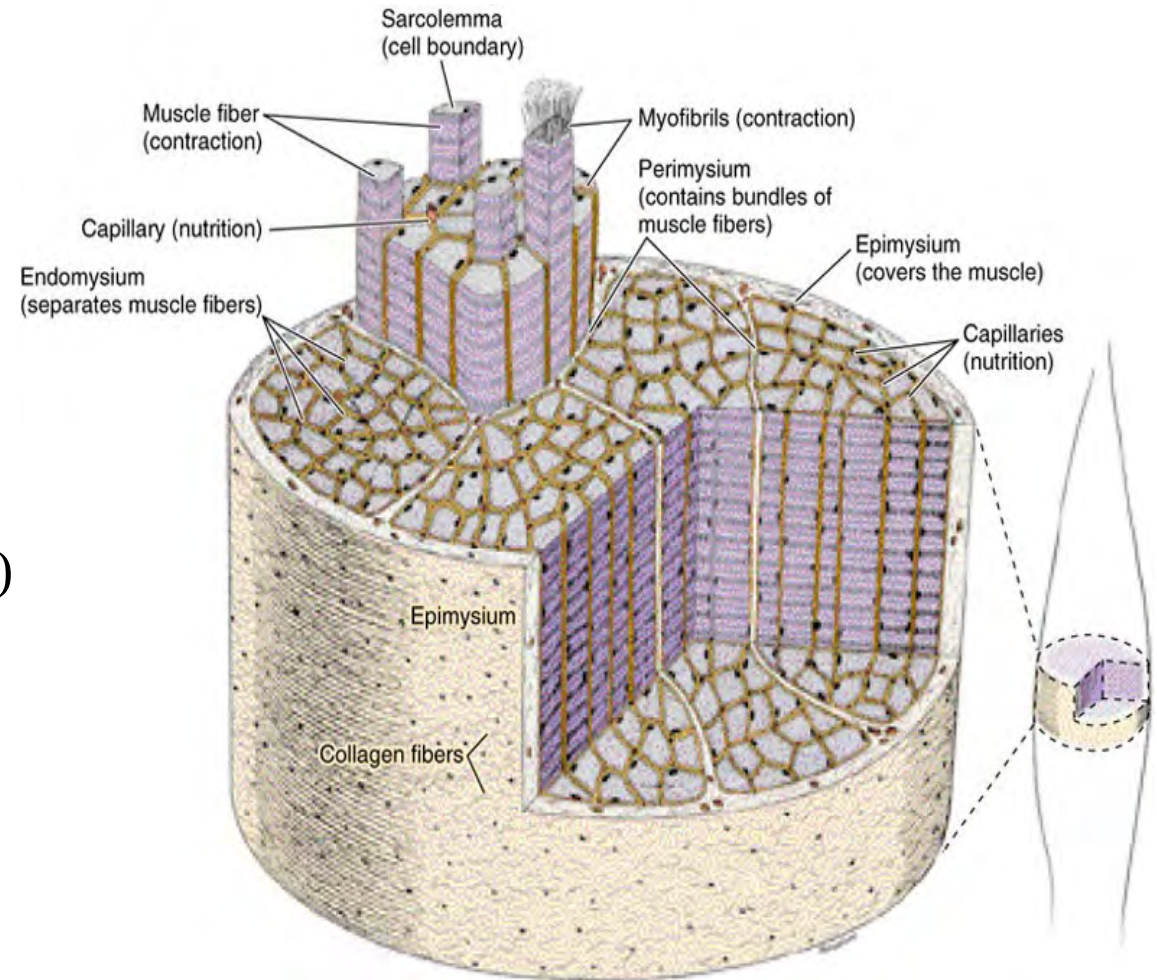
Myofilament → myofibril → muscle fiber → muscle fascicle → biceps brachii muscle

endomysium

perimysium

epimysium

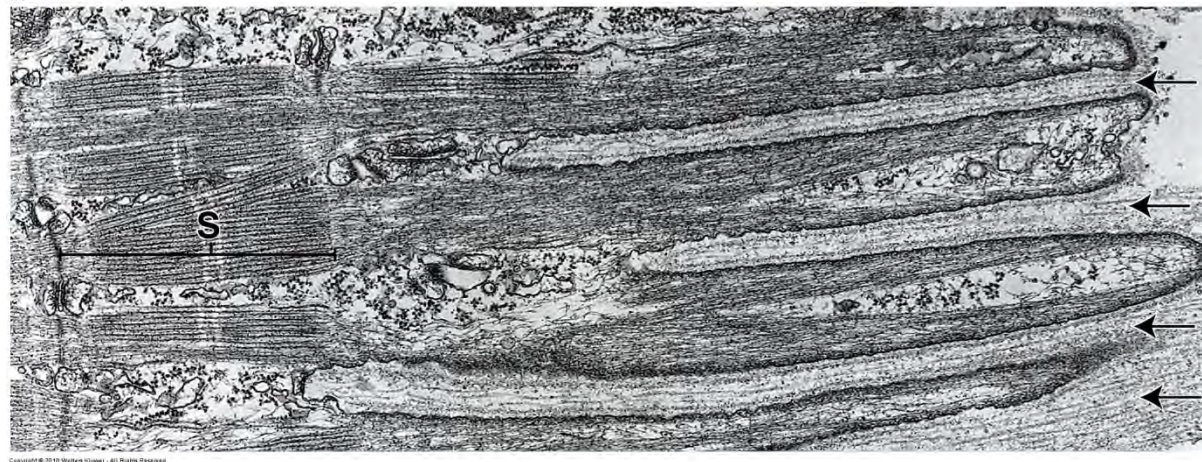
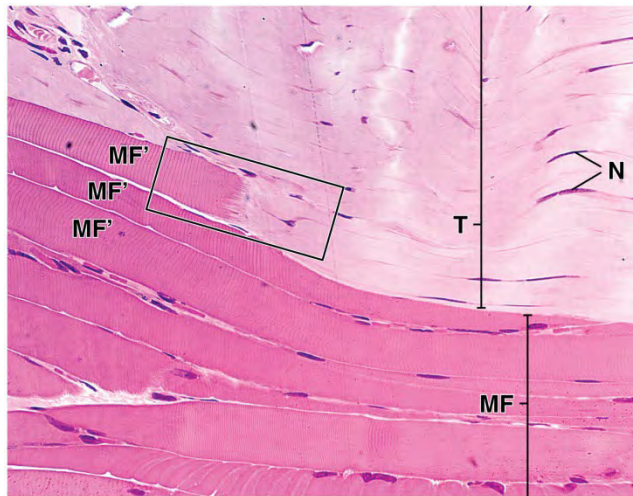
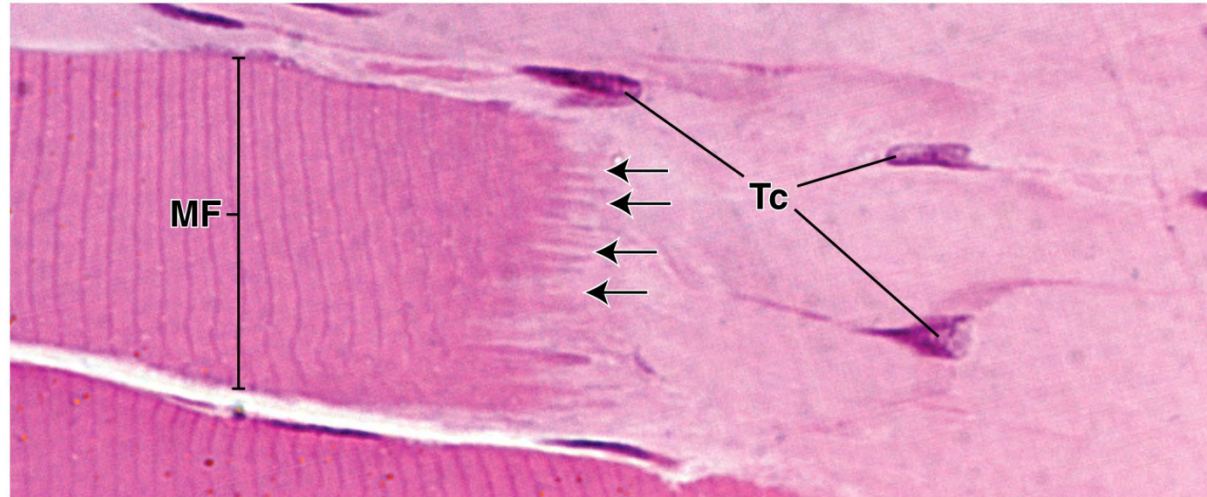
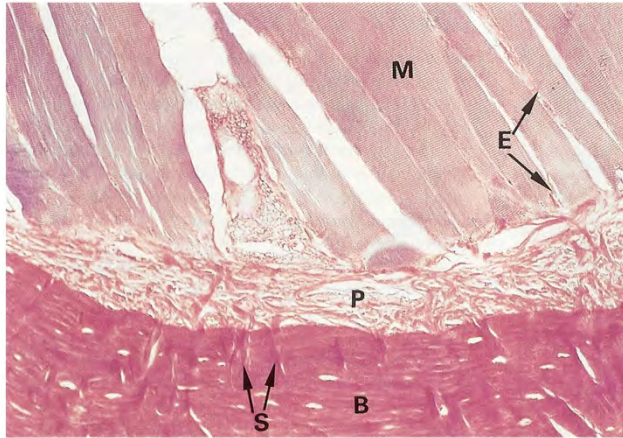
- Connection between fibers
 - Epimysium :
dense irregular C-T collagenous
 - Perimysium :
less dense irregular C-T. collagenous
 - Endomysium :
reticular fiber + basal lamina
- Muscle attachment to bone
 - Direct or Indirect (via 힘줄, 널힘줄)



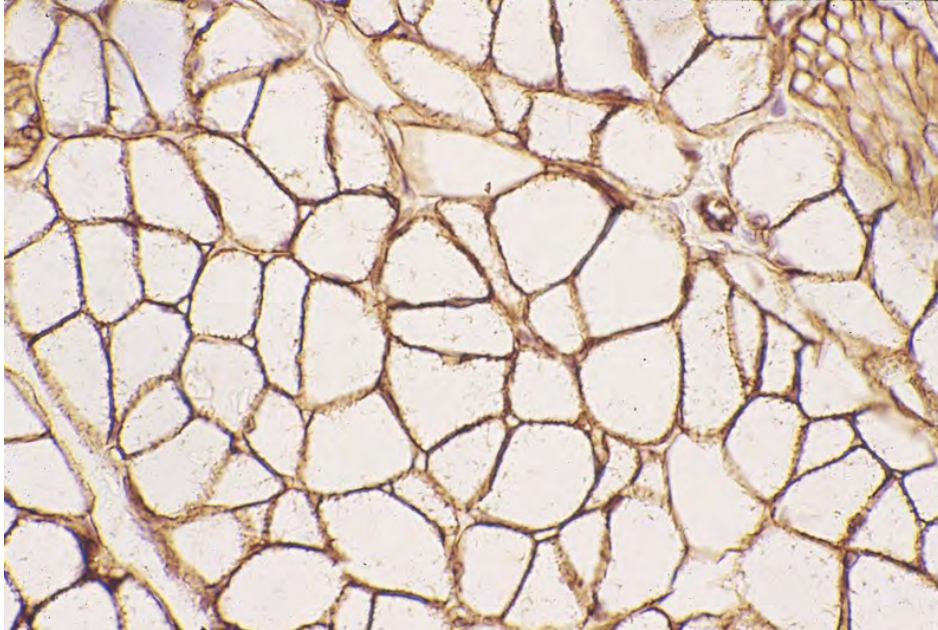
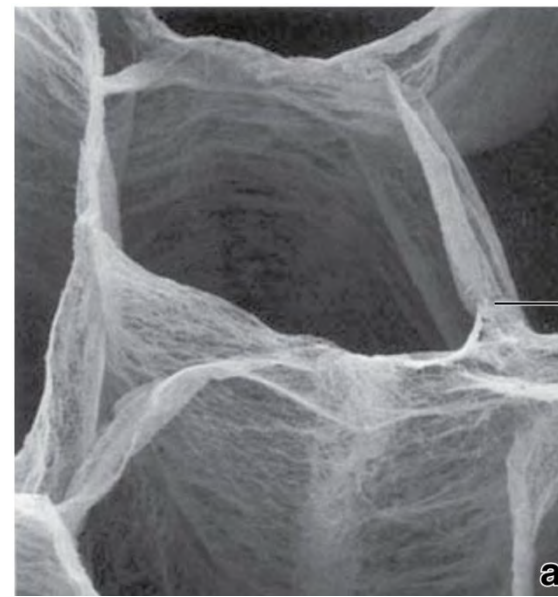
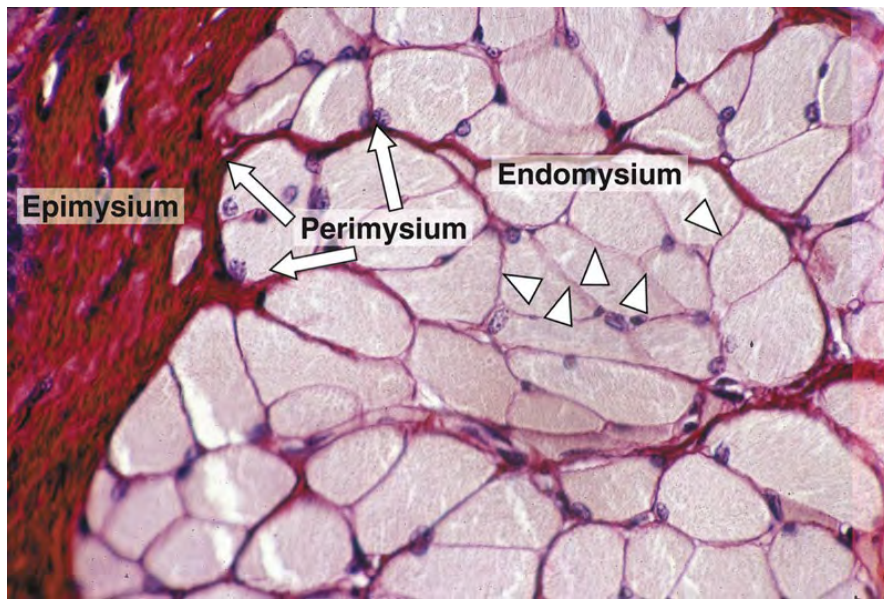
Direct Attachment
(to skull)

Indirect Attachments

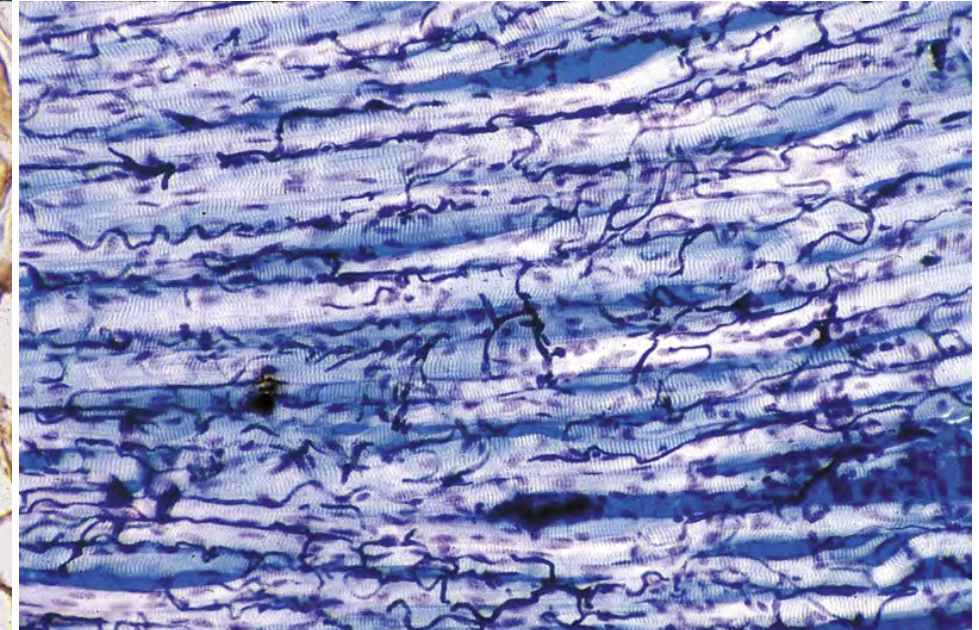
- Connection between muscle and bone
 - Actin filament ---linker proteins---integrin---
basal lamina (laminin---Entactin---type IV collagen)---reticular fiber---
collagen fibers of tendon or aponeurosis---
collagen fiber of periosteum---Sharpey's fiber---bone matrix



Myotendinous junction



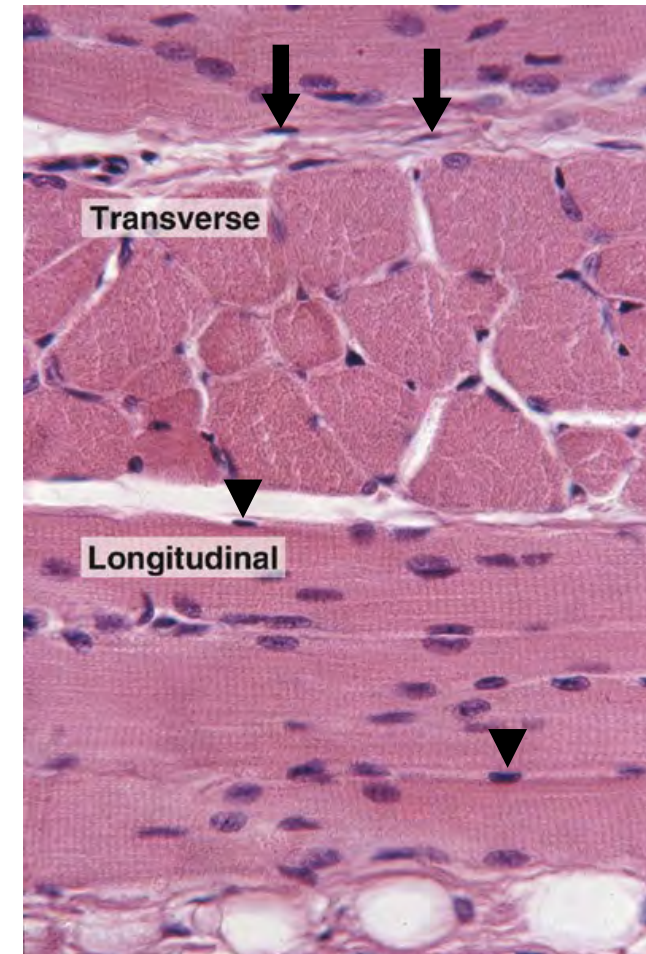
Laminin in endomysium



Blood vessel in skeletal muscle

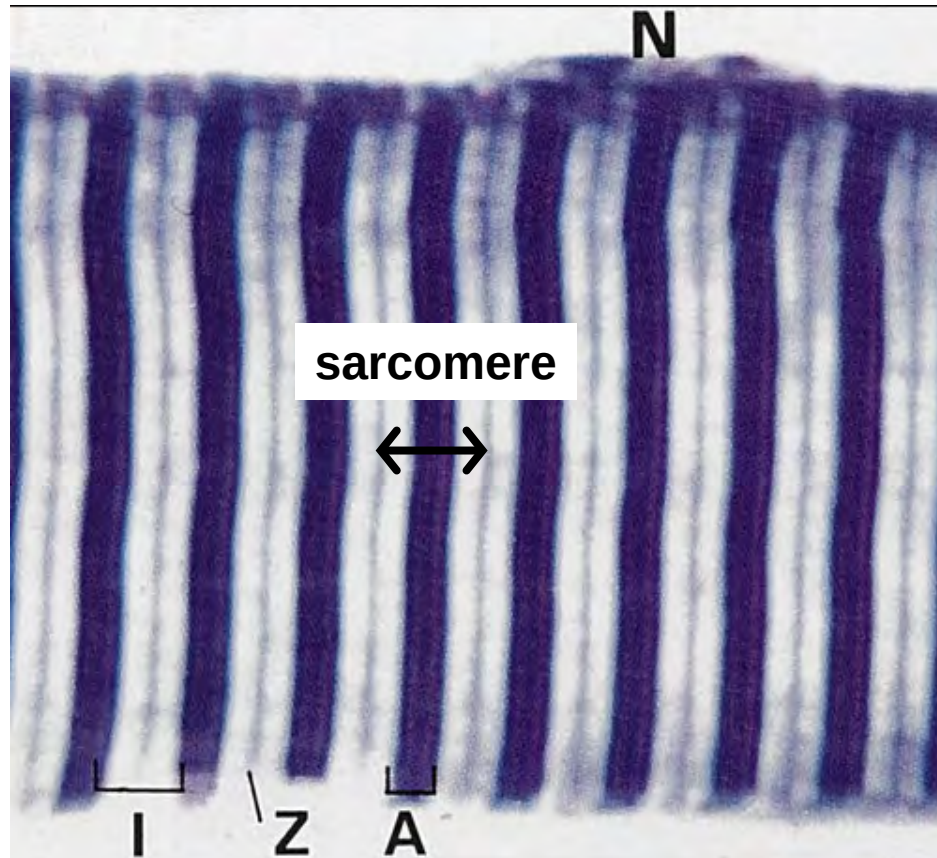
Cells in Skeletal Muscle

- Muscle cell
 - Shape: long (up to 50 cm), cylindrical, multinucleated
 - Cell diameter
 - 30 ~ 150 μm
 - Thicker $\rightarrow$ stronger
 - * more number of cells $\rightarrow$ stronger
 - Striation : from myofibrils
- Satellite cell
 - myogenic cell (arrowheads)
 - between basal lamina and muscle cell.
 - small, heterochromatic single nucleus
 - become muscle cell when needed for muscle fiber growth and repair throughout life
- Endothelial cell (red arrows)
- Fibroblasts (black arrows)

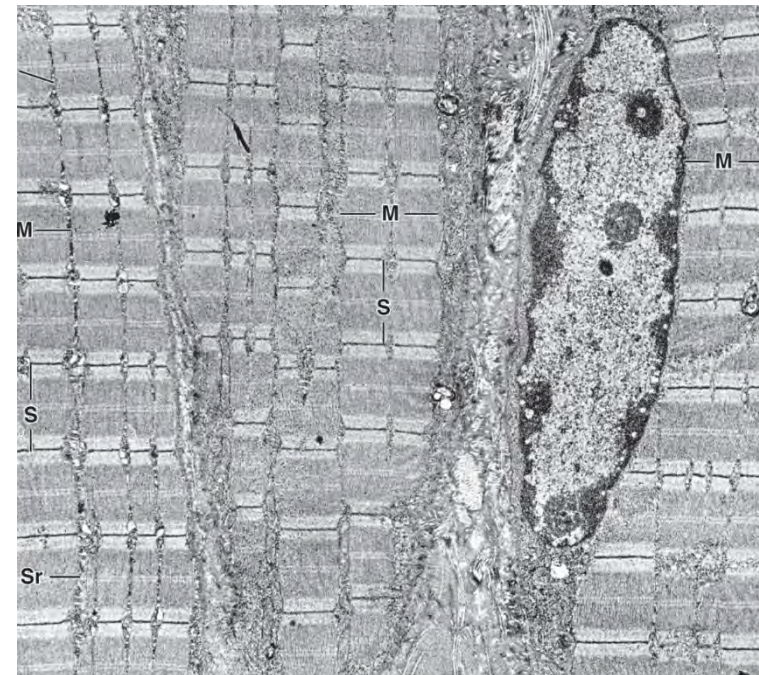
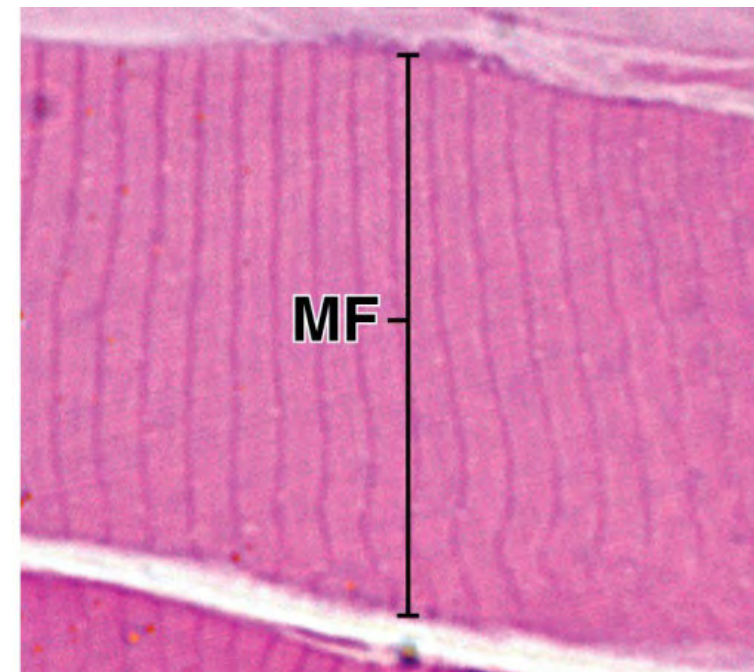


Striations

Sarcomere ($2.5\ \mu\text{m}$, Z line ~Z line)

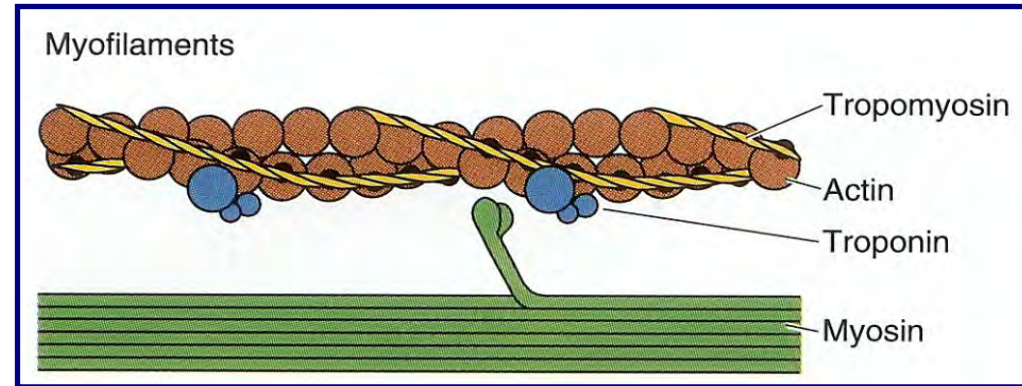
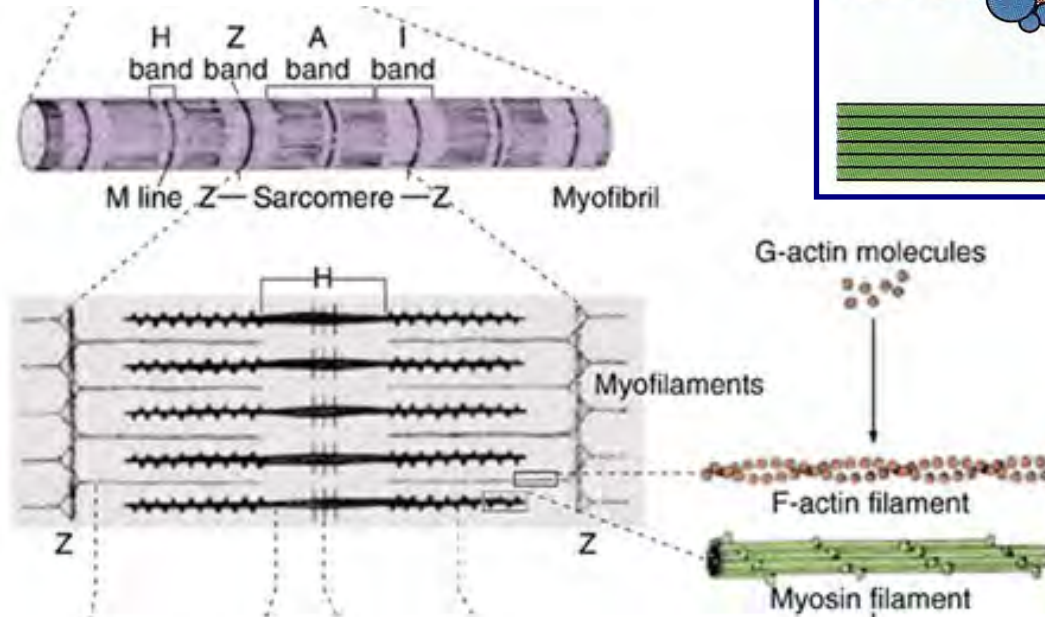


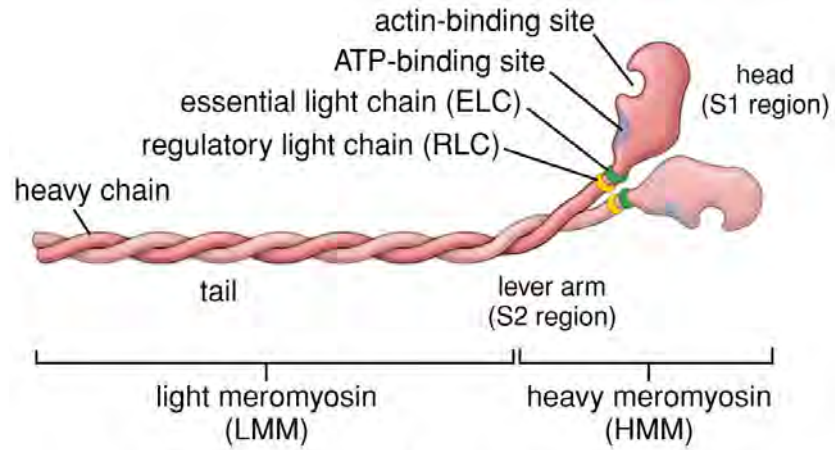
- I = I band (isotropic in polarized light)
- A = A band (anisotropic in polarized light)
- Z = Z disk or Z line (Zwischenscheiben)
- M = M line (Mittelscheibe)
- N = nucleus
- H = H band (Heller band)



Myofilaments in Skeletal Muscle

- Thin filament
 - Actin molecule: G-actin
 - Actin filament: F-actin: 6 nm diameter, 1.0 μm long
- Thick filament
 - Myosin molecule : 3 nm diameter
 - Myosin filament: 15 nm diameter, 1.5 μm long, 200~300개의 myosin molecules
- Arrangement in mammals
 - 1 thick filament + 6 thin filaments

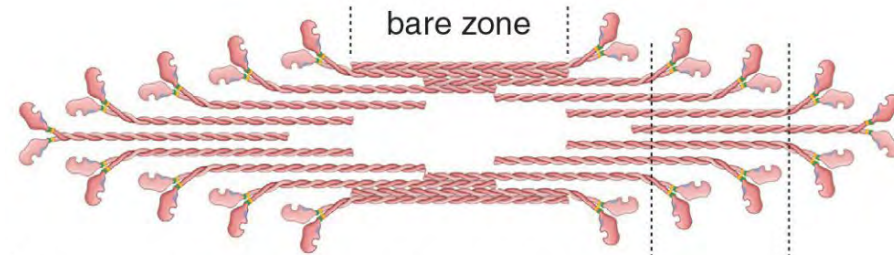




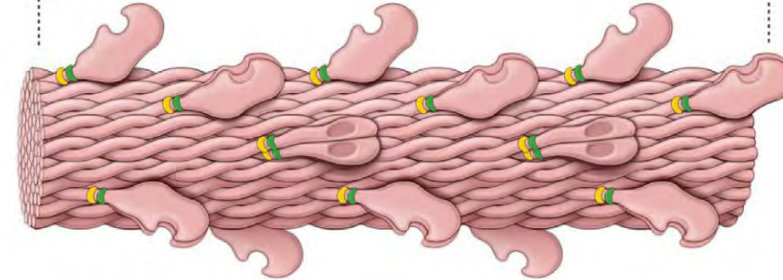
Myosin II molecule



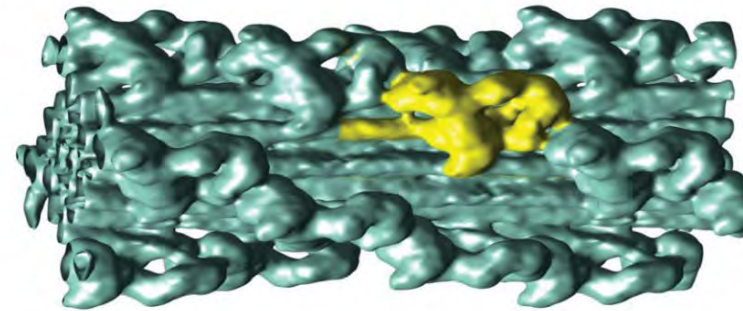
a. Nucleation of thick filament assembly



b. Assembly of myosin in a bipolar thick filament



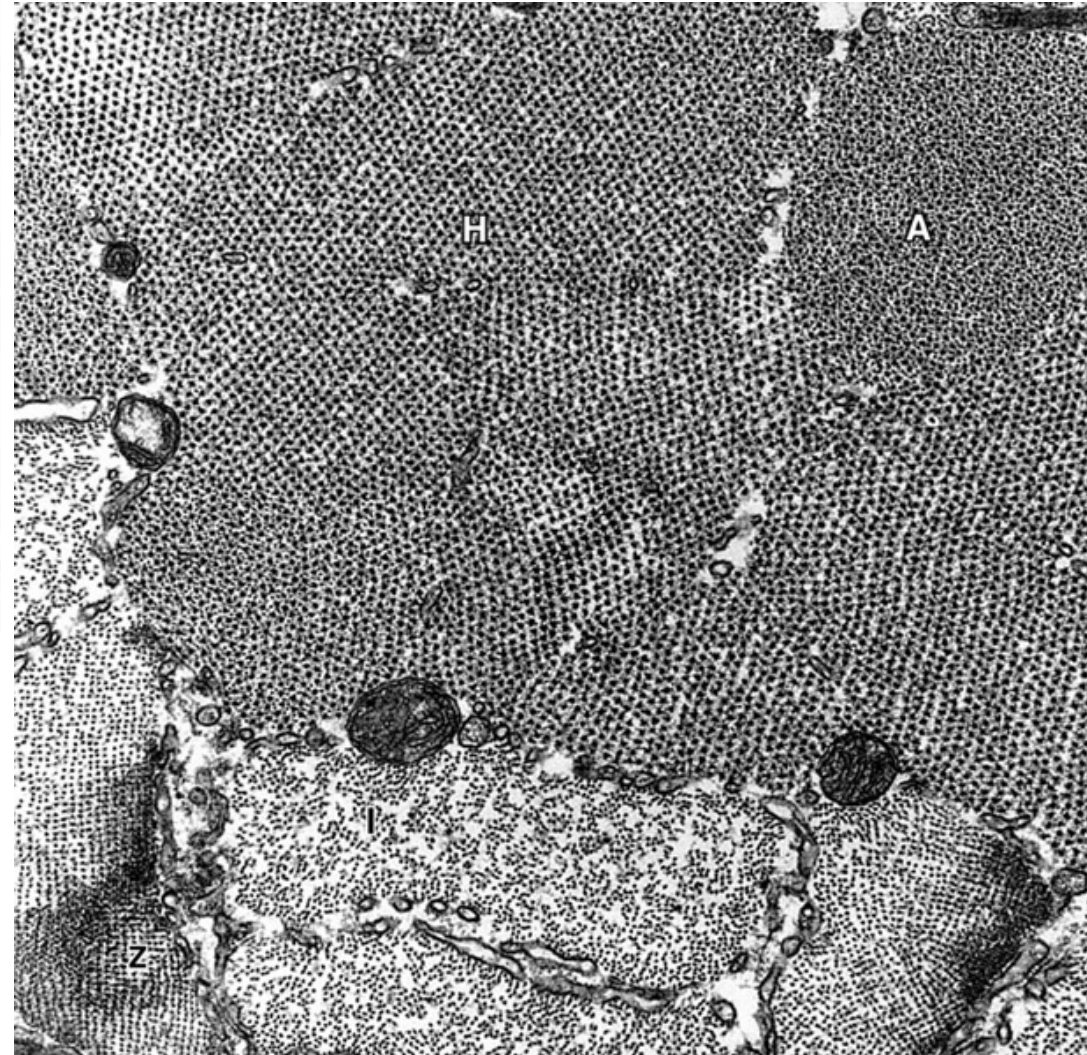
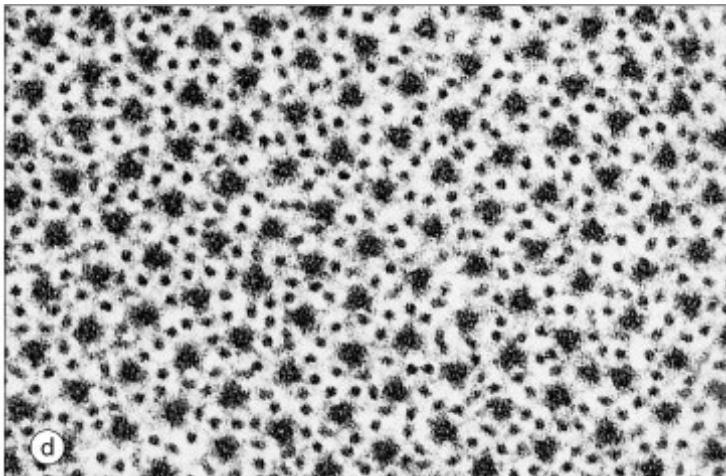
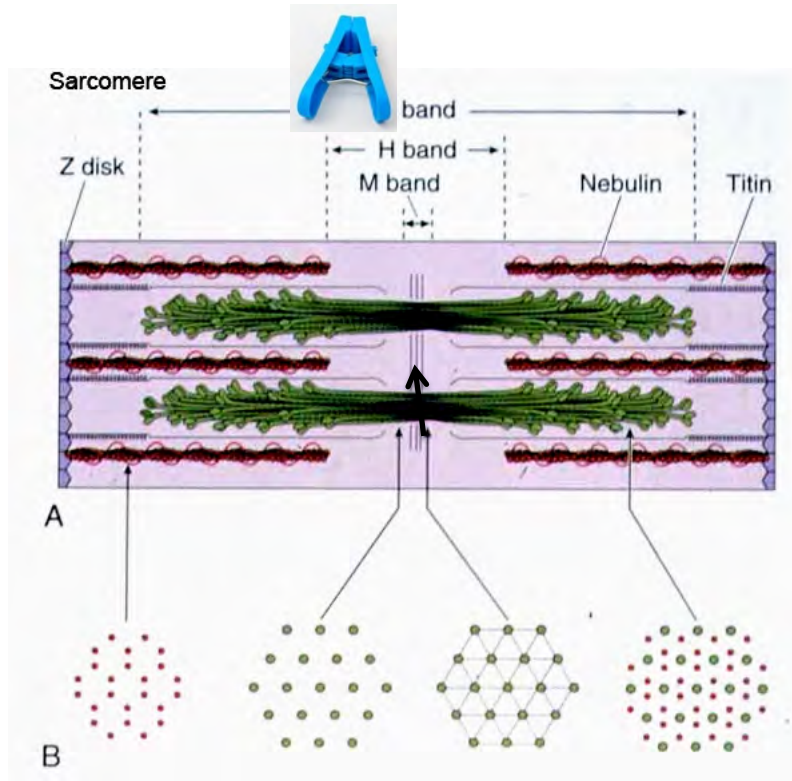
c. Myosin thick filament



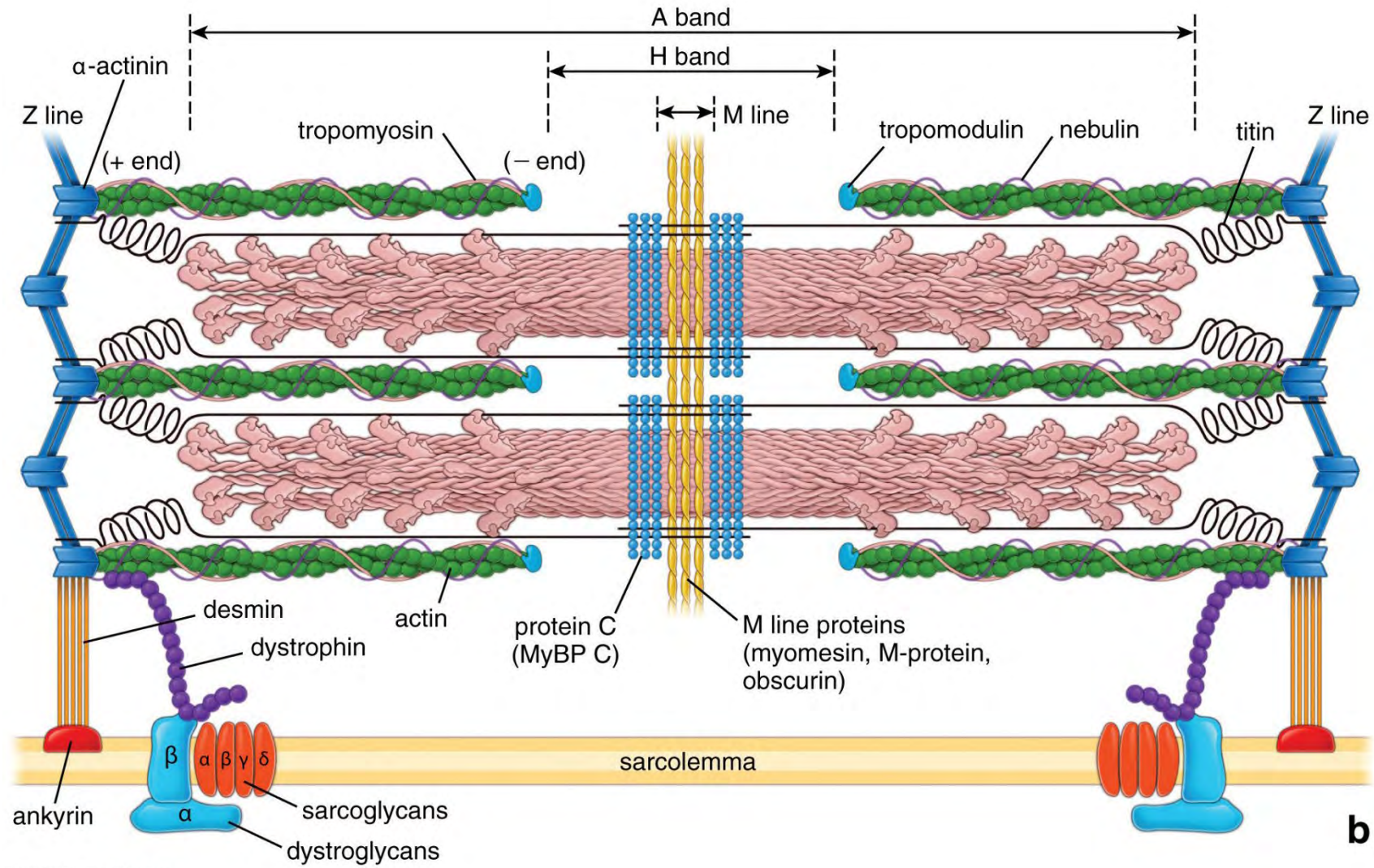
d. Cryo-electron micrograph 3D reconstruction

Myosin II thick filament

Arrangement of myofilaments in myofibril

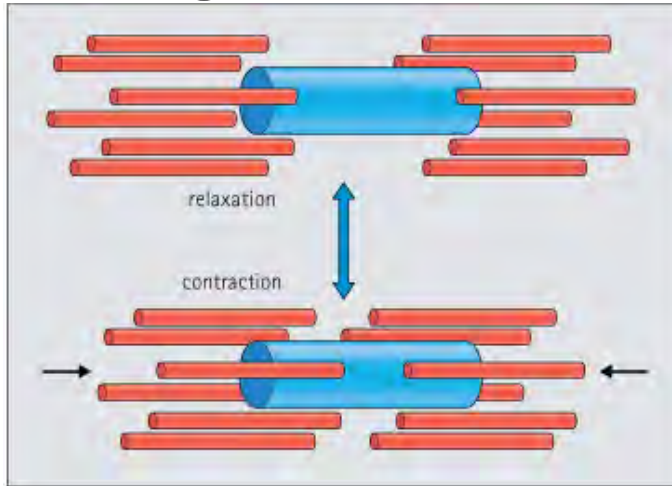


Proteins in Sarcomere

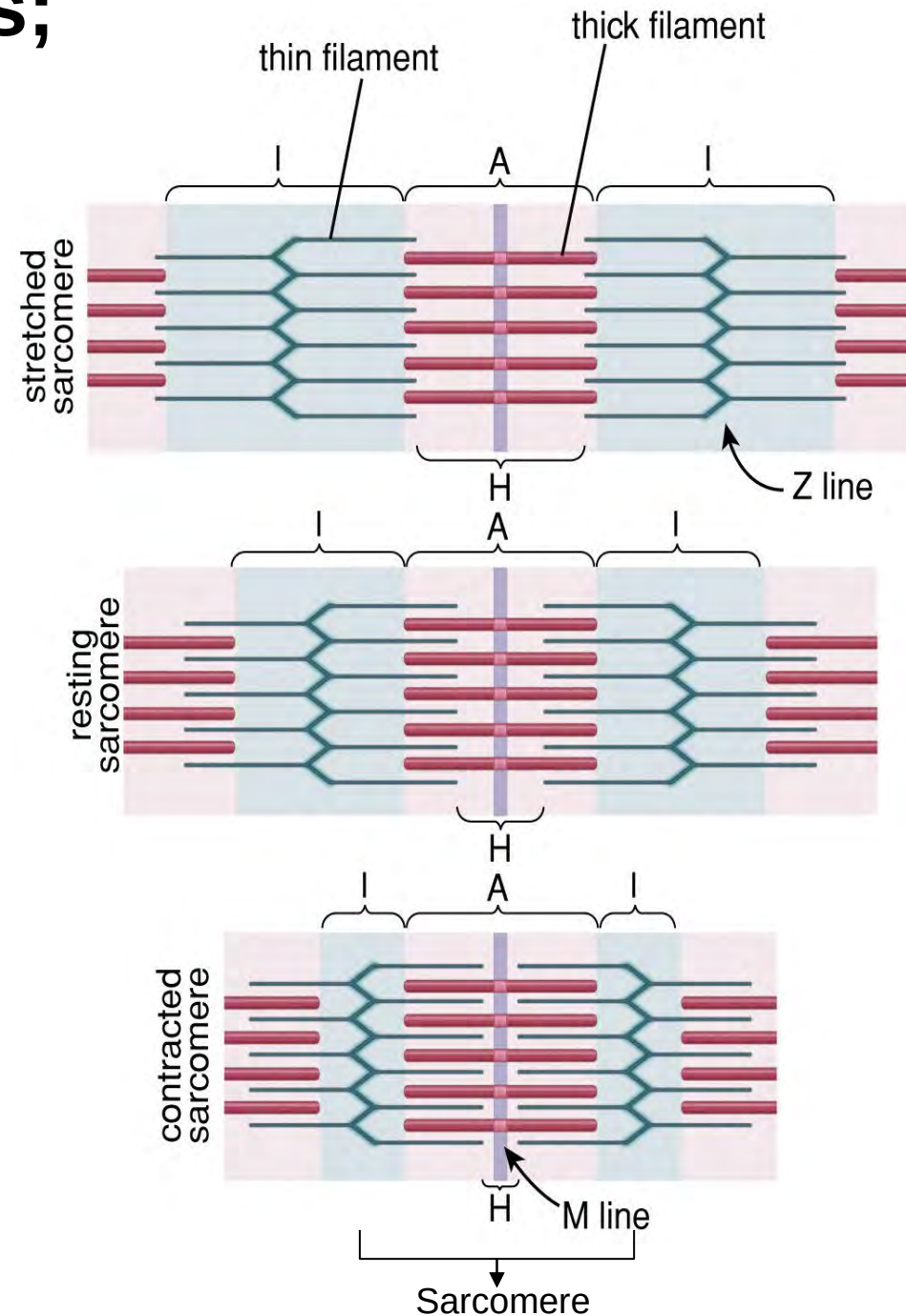


Contraction in muscles;

Sliding filament theory

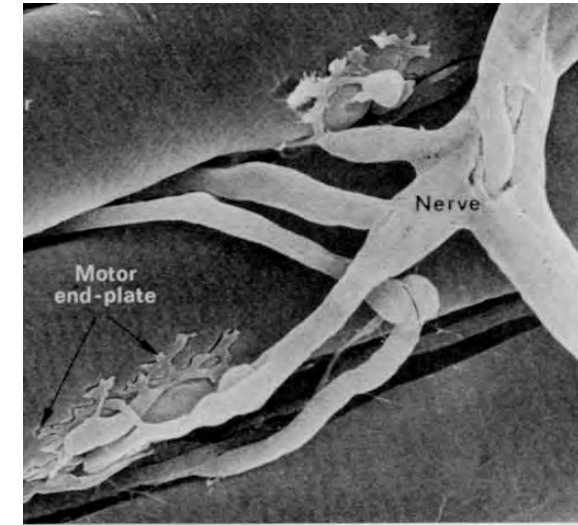


- Skeletal muscle
 - All muscle cells are innervated
 - All or none
- Cardiac muscle
 - Some cells are innervated
 - can contract w/o nerve stimulus
- Smooth muscle
 - Some cells are innervated (individual)
 - can contract w/o nerve stimulus

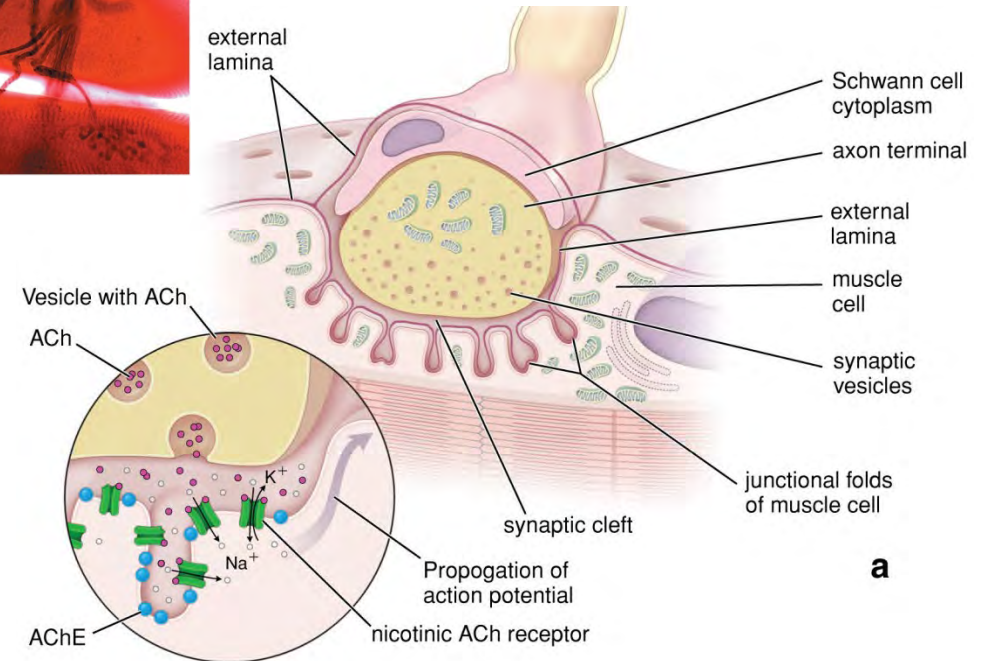
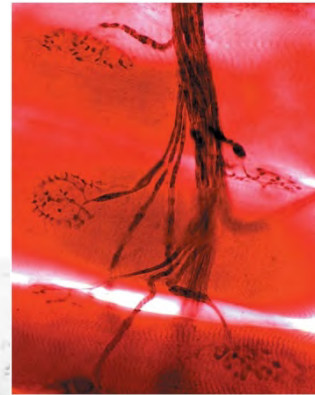


Neuromuscular junction (= motor end plate)

1. Depolarization of neuron
2. depolarization of axon terminal
3. opening of the voltage-gated Ca^{2+} channel in axon terminal
4. influx of Ca^{2+} in axon terminal
5. fusion of synaptic vesicle and presynaptic membrane — **Votulinum toxin**
6. release of acetylcholine(ACh) into synaptic cleft
7. binding to ACh receptor on postsynaptic membrane
8. opening of ligand-gated ion channel (ACh receptor)
9. Influx of Na^+ ions in the muscle cell

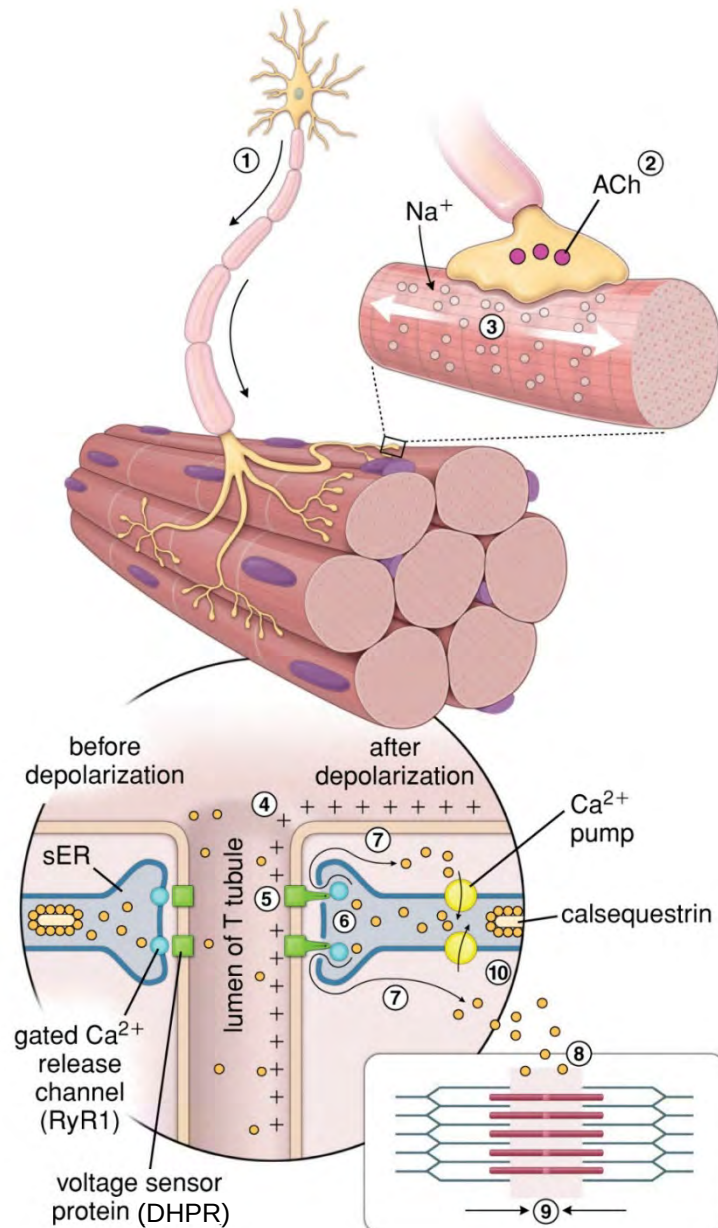


ig micrograph of a motor nerve and two end plates on adjacent muscle fibers. (Microgr
ira. 1981. J. Neurocytol. 10:101.)

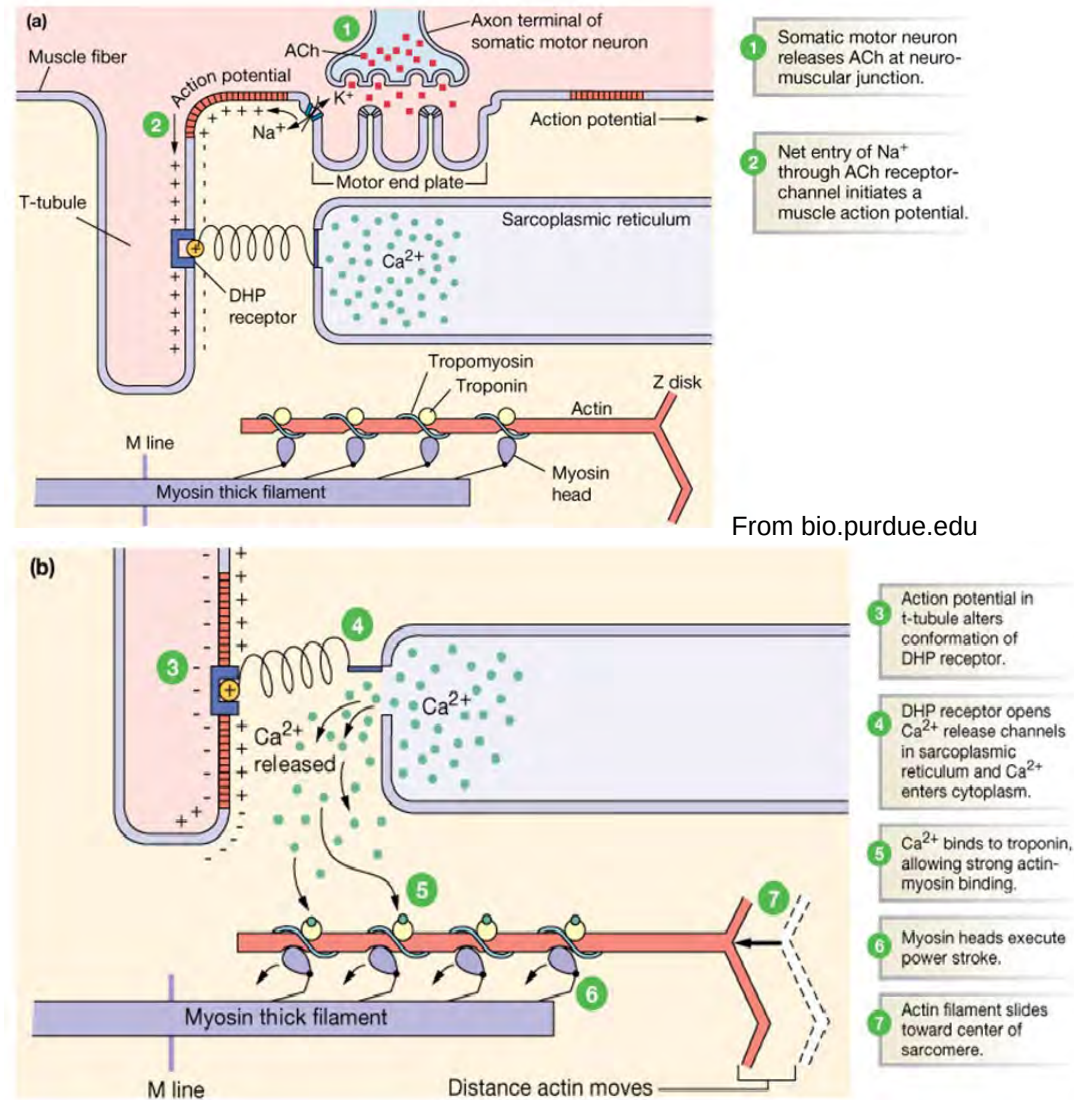


a

10. depolarization of sarcolemma
11. transmission of action potential through T-tubule
12. spread to terminal cisternae(TC) of SR
13. release of Ca^{2+} into cytosol

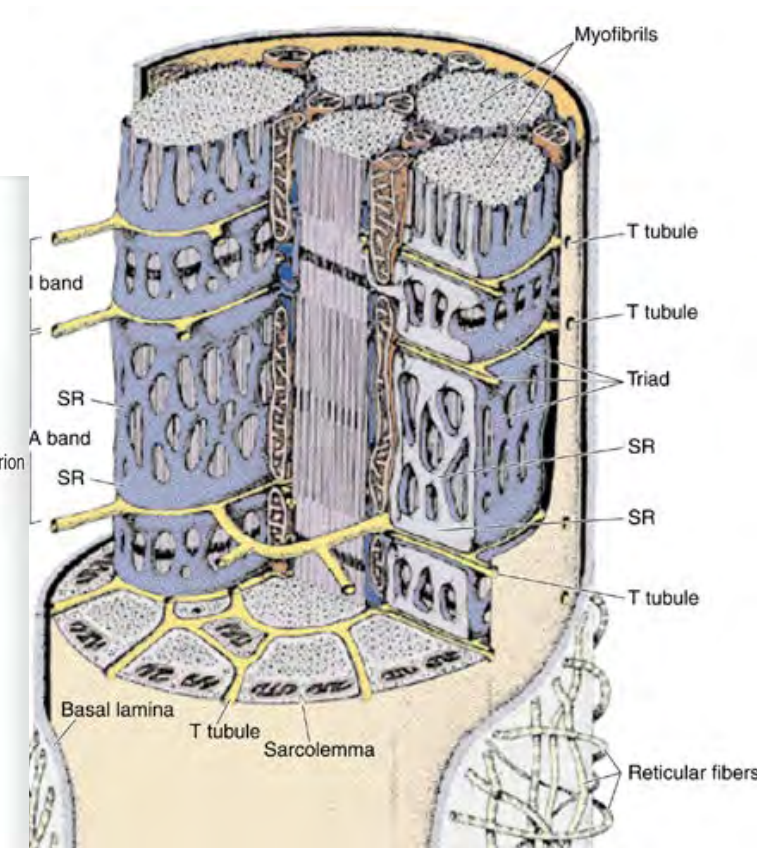
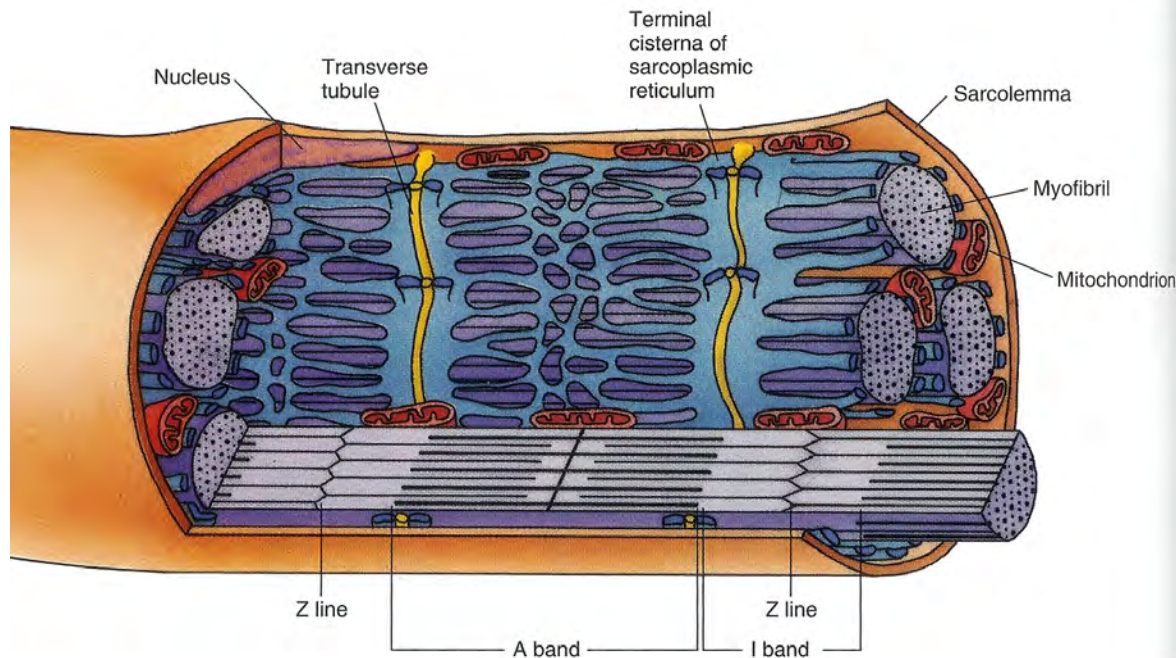
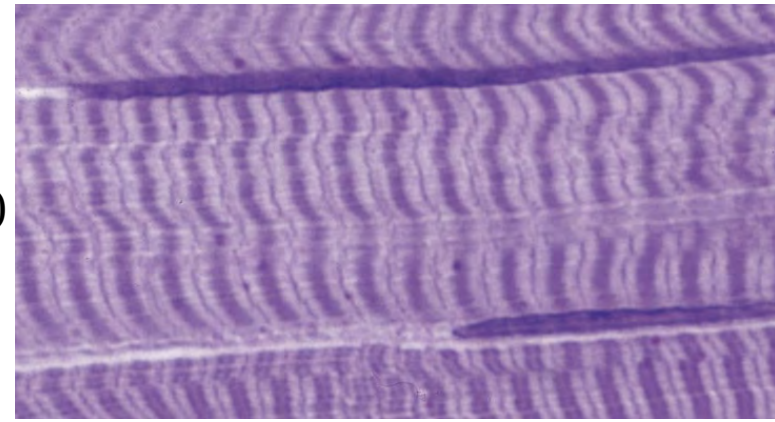


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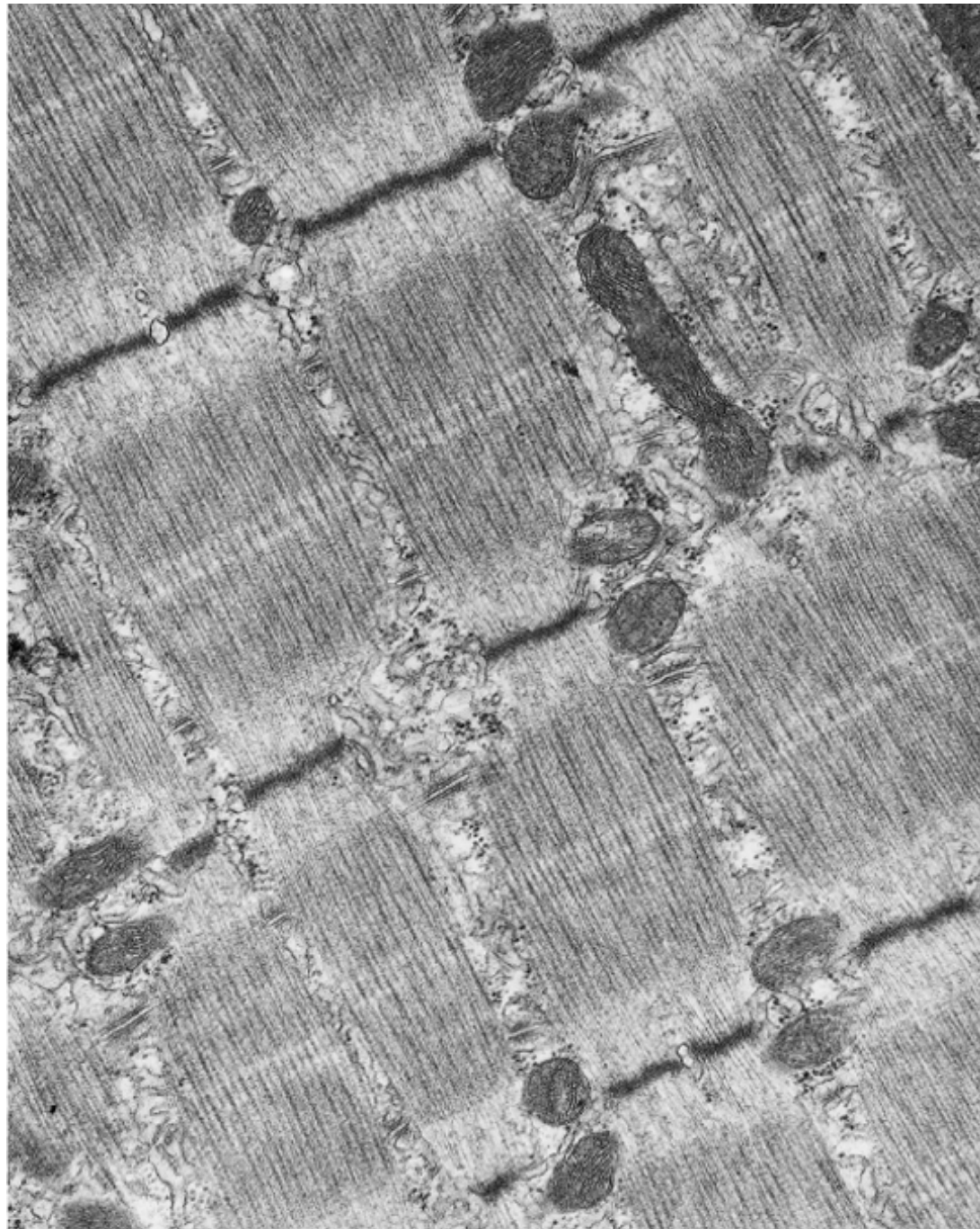
Structures in skeletal muscle cell

- T-tubule
 - Intracellular cell membrane (phospholipid bilayer)
 - conduct depolarizing waves
 - in the ends of myosin filaments
- Sarcoplasmic Reticulum: Smooth ER
 - form meshwork around myofibril
 - Intracellular Ca^{2+} reservoir
- Myofibrils
- Sarcosome

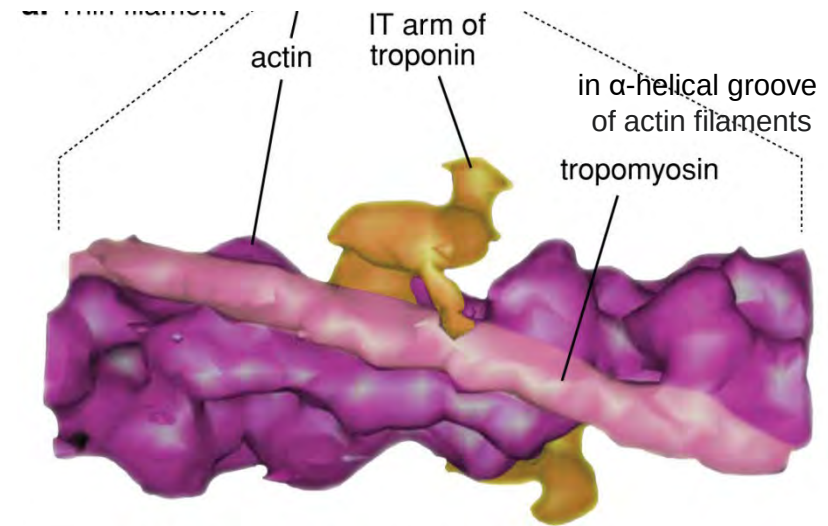
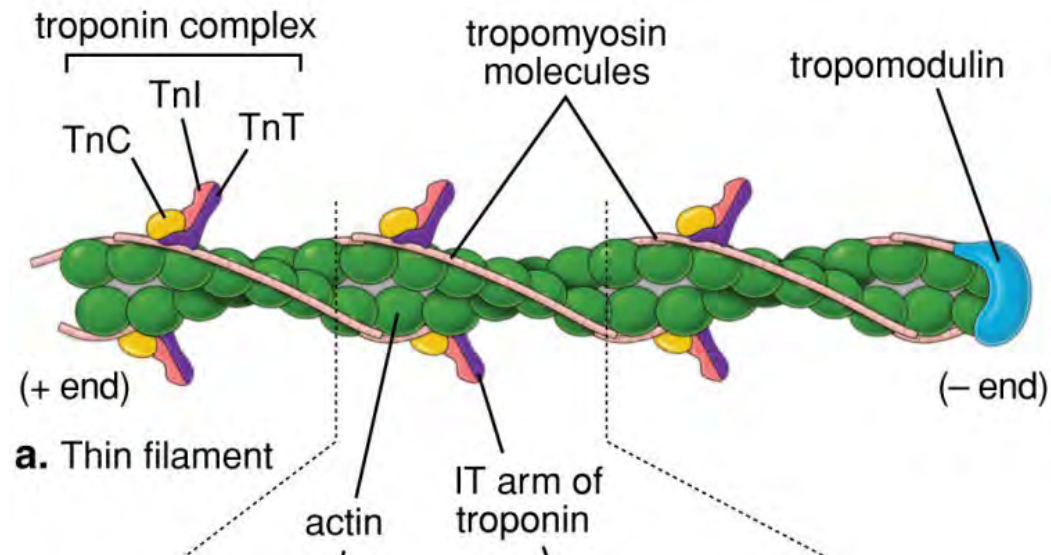
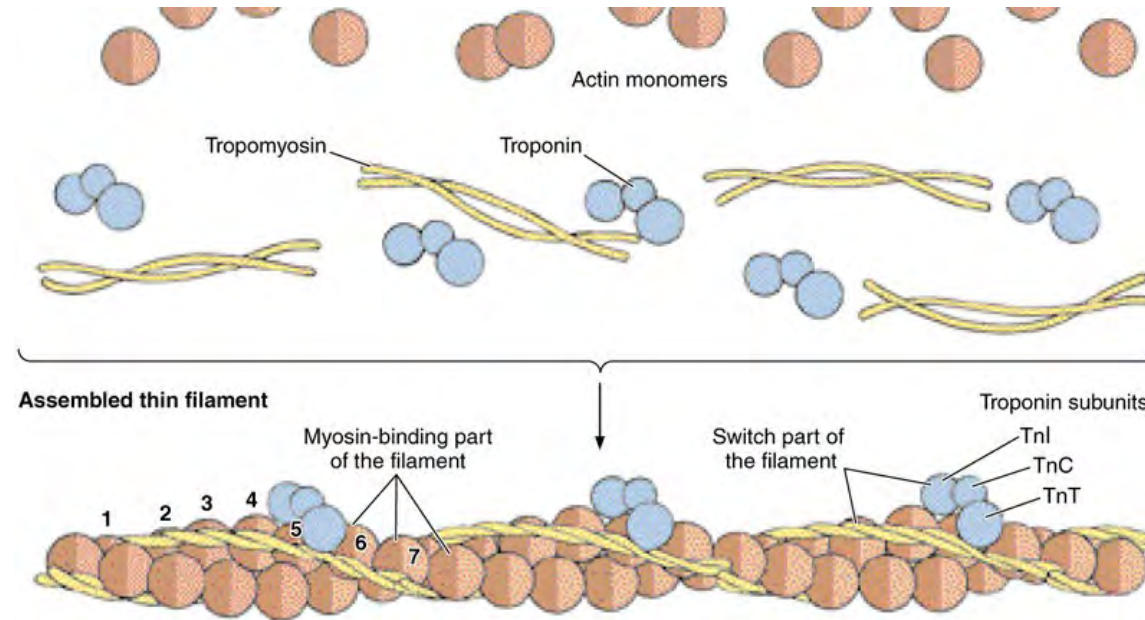


Triad in skeletal muscle

= one tubule + two terminal cisternae



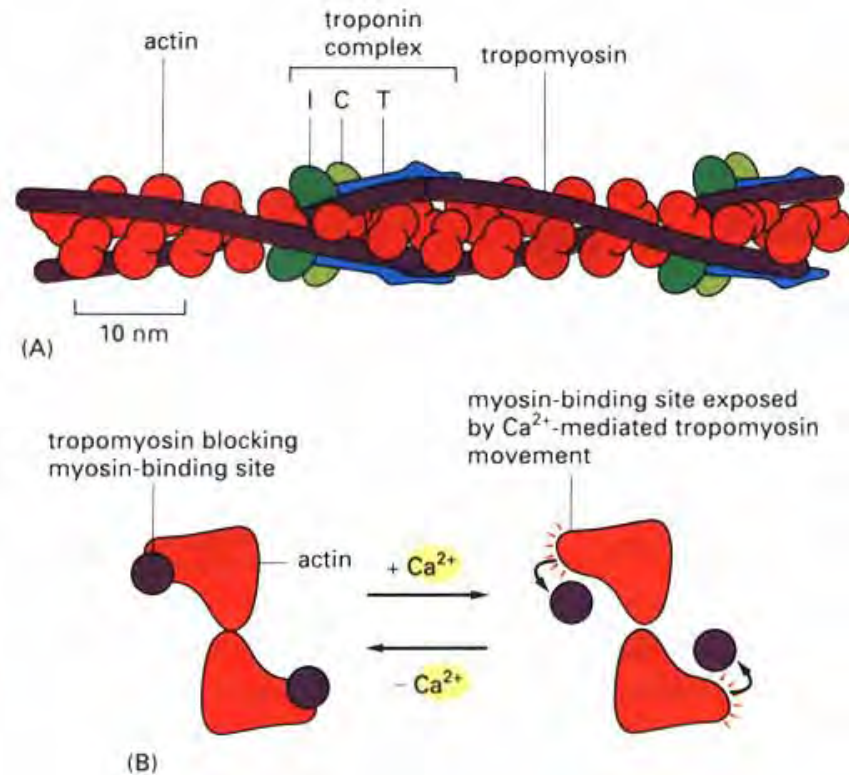
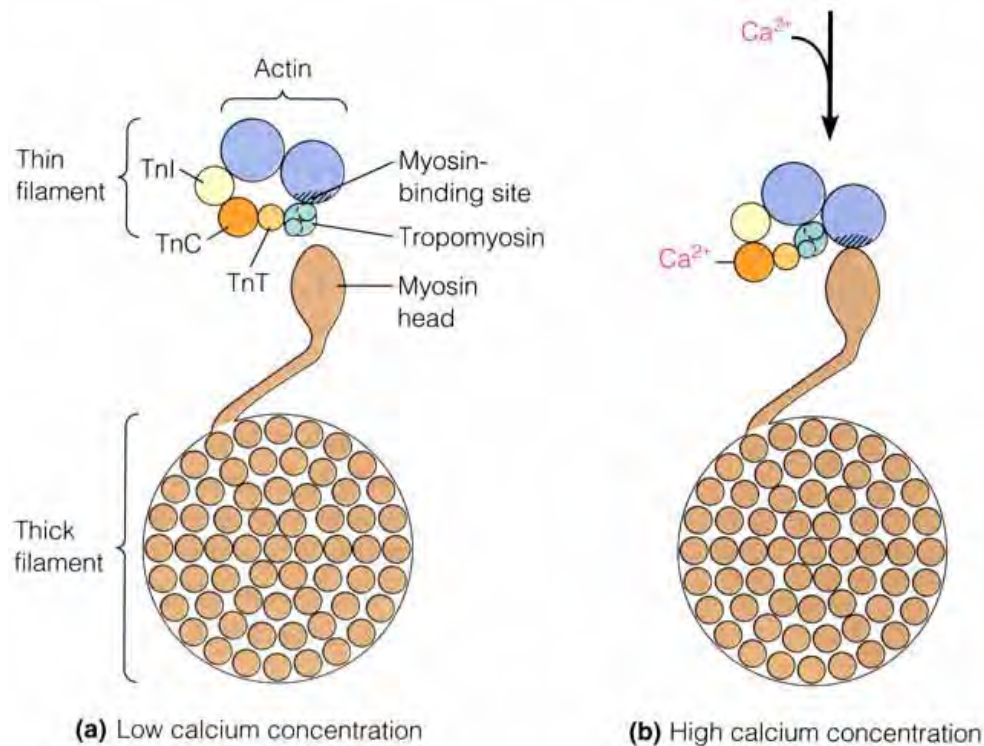
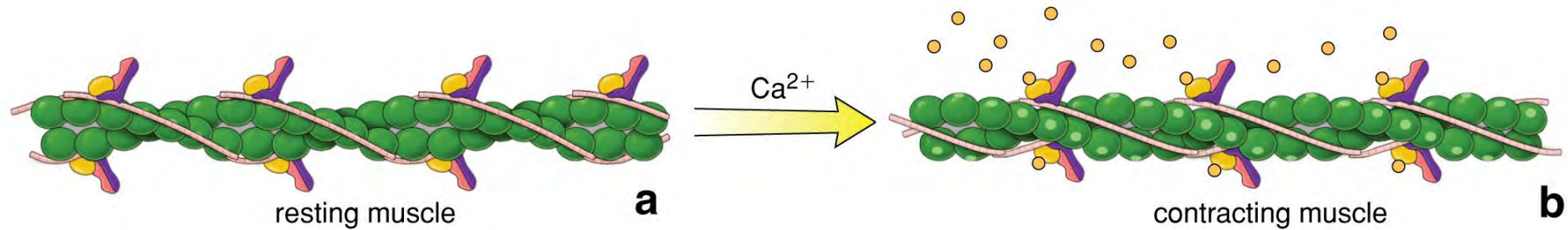
Actin, Tropomyosin, Troponin



b. Cryo-electron micrograph 3D reconstruction

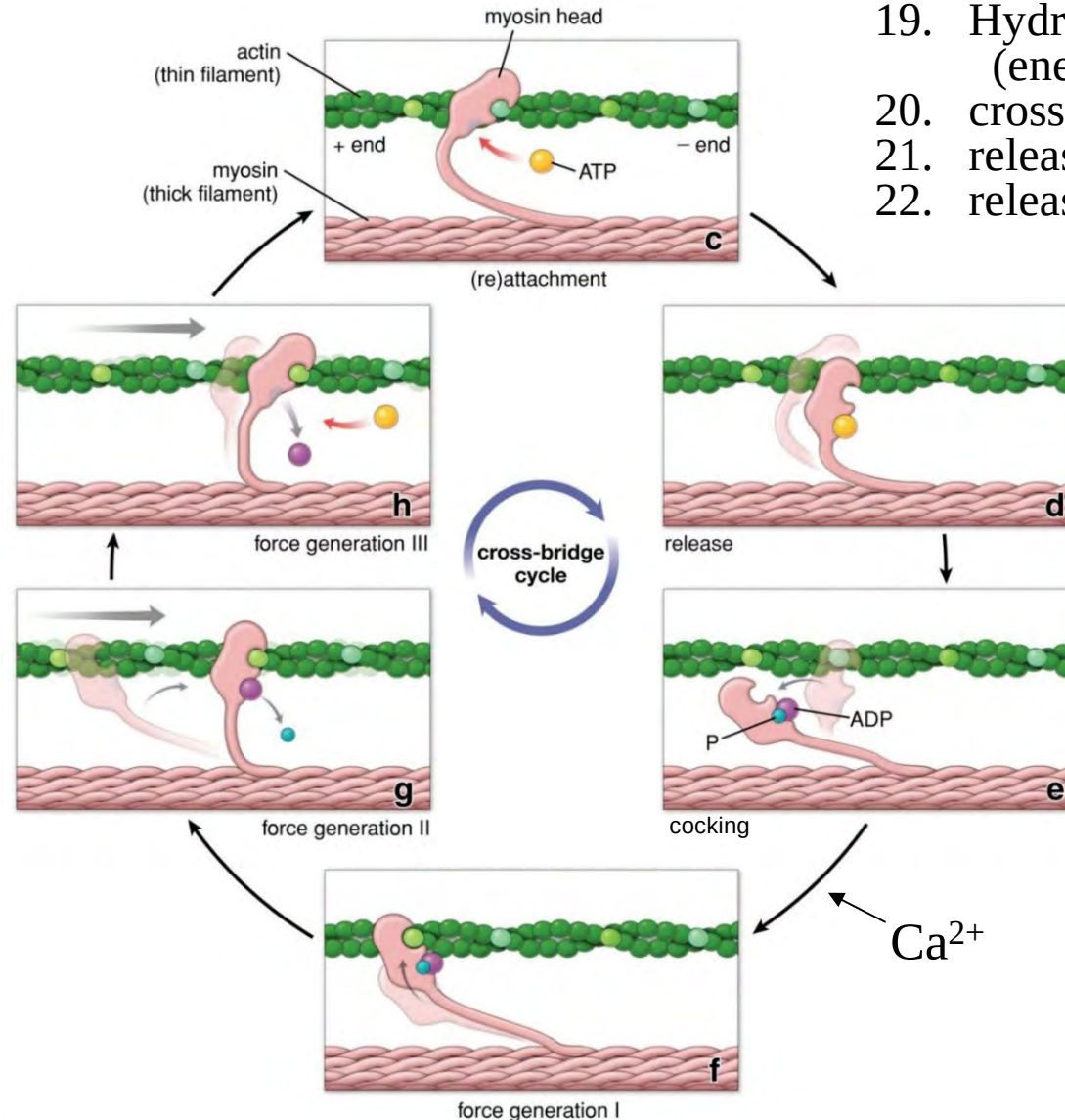
Actin-Myosin interface

14. Ca^{2+} binds to TnC
15. conformational change in TnI
16. positional shift of tropomyosin
17. unmasking of myosin-binding site on actin



Contraction cycles

Repeated contraction & release



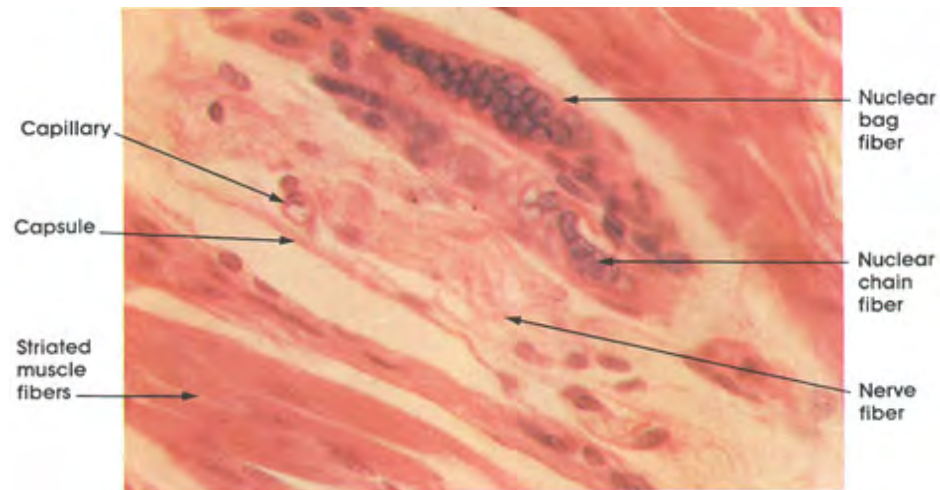
18. ATP binding on myosin head ($\rightarrow$ detach)
19. Hydrolysis of ATP $\rightarrow$ ADP + Pi
(energy release $\rightarrow$ change angle of head)
20. cross-bridge: contact of myosin head to actin
21. release of Pi $\rightarrow$ **powerstroke**
22. release of ADP $\rightarrow$ **powerstroke**

Relaxation in skeletal muscle

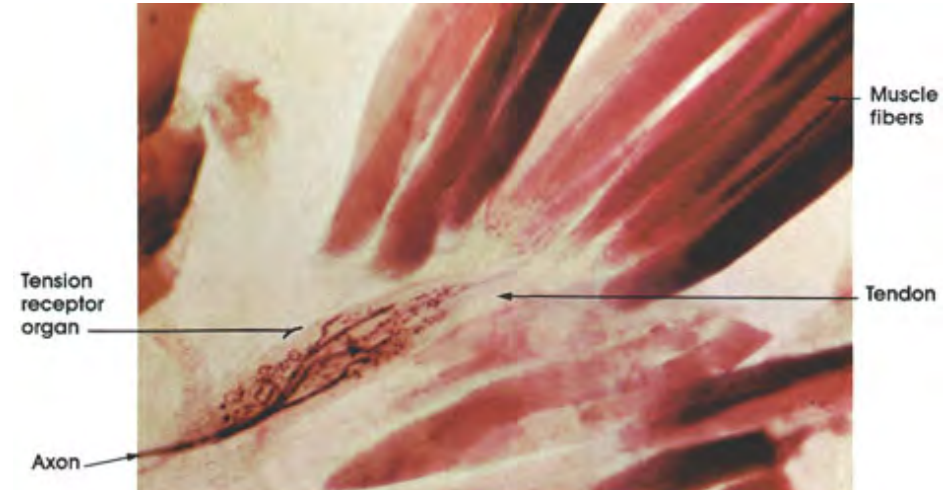
- Skeletal muscle
 - Single stimulus → single response
 - Outside of sarcoplasm:
 - acetylcholinesterase: Ach → acetate + choline
 - In the sarcoplasm
 - In Ca^{2+} pump of Terminal cisternae
 - influx of Ca^{2+} into SR
 - decreased level of Ca^{2+}
 - release Ca^{2+} from TnC
 - masking of myosin-binding sites in actin filament by tropomyosin

Sensory receptors in skeletal muscle

- Encapsulated sensory receptor : proprioceptors
- Organs
 - Muscle spindle (between muscle fibers): muscle fibers + nerve fibers
 - detect stretch in muscle length
 - Golgi tendon organ (between collagen fibers in tendon): nerve fibers
 - detect tension in tendon
- A component of Reflex system
 - Upto spinal cord: spinal reflex systems
 - Upto cerebellum or cerebral cortex



Muscle spindle 20 μm

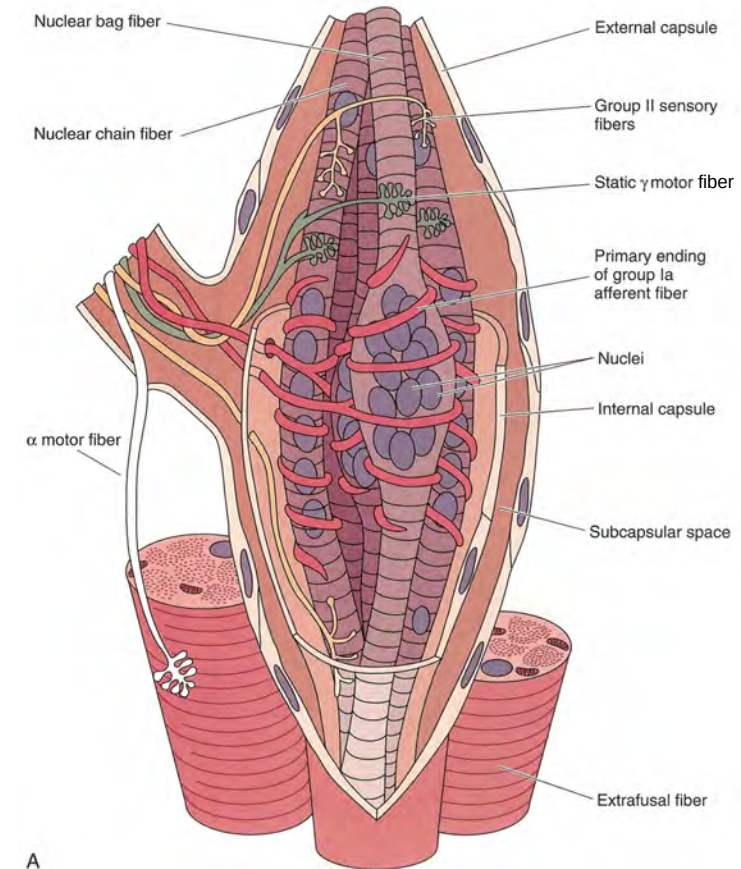
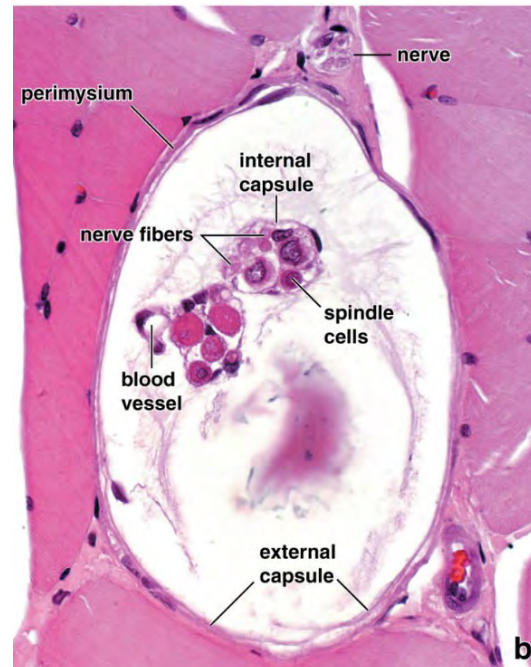
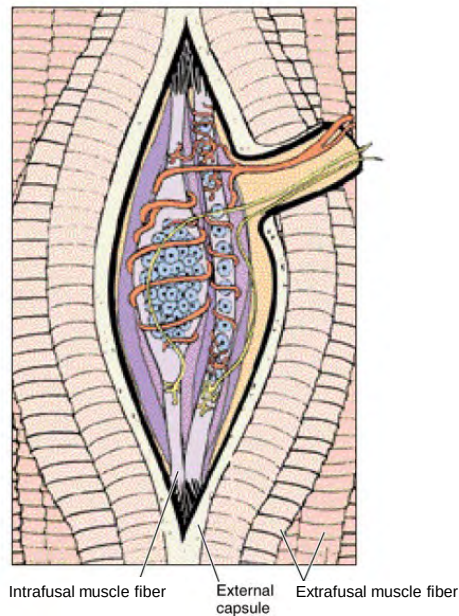


Golgi tendon organ 100 μm

Muscle spindle

- Function: constantly monitors
 - Alteration in muscle length and alteration rate (fast or slow)
- Structures
 - Capsule (external and internal)
 - Intrafusal (muscle) fiber : 8~10 muscle fibers
 - Nuclear bag fiber (static and dynamic)
 - Nuclear chain fiber
 - Nerve fibers: sensory (Ia, II), motor (γ)
 - Fluid (GAG-rich)

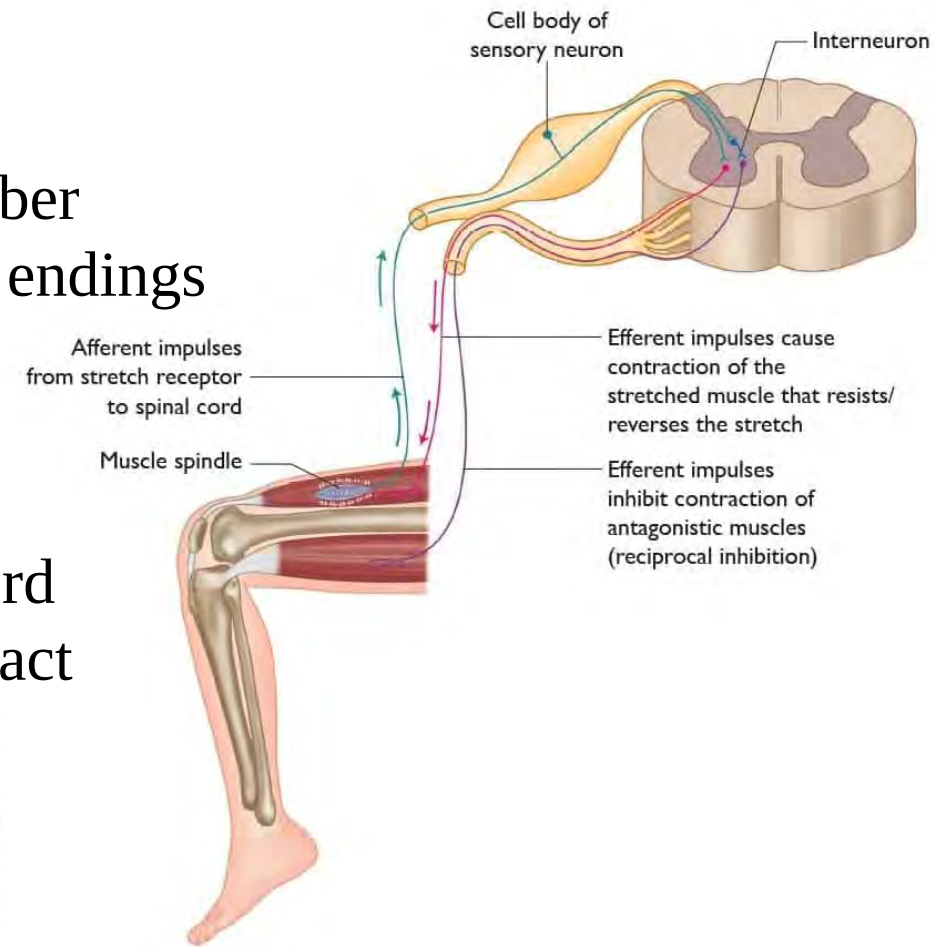
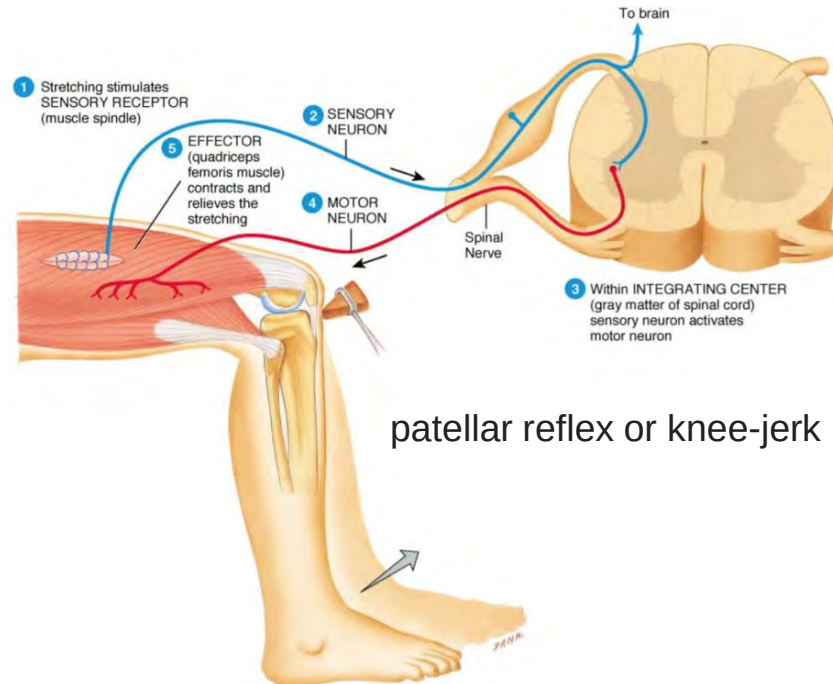
Fusiform = spindle shape



- # Stretch reflex

Stretch of muscle

- stretch of intrafusal muscle fiber
- stimulation of sensory nerve endings (Ia and II nerve fiber)
- dorsal root ganglion
- (interneuron in spinal cord)
- α -motor neurons in spinal cord
- extrafusal muscle fiber contract



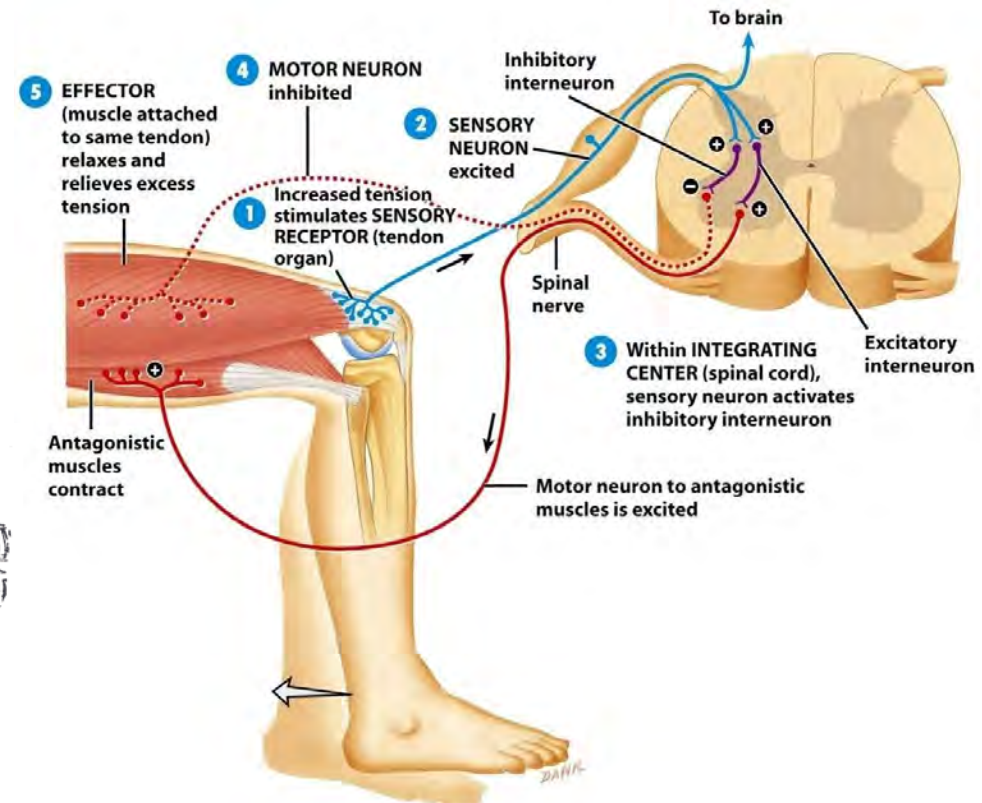
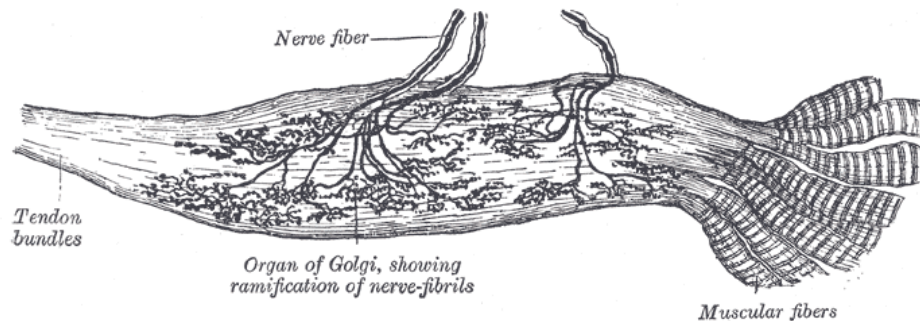
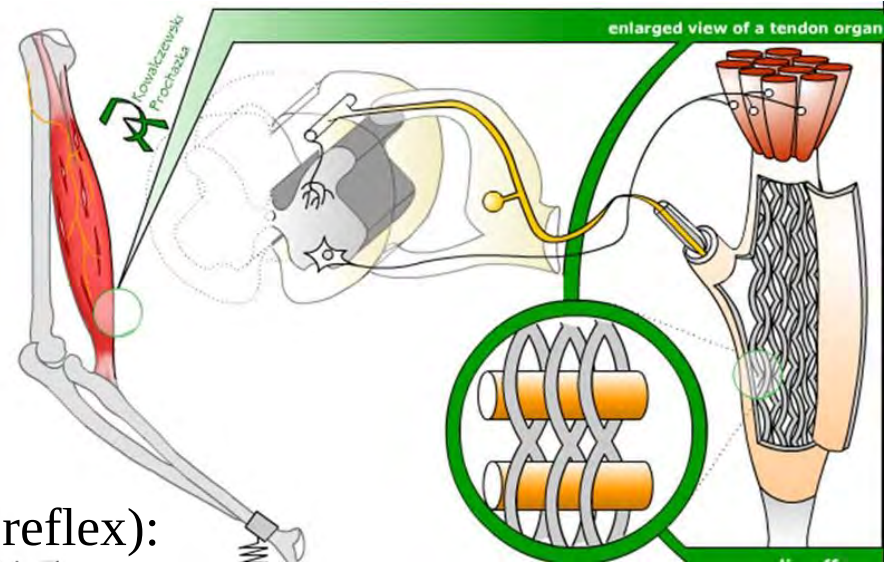
Exercise-related muscle cramps (ERMC)

errors in nerve-muscle reflex arc

- Golgi tendon organs: inhibited
- Muscles spindles : hyperactive

Golgi tendon organ

- Function: detect tension in tendon
- capsule + type Ib nerve fibers
- Golgi tendon reflex (=inverse stretch reflex):
 - Contraction of muscle
 - tension in tendon
 - tendon organs (Ib fiber)
 - inhibition of α -motor neuron
 - muscle relaxation



Exercise-related muscle cramps (ERMC)

- Etiology
 - Traditionally, known to arise from dehydration, electrolyte imbalances (including magnesium, potassium and sodium), accumulation of lactic acid, or low cellular energy levels. → however, minimal scientific value
 - Recently, most likely involves the nerve-muscle reflex arc in skeletal muscles
 - CNS fatigue or overuse of feedback communication with muscles
 - the loss of inhibitory activity of the central nervous system (CNS) reflexes
 - Golgi tendon organs may become inhibited and muscles spindles can become hyperactive
 - sustained activation of the muscle
- influencing factors
 - poor flexibility
 - prolonged sitting, poor or abnormal posture or inefficient biomechanics (all of which may be related to).
 - Abnormal mechanical load
 - increased body weight, improper footwear, eccentric muscle contraction and other musculoskeletal injuries
 - Aging

Types of Nerve fiber

Motor fiber types

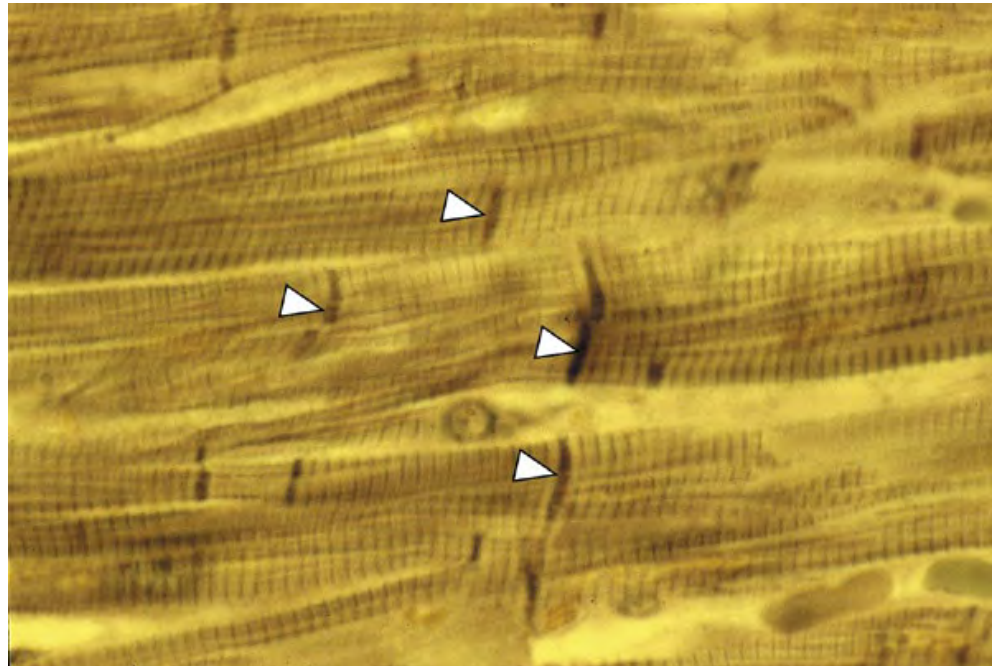
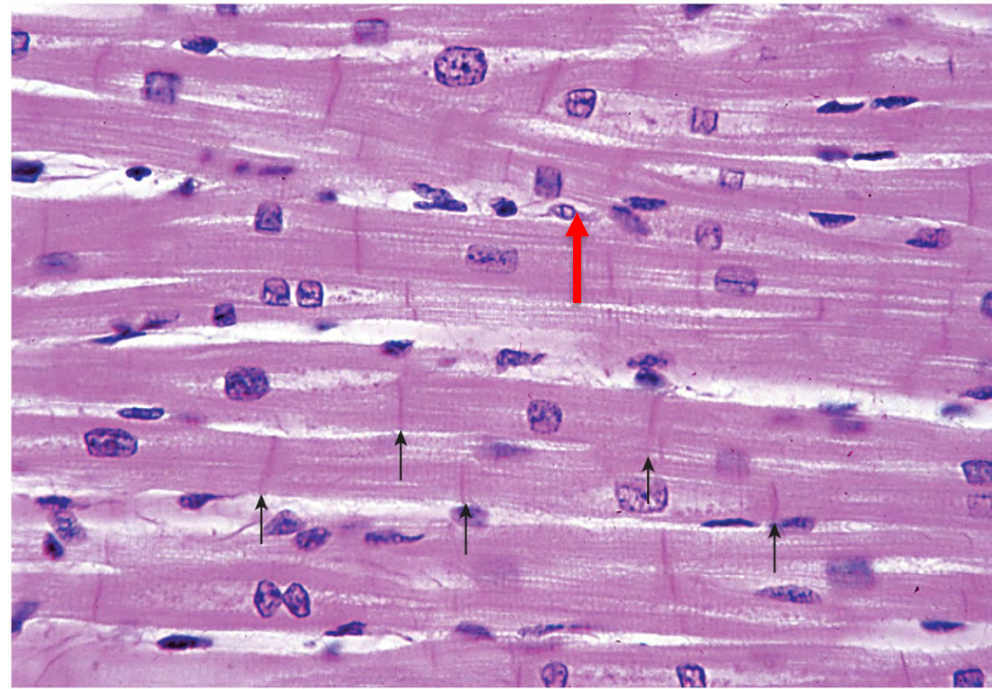
Type	Erlanger-Gasser Classification	Diameter	Myelin	Conduction velocity	Associated muscle fibers
α	A α	13-20 μm	Yes	80–120 m/s	Extrafusal muscle fibers
γ	A γ	5-8 μm	Yes	4–24 m/s ^{[12][13]}	Intrafusal muscle fibers

Sensory fiber types

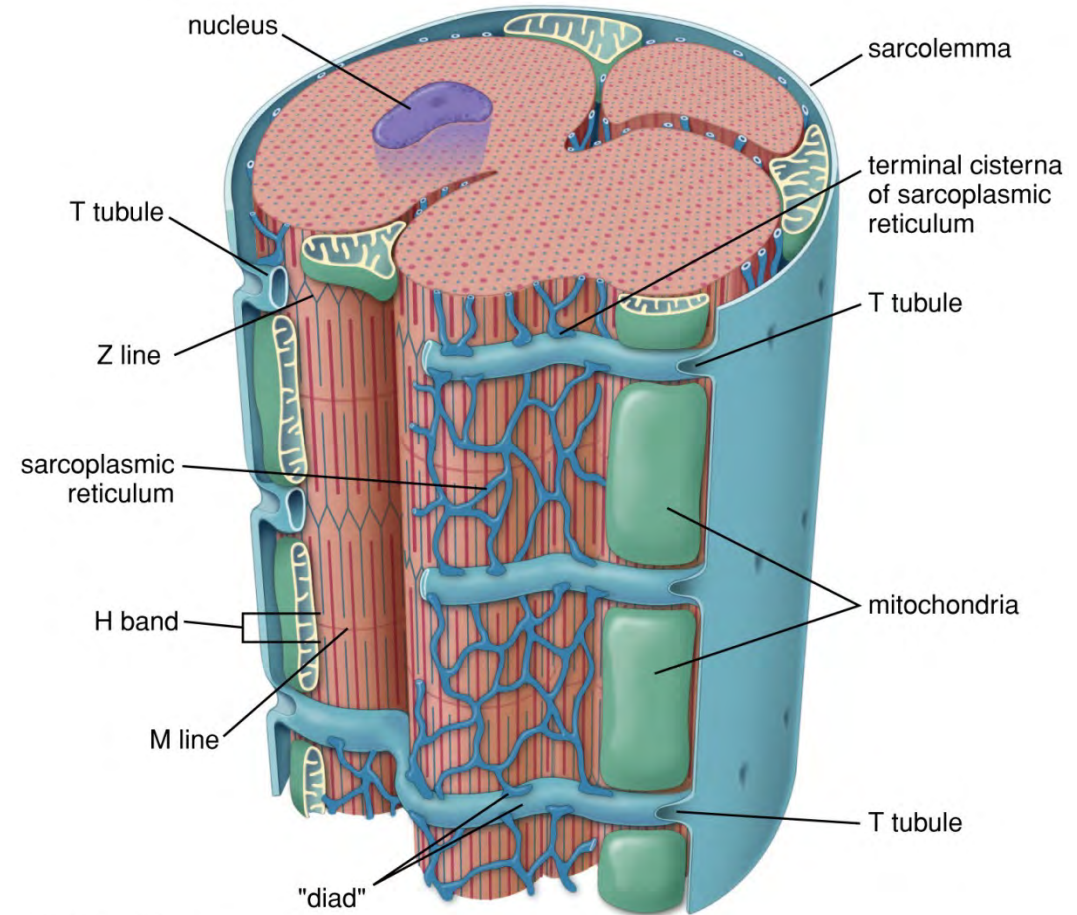
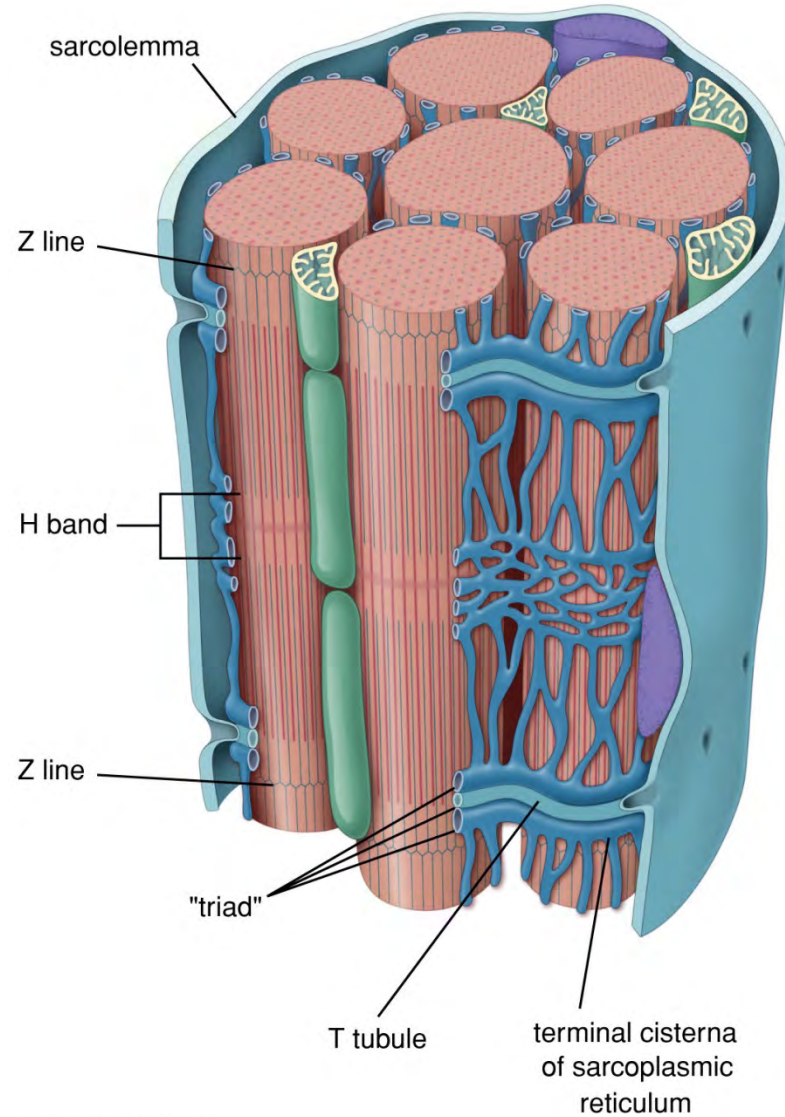
Type	Erlanger-Gasser Classification	Diameter	Myelin	Conduction velocity	Associated sensory receptors
Ia	A α	13-20 μm	Yes	80–120 m/s	Primary receptors of muscle spindle
Ib	A α	13-20 μm	Yes	80–120 m/s	Golgi tendon organ
II	A β	6-12 μm	Yes	33–75 m/s	Secondary receptors of muscle spindle All cutaneous mechanoreceptors
III	A δ	1-5 μm	Thin	3–30 m/s	Free nerve endings of touch and pressure Nociceptors of neospinothalamic tract Cold thermoreceptors
IV	C	0.2-1.5 μm	No	0.5-2.0 m/s	Nociceptors of paleospinothalamic tract Warmth receptors

Cardiac Muscle

- Cell size: 15 μ m dia, 80 μ m long
- 1-2 central nuclei
- Striation
- Basal lamina + Reticular fiber
- Intercalated disks :
 - cell-cell junctions
- highly vascularized
- Myoglobin and Glycogen
- Rich Mitochondria (halo)
- T-tubule: large diameter
- SR: less in number and small terminal → Not triad but diad



Skeletal muscle cell vs Cardiac muscle cell; nucleus, T tubule, SR and mitochondria

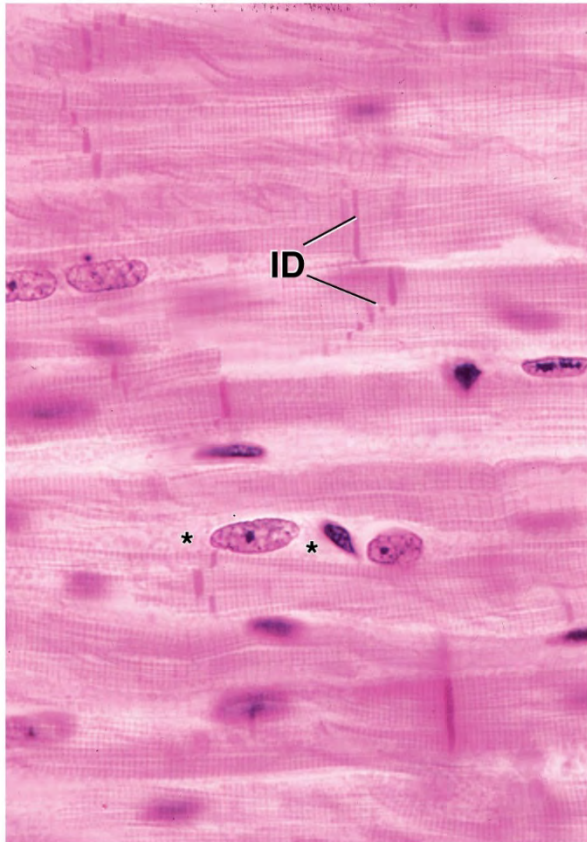


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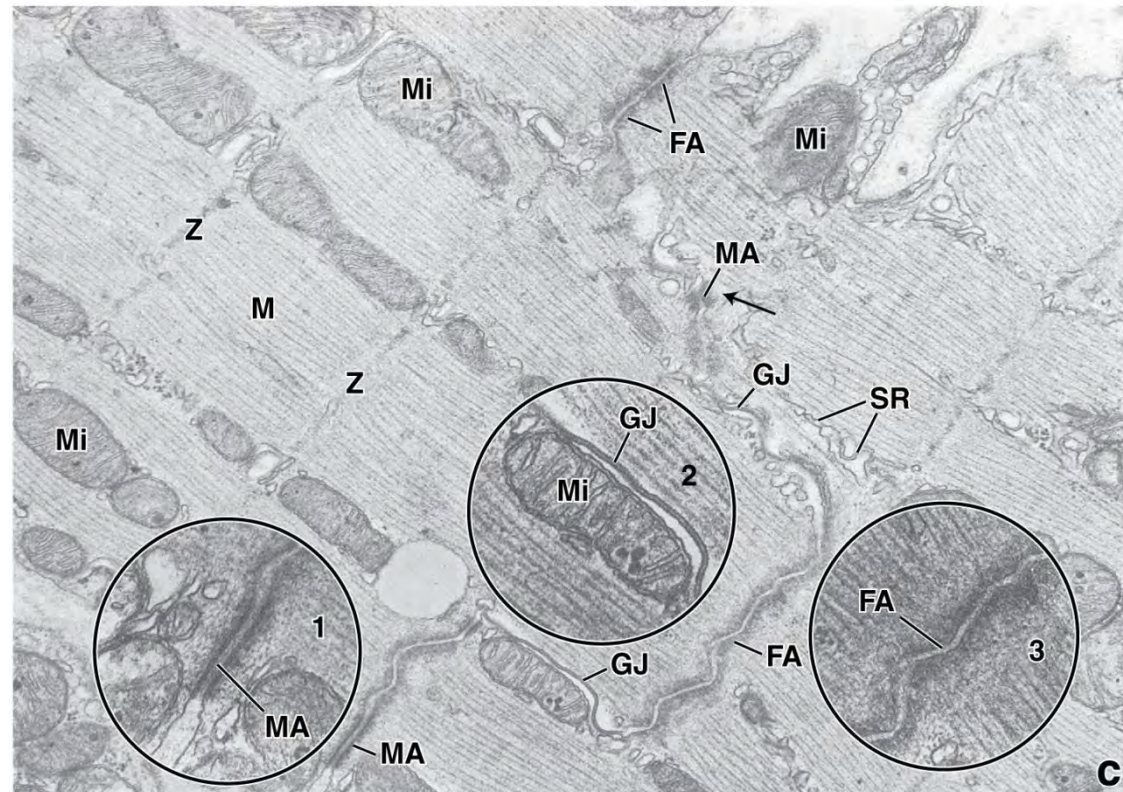
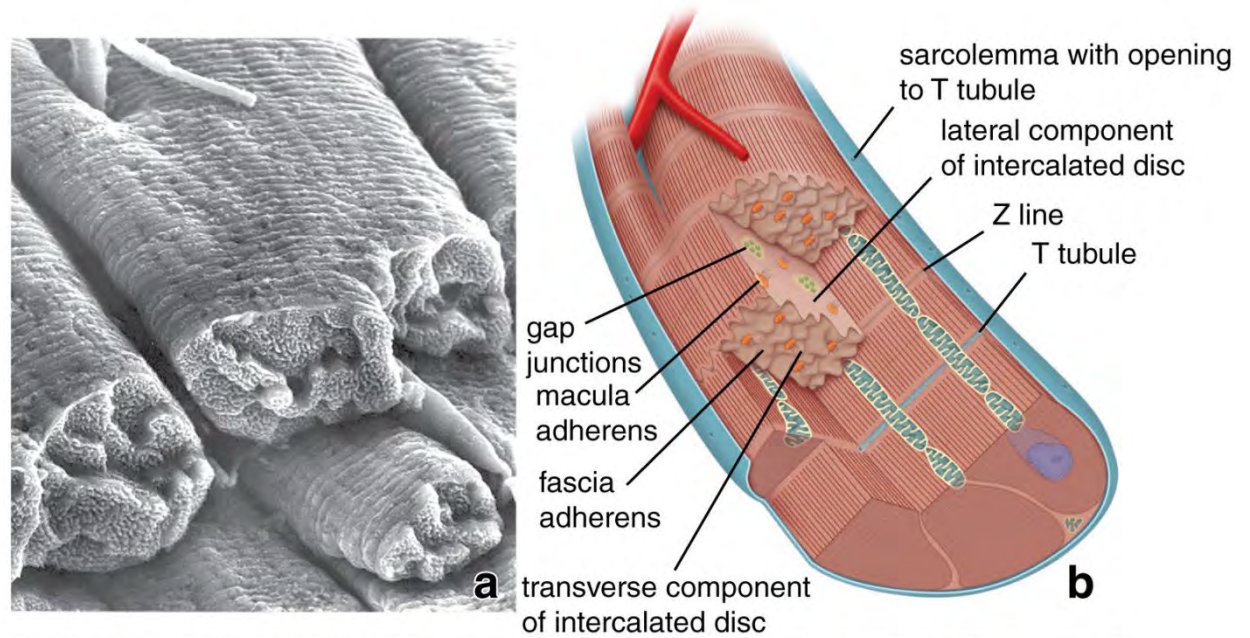
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Intercalated disk

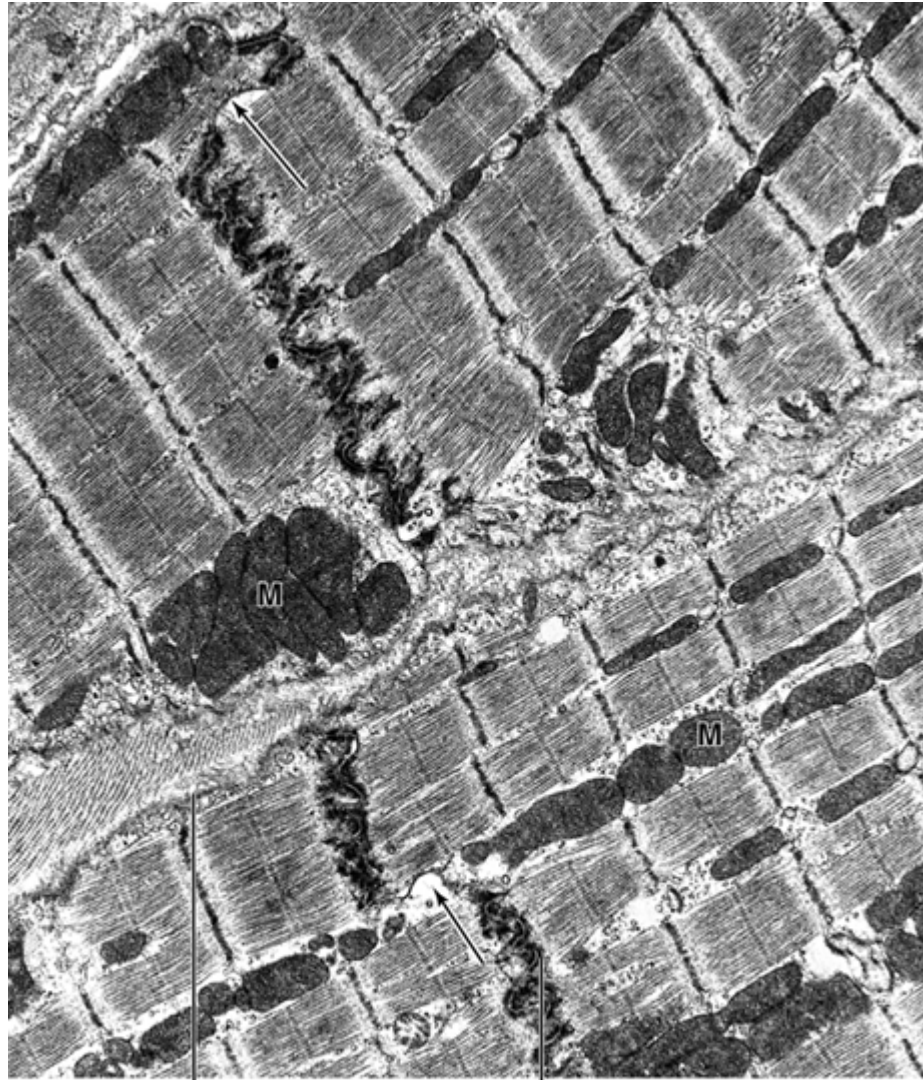
- Transverse portion (가로부분)
 - cells joined end-to-end
 - desmosome, fascia adherens
- Lateral portion (세로부분)
 - gap junction



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Intercalated disc in Cardiac muscle



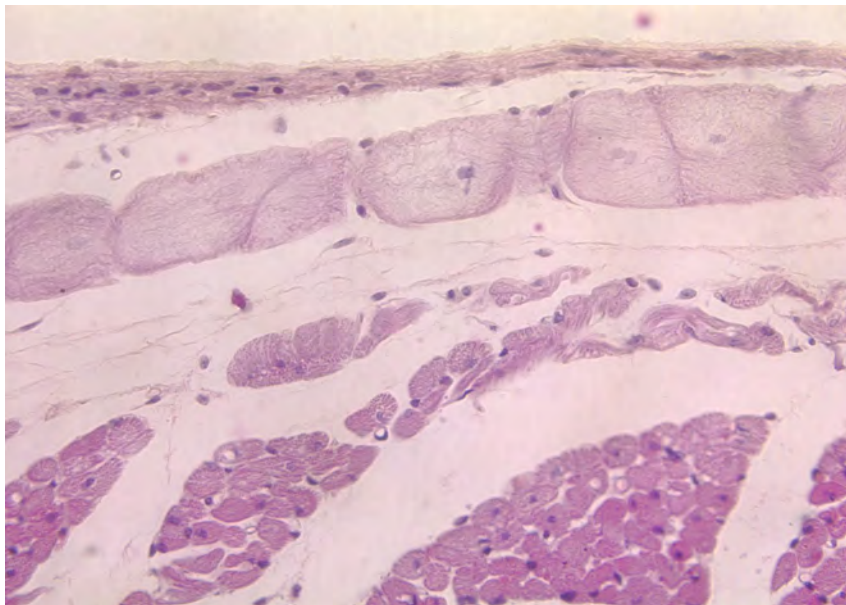
Fibrils of
reticular
fibers

Intercalated
disc

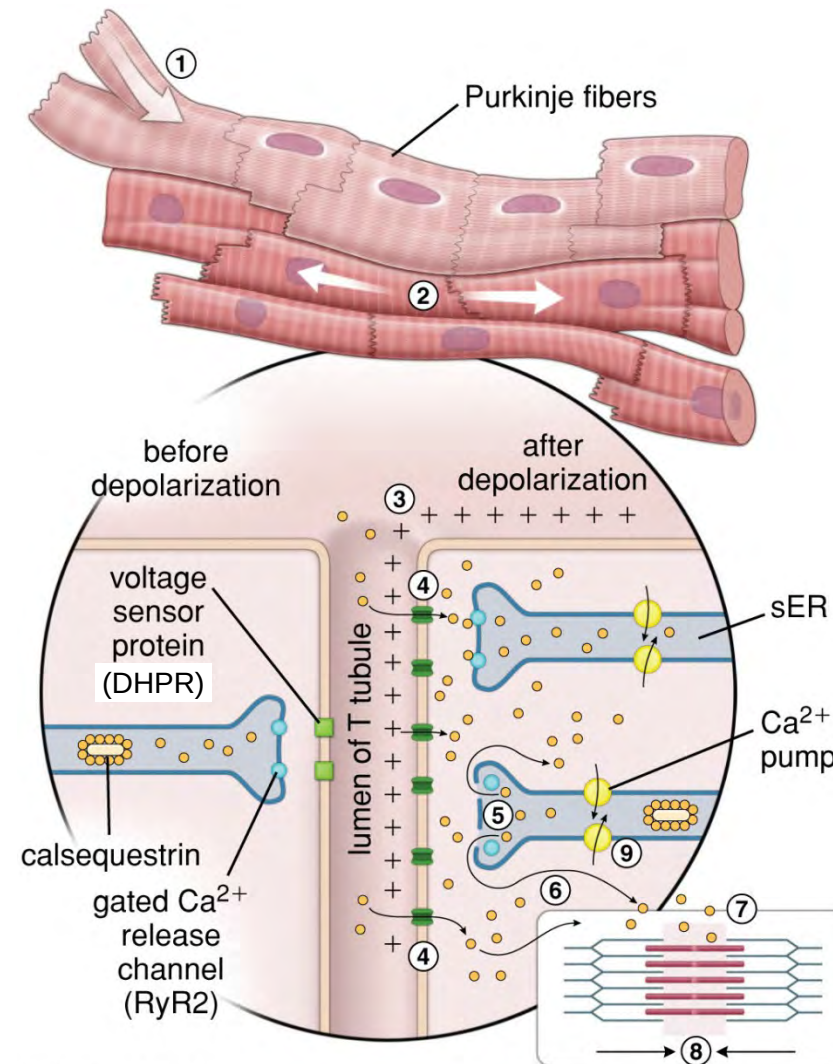


• Purkinje fibers

- Large specialized cardiac muscle cells
- in atrial and ventricular walls
- Nucleus: one or two nuclei
- Cytoplasm: Pale pink: Glycogen & mitochondria
- Striations: weak, poor myofibrils in cytoplasm
- Cell-cell junction: **intercalated disk**
(via gap junctions and desmosomes)
- Function: conduct electric impulse to cardiac muscle cells: contact with cardiac muscle cells



T-tubule: major source of Ca^{2+}
SR: minor source of Ca^{2+}



Impulse generating & impulse-conducting system

- initiate and regulate the heart beat
- specialized cardiomyocytes

1. Sinoatrial (SA) node

- the pacemaker of the heart do not contract robustly.

2. Atrioventricular (AV) node

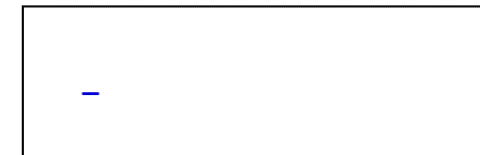
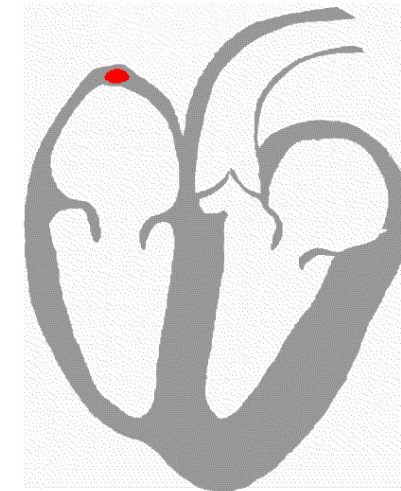
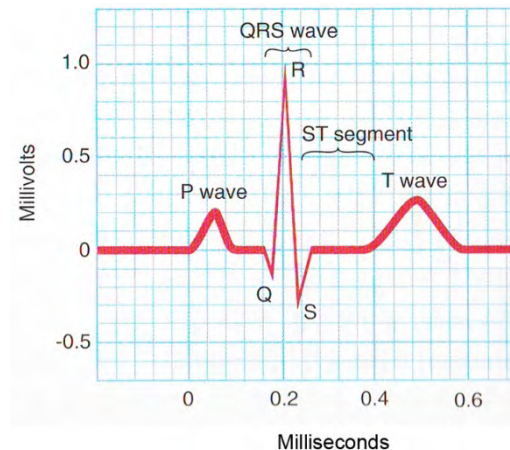
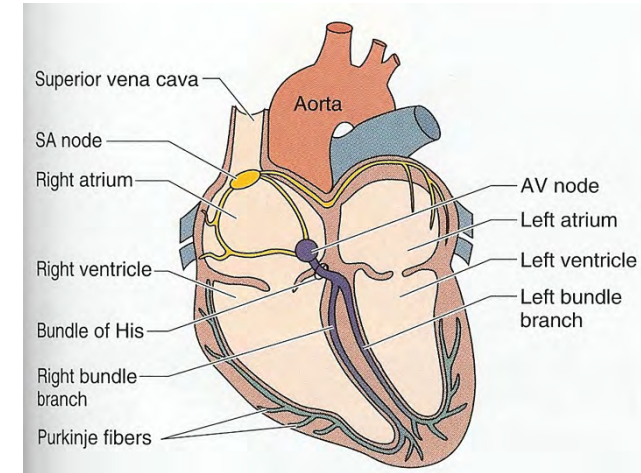
- between the atria and the ventricles

3. Atrioventricular bundle (AV bundle , or bundle of His)

- a band of conducting tissue
- radiating from the AV node into the interventricular septum

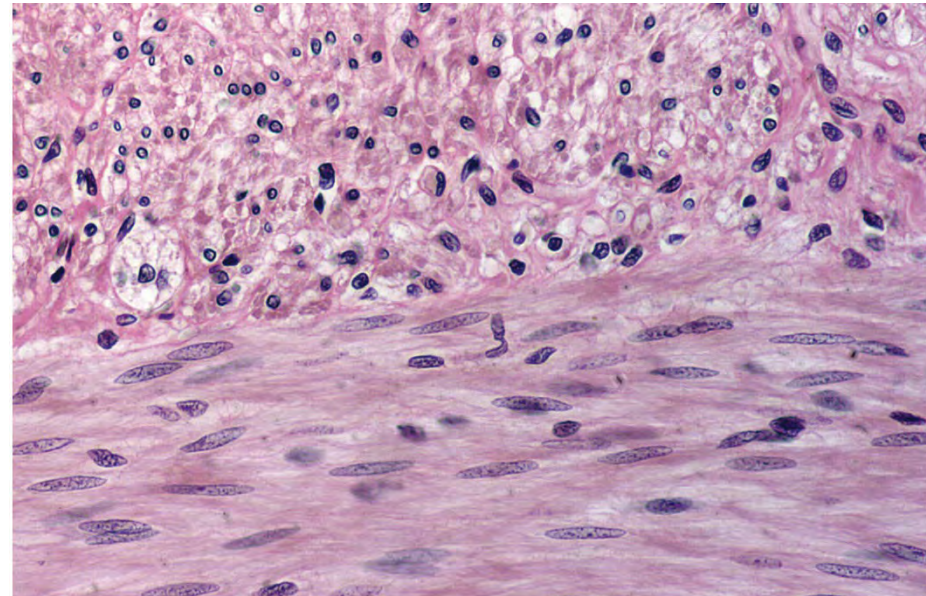
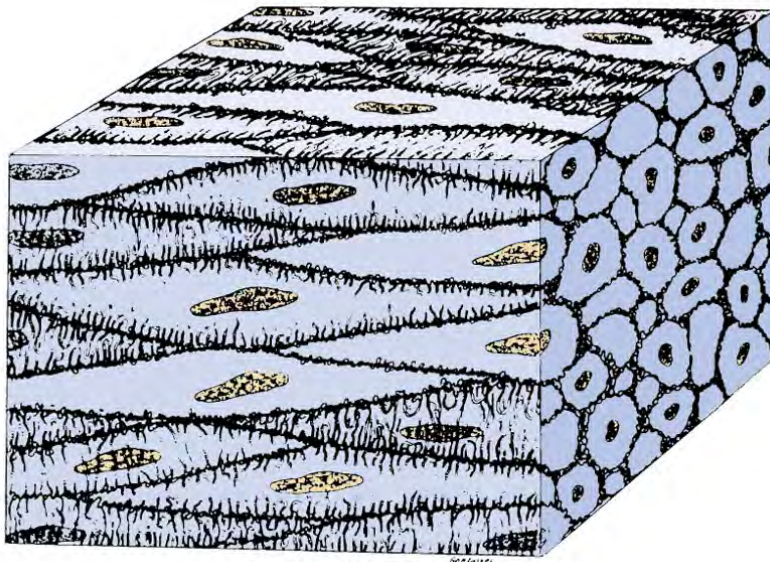
4. Purkinje fibers

- at endocardium



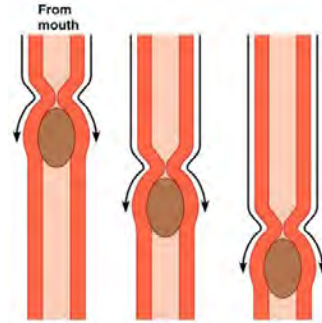
Smooth Muscle

- 20 μm ~ 500 μm long, 5~10 μm dia
- Spindle shape (fusiform), taper
- One central nucleus
- Cell division
- No striation (No microfibril: irregular arrangement of actin and myosin filaments)
- Basal lamina + Reticular fiber
- Location :
 - walls of hollow viscera, blood vessel, compound glands, respiratory tract, dermis of skin, digestive tract, reproductive tract, ureter and bladder

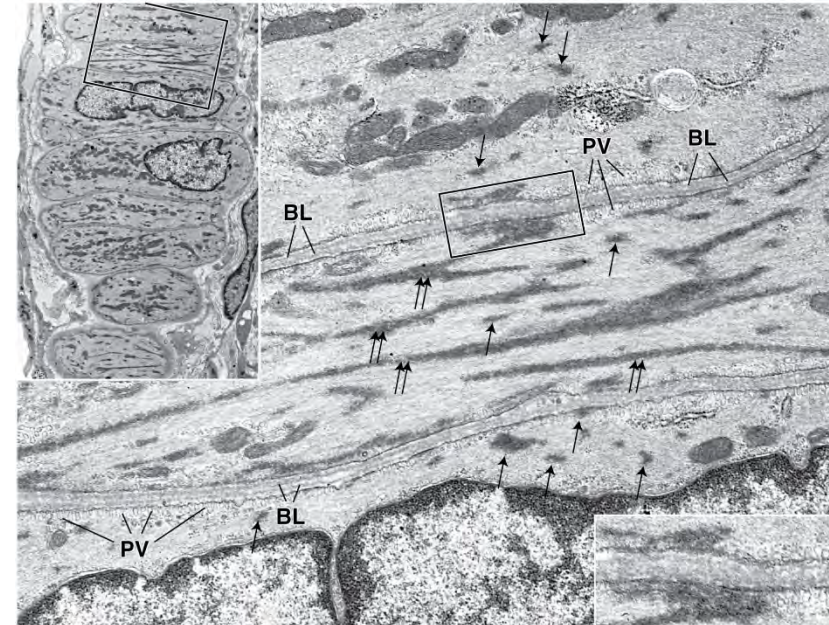
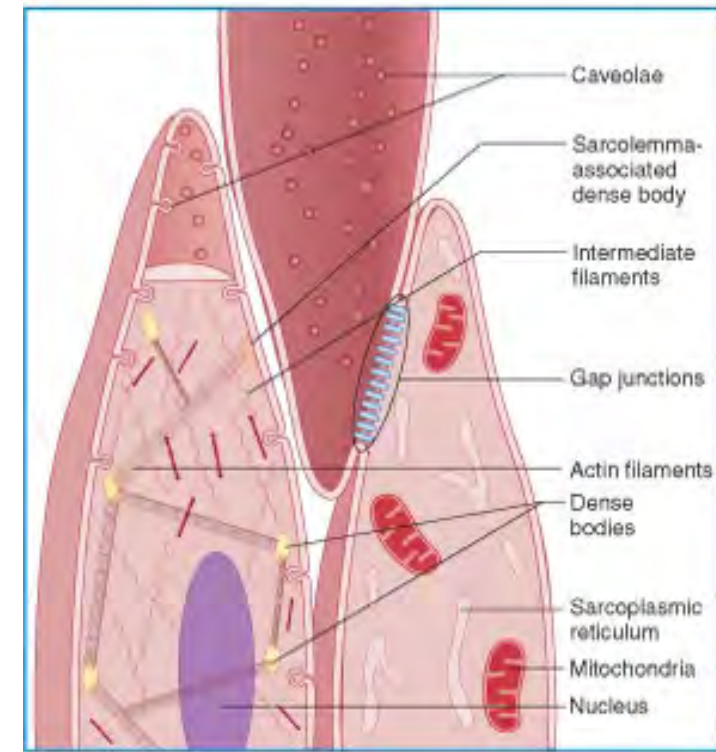


Smooth Muscle

- Structures
 - Many gap junctions
 - No T-tubule but Caveola(e)
 - Dense bodies connected by actin-myosin filaments



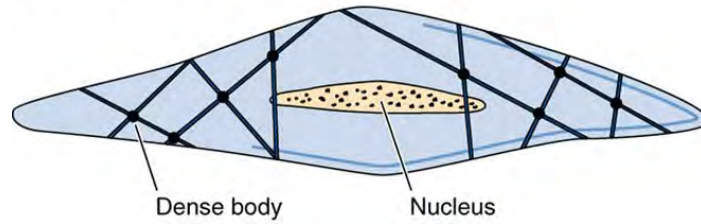
- Type
 - Single-unit smooth muscle (common)
 - nerve supply on a few cells
 - Gap junctions
 - induce wave contraction ‘Peristalsis’
 - Multiunit smooth muscle :
 - as distinct individual organs/structures
 - Each cell innervated by a nerve fiber



Smooth muscle cell

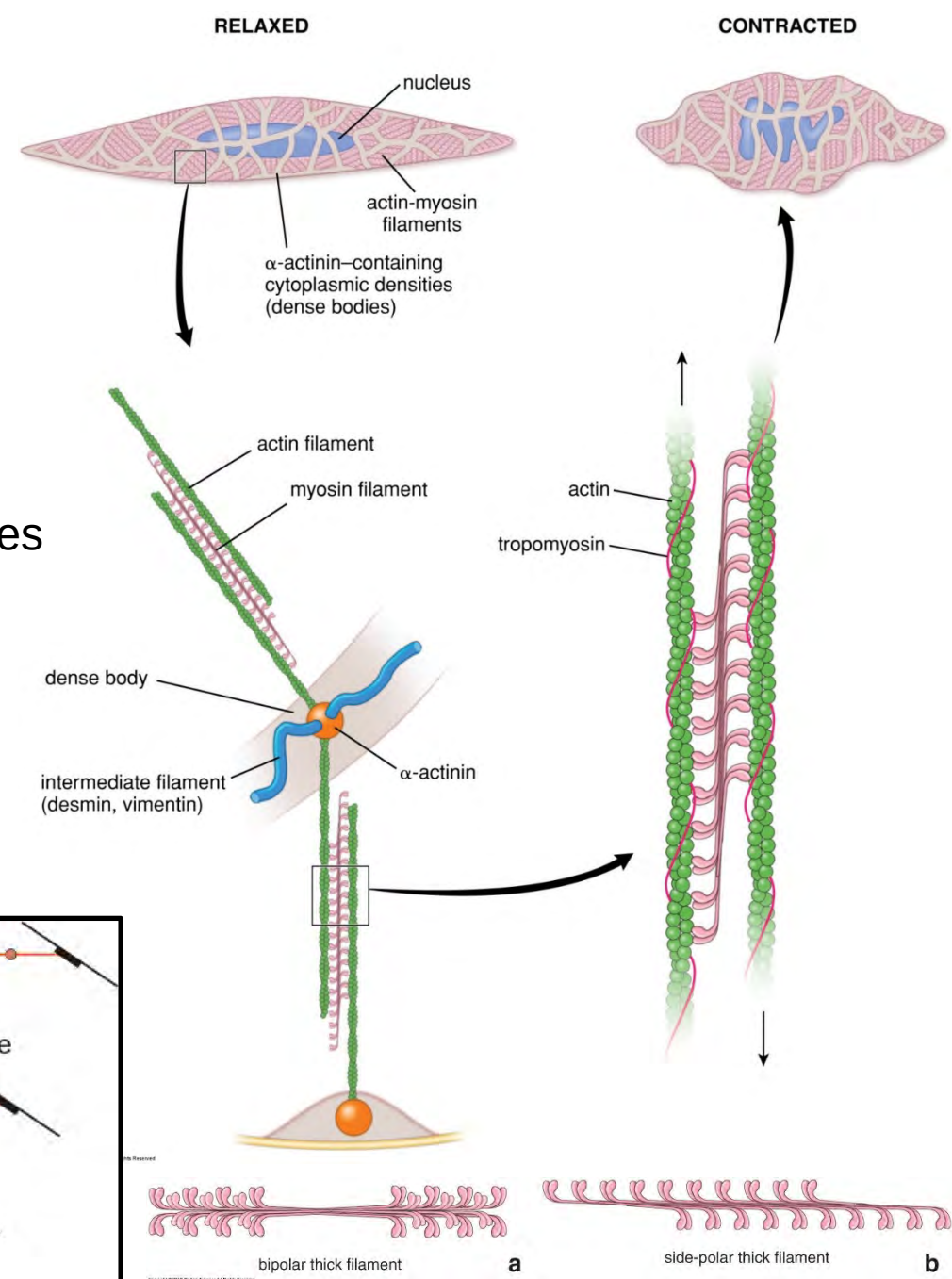
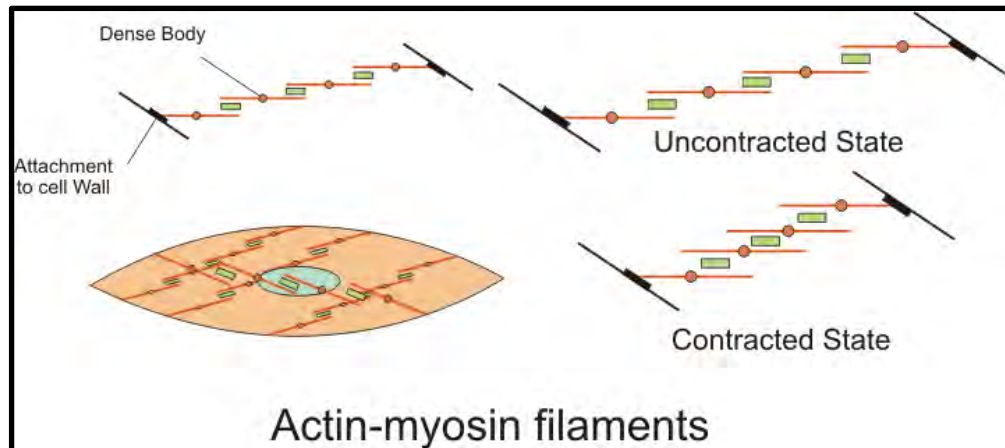
No troponin in Smooth muscle

Relaxed smooth muscle cell



Dense body

- α -actinin-rich electron dense structures
- Dense body in cytoplasm
- Dense body on cell membrane (desmosome)
- Actin filaments and intermediate filaments bind
- irregular arrangement of actin and myosin filaments



Contraction in Smooth Muscle

- release of Ca^{2+} into cytosol
- Ca^{2+} binds to Calmodulin
- Calmodulin binds to Caldesmon and activates myosin light chain kinase
- conformational change in Caldesmon and myosin filament
- Exposing of myosin-binding site on actin filament, unfolding of myosin light chain and unmasking the actin-binding site in myosin head
- binding of myosin head with actin filament
- Release of P_i and ADP in order (in Cross-bridge cycling)
- “contraction”

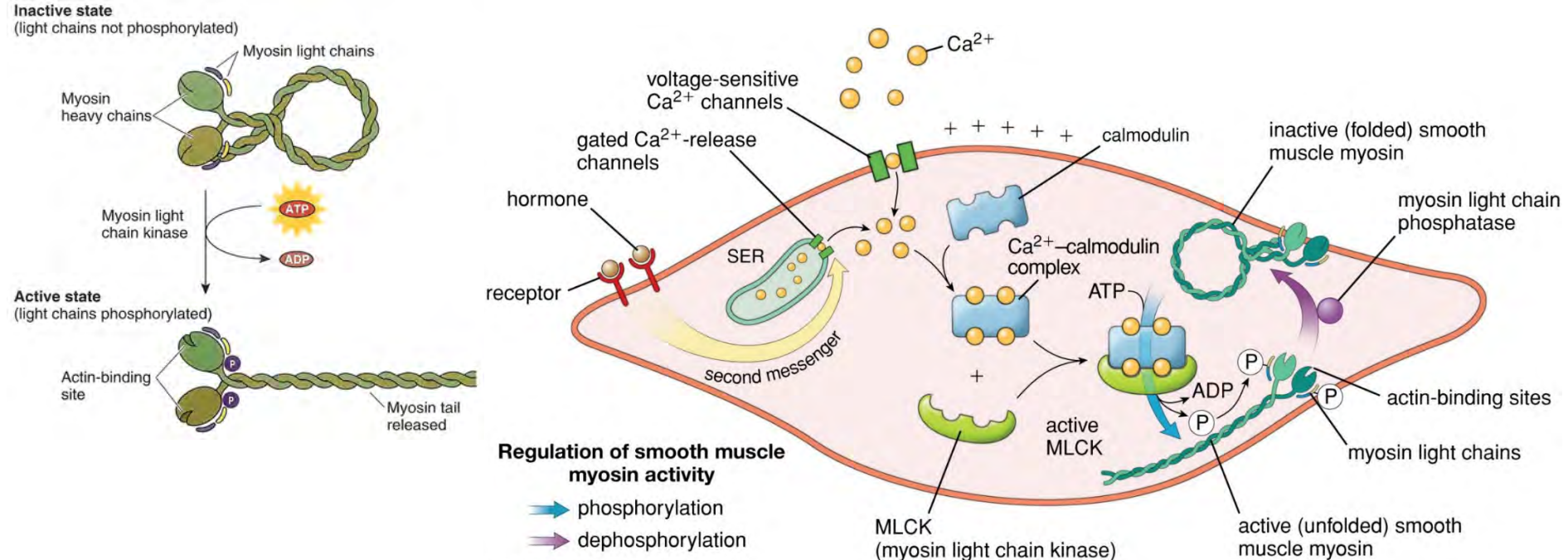


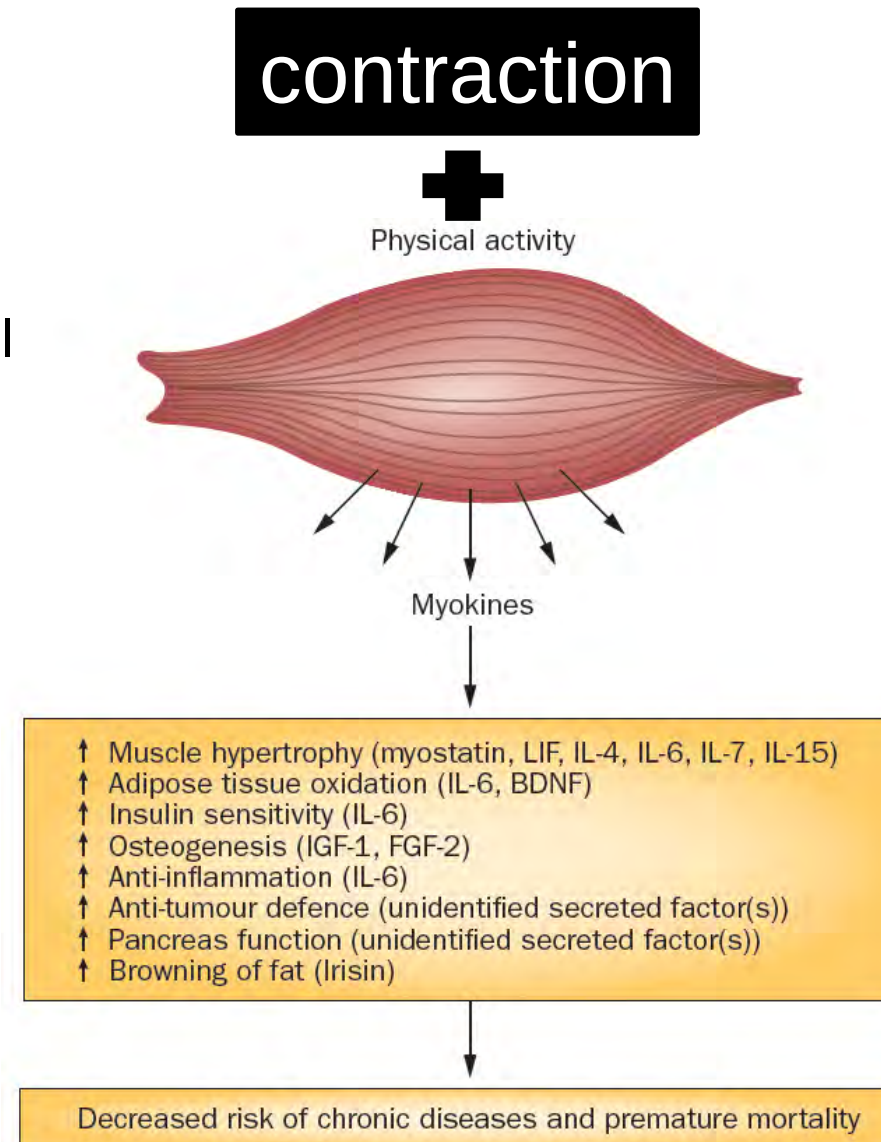
TABLE 8–3 Comparison of the Three Types of Muscle

FEATURE	SKELETAL MUSCLE	CARDIAC MUSCLE	SMOOTH MUSCLE
Sarcomeres/ Striations	Yes	Yes	No
Nuclei	Multinucleated; peripherally located	One (or two); centrally located	One; centrally located
Sarcoplasmic reticulum	Well-developed with terminal cisterns	Poorly defined; some small terminals	Some smooth endoplasmic reticulum
Size (diameter)	30-150 μm	10-15 μm	5-10 μm
T tubules	Yes; small, involved in triad formation	Yes; large, involved in diad formation	No
Cell junctions	No	Intercalated disks	Nexus (gap junctions)
Contraction	Voluntary; “all or none” Movement/ stabilization	Involuntary; rhythmic and spontaneous Pumping	Involuntary; slow and forceful; not “all or none” Peristalsis
Calcium control	Calcineurin in terminal cisternae	Calcium from extracellular sources	Caveolae
Calcium binding	Troponin C	Troponin C	Calmodulin
Regeneration	Yes, via satellite cells	No	Yes
Mitosis	No	No	Yes
Nerve fibers	Somatic motor	Autonomic	Autonomic
Connective tissue	Epimysium, perimysium, and endomysium	Connective tissue sheaths and endomysium	Connective tissue sheaths and endomysium
Distinctive features	Long; cylinder-shaped; many peripheral nuclei	Branched cells; intercalated disks; single nucleus	Fusiform cells with no striations; one nucleus

Regeneration of Muscle

- Skeletal muscle: satellite cell
- Cardiac muscle: scar tissue by fibroblast
- Smooth muscle; smooth muscle cell

Function of muscle

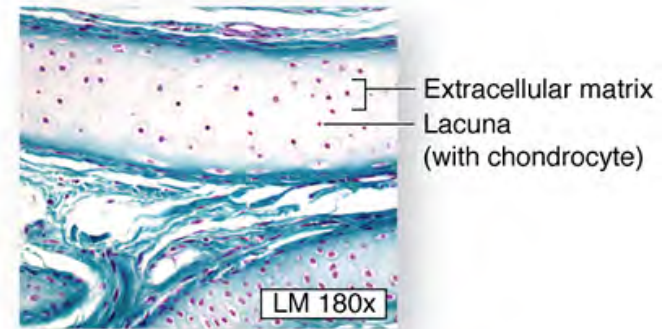
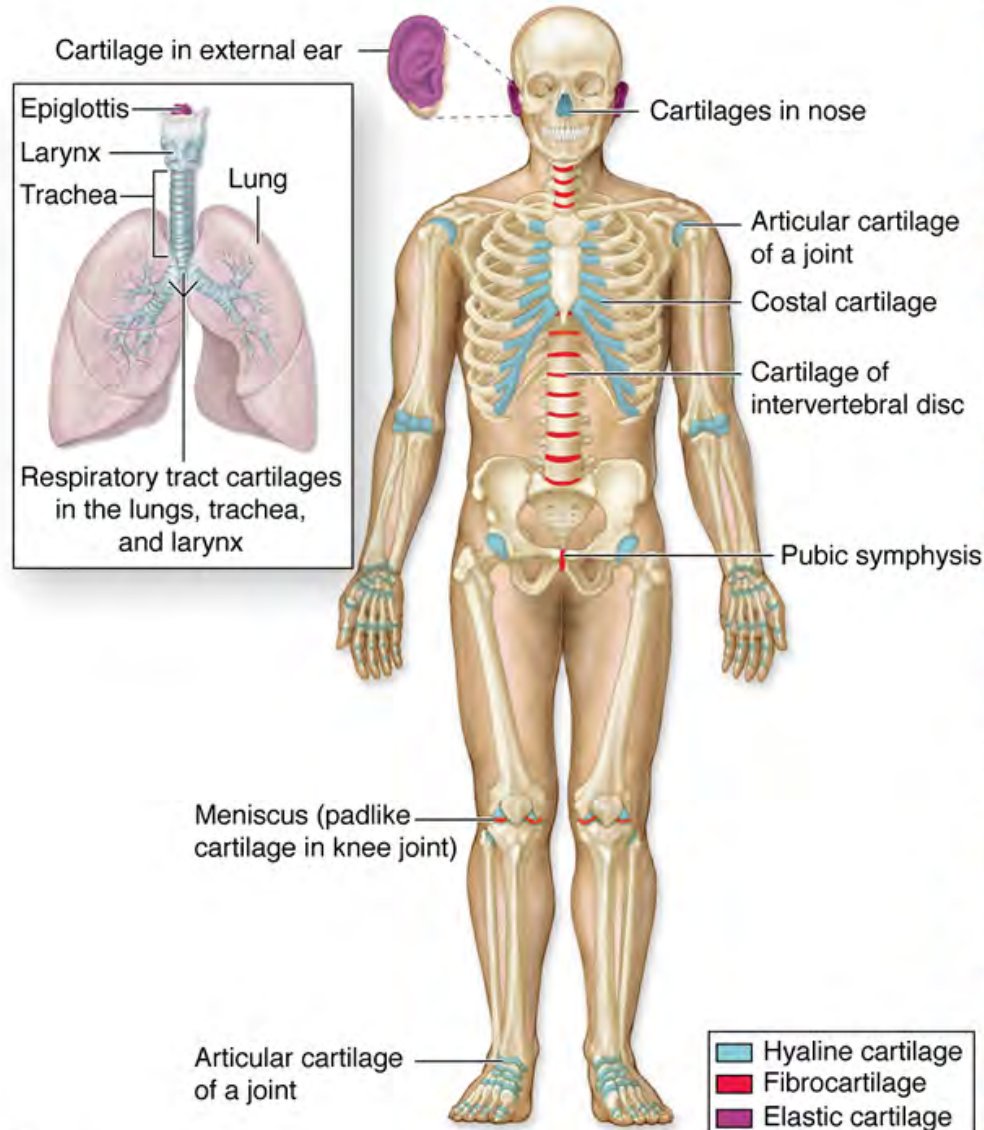


Cartilage and bone

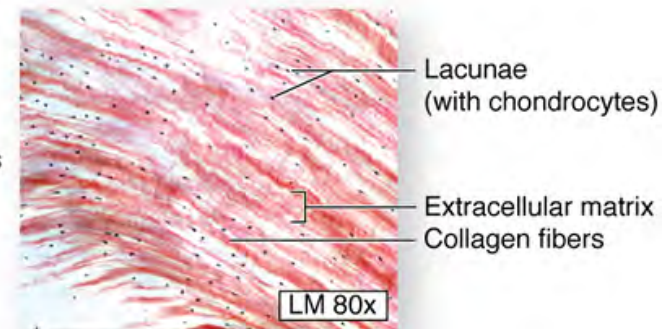
- Two of specialized connective tissues
- Main function: support
- Composed of
 - Cells
 - Extracellular matrix
 - Fiber
 - Ground substance
 - Fluid

			Bone	Cartilage
Cells			Osteoblast, Osteocyte, osteoclast	Chondroblast, Chondrocyte
Extra-cellular Matrix	inorganic matrix	Mineralization Mechanical properties	67% Hard	0% Gel-like
	organic matrix	Fiber	Collagen fiber	Collagen fiber or fibril Elastic fiber
		Ground substance	GAG, proteoglycan, glycoprotein	GAG, proteoglycan, glycoprotein, hyaluronic acid, elastin
Structures contained	Blood vessel Nerve Lymphatics		Yes Yes Yes	No No No
Function			Support, Protection Mineral store, Motion Blood cell formation	Support, Shock absorption, Elasticity Sliding
Growth			Interstitial growth, Appositional growth	Interstitial growth Appositional growth
subtypes			Compact bone, Spongy bone, Lamellar bone, Woven bone	Hyaline cartilage, Elastic cartilage, Fibrous cartilage
coverage			Periosteum, Endosteum	Perichondrium

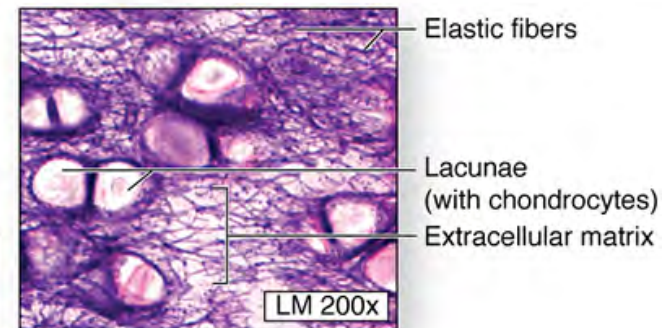
Cartilage



b Hyaline cartilage



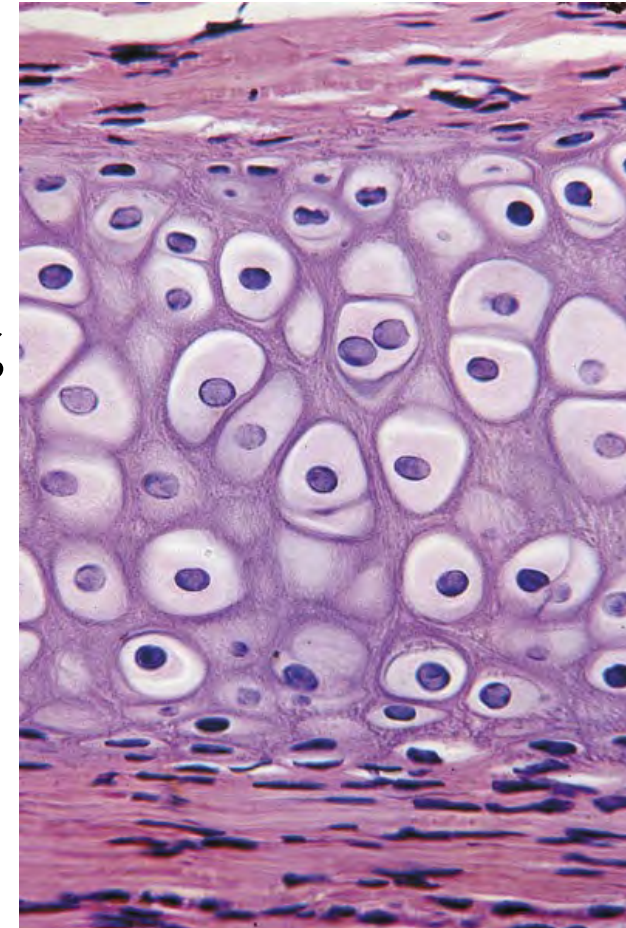
c Fibrocartilage

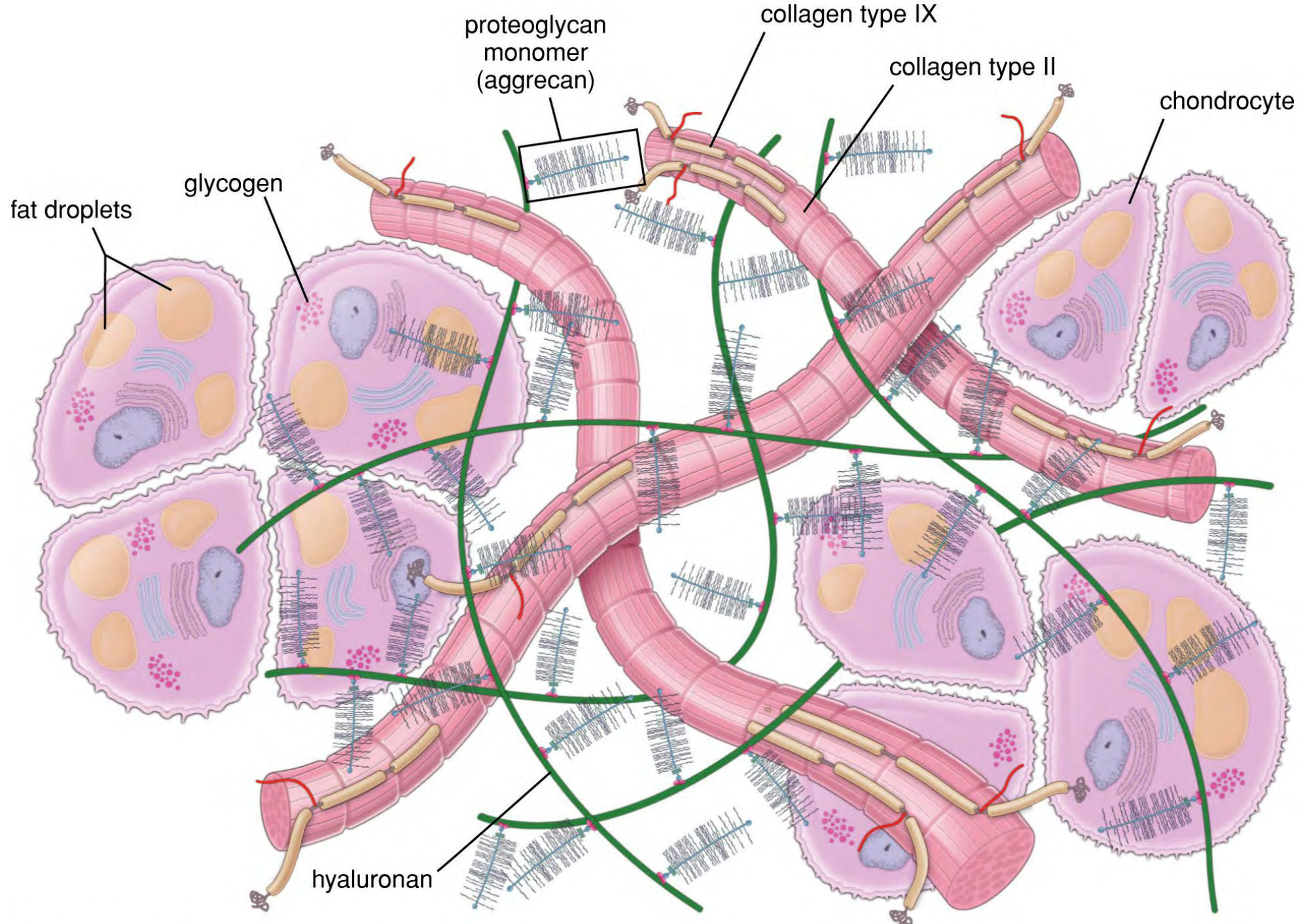


d Elastic cartilage

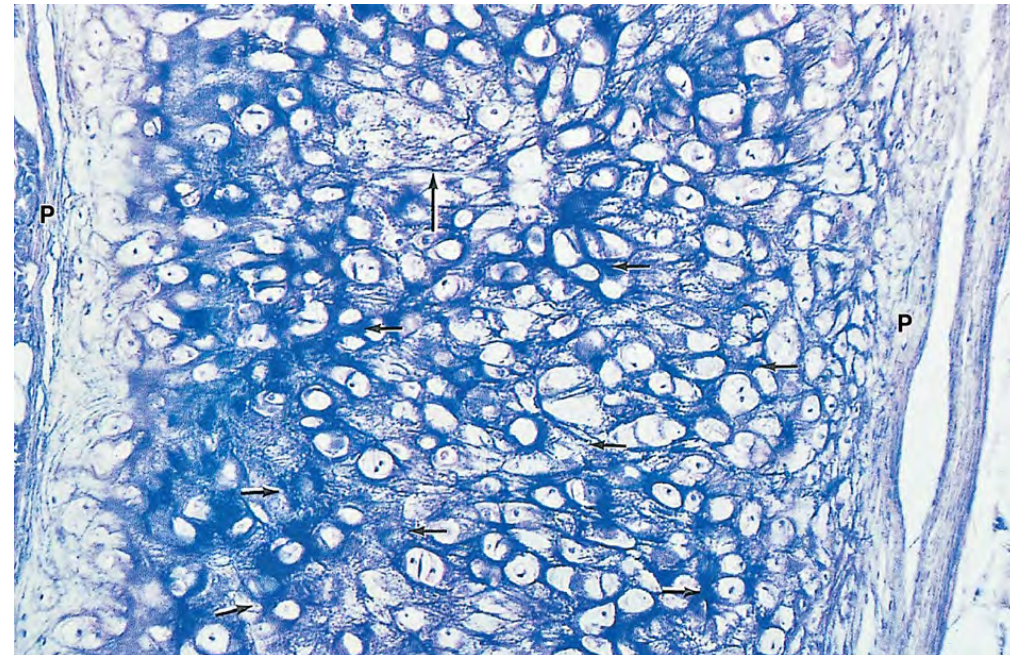
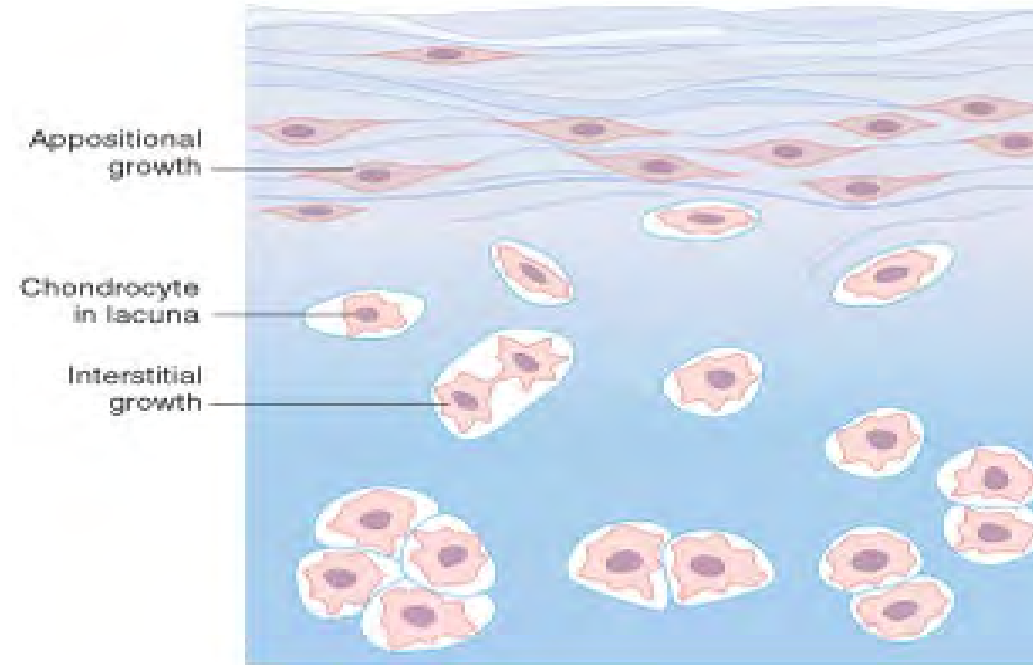
Cartilage

- Function
 - Solid → Supporting
 - Gel like and flexible → Shock-absorbing
 - Smooth surface → sliding
 - Long bone formation
- Components
 - Cell in lacuna
 - Fiber: collagen fiber, elastic fiber
 - Ground substance: proteoglycan, GAGs, Hyaluronic acid, glycoprotein, collagen
 - Extracellular fluid: water (75% in wet weight) in fresh cartilage
 - forms a gel with the components of the ground substance
- grow rapidly → important in development
- also is formed very early during the repair of bone fractures.



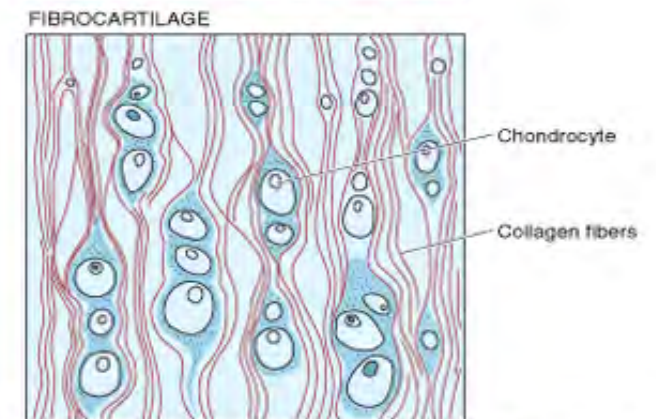
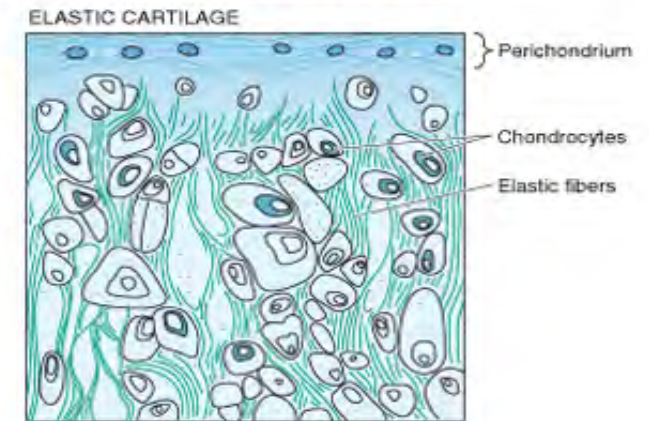
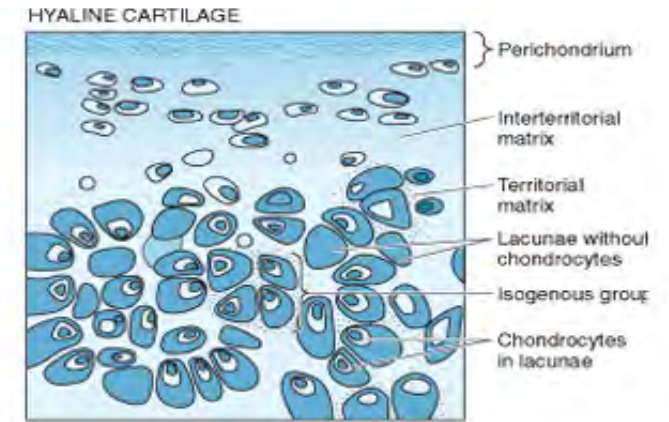


- Origin: undifferentiated mesenchymal cells
- Characteristics
 - Solid and gel-like matrix
 - Collagen fibril-GAG-core protein-hyaluronic acid-core protein-GAG-collagen fibril
 - No blood vessel (nourishment: diffusion through the water in matrix)
 - No nerve
 - No lymphatic vessel
 - Covered by perichondrium
 - Outer fibrous layer : fibroblast, type I collagen, blood vessels
 - Inner cellular layer : chondrogenic cells
 - Isogenous cell group (<8)



- Classification

- Hyaline cartilage: collagen type II (40% in dry weight)
- Elastic cartilage: collagen type II + elastic fiber
- Fibrous cartilage: collagen type I
- * Type II collagen: easy to bind proteoglycan

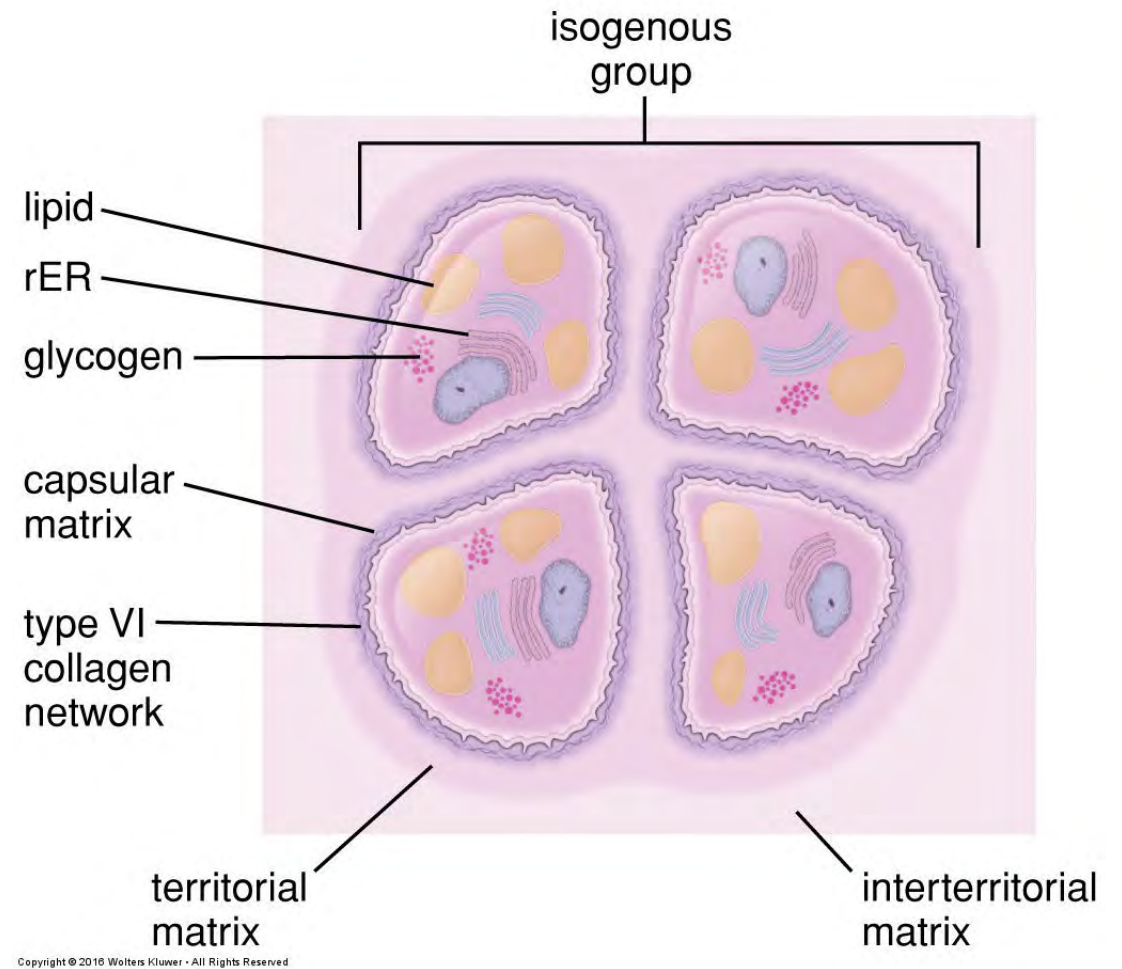


Classification of cartilage

Type	Identifying Characteristics	Perichondrium	Location
Hyaline	-Type II collagen (40%), -basophilic matrix, -chondrocytes usually arranged in groups	present in most places (exceptions: articular cartilages and epiphyses)	Articular ends of long bones, nose, larynx, trachea, bronchi, ventral ends of ribs
Elastic	-Type II collagen -elastic fibers	present	Pinna of ear, walls of auditory canal, auditory tube, epiglottis, cuneiform cartilage of larynx
Fibrous	-Type I collagen, Type II collagen -acidophilic matrix; -chondrocytes : parallel rows between bundles of collagen; - - always associated with dense regular CT or hyaline cartilage	absent	Intervertebral disks, articular disks, pubic symphysis, insertion of some tendons

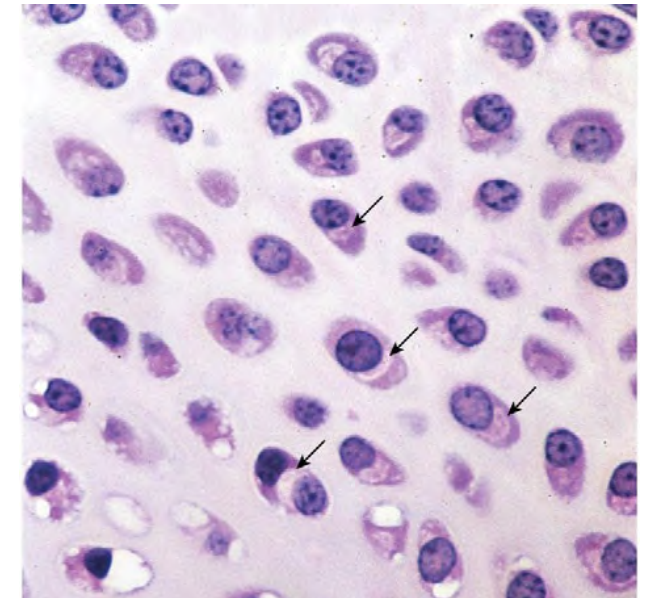
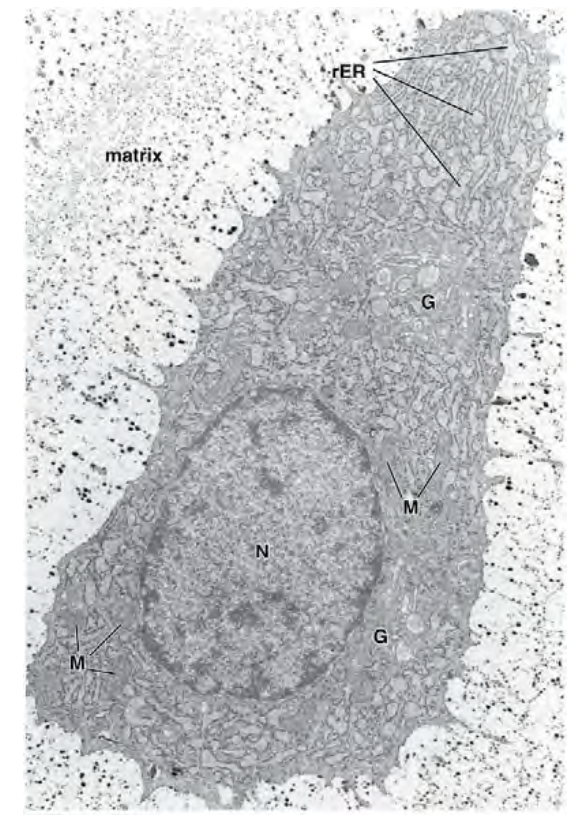
Hyaline cartilage

- Bluish white, transparent
- Cells
- Extracellular matrix
 - Fibril : Type II collagen, type IX, X, XI
 - same refractive index as the ground substance.
 - Ground substance
 - Proteoglycan (especially aggrecan)
 - Core protein
 - GAGs:
 - » Chondroitin 4-sulfate
 - » Chondroitin 6-sulfate
 - » Keratan sulfate
 - Hyaluronic acid : 200 proteoglycan
 - Glycoprotein: chondronectin (bind to typeII collagen, C4S, C6S, HA, integrins)
 - Classification of ECM
 - Pericellular matrix (capsular Mat.): type VI collagen, integrin to attach to the matrix
 - Territorial matrix : 50 μm band, rich chondriotin sulfate, poor in collagen
 - Interterritorial matrix :richer type II collagen, poorer proteoglycans



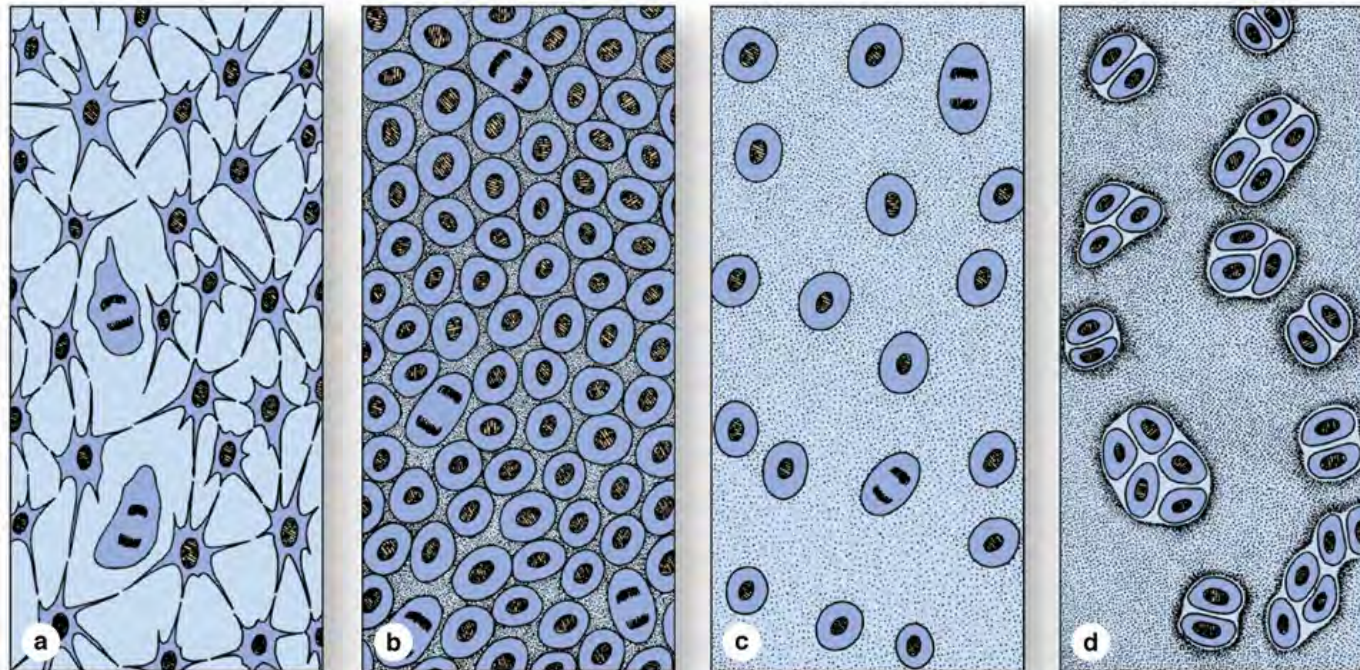
Cells in cartilage

- Chondrogenic cells
 - Origin: mesenchymal cells
 - Spindle shape
 - Ovoid nucleus, 1~2 nucleoli
 - Sparse cytoplasm
 - Differentiate into chondroblasts and osteoprogenitor cells
- Chondroblast
 - Origin:
 - chondrogenic cells in perichondrium
 - mesenchymal cells within chondrification center
 - Plump, basophilic
 - r-ER, Golgi complex, mitochondria, secretory vesicles
- Chondrocyte
 - chondroblasts in cartilage lacunae
 - Still proliferate → isogenous cell group
 - Ovoid to round shape
 - large nucleus and prominent nucleolus
 - Young and old chondrocyte



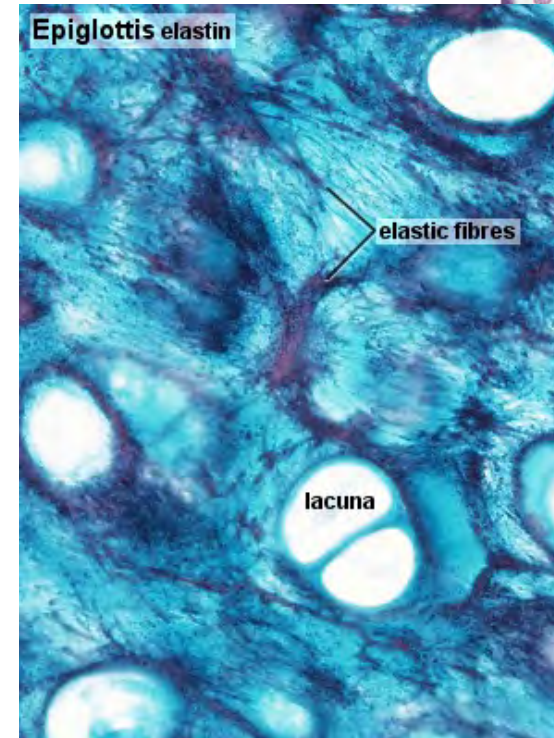
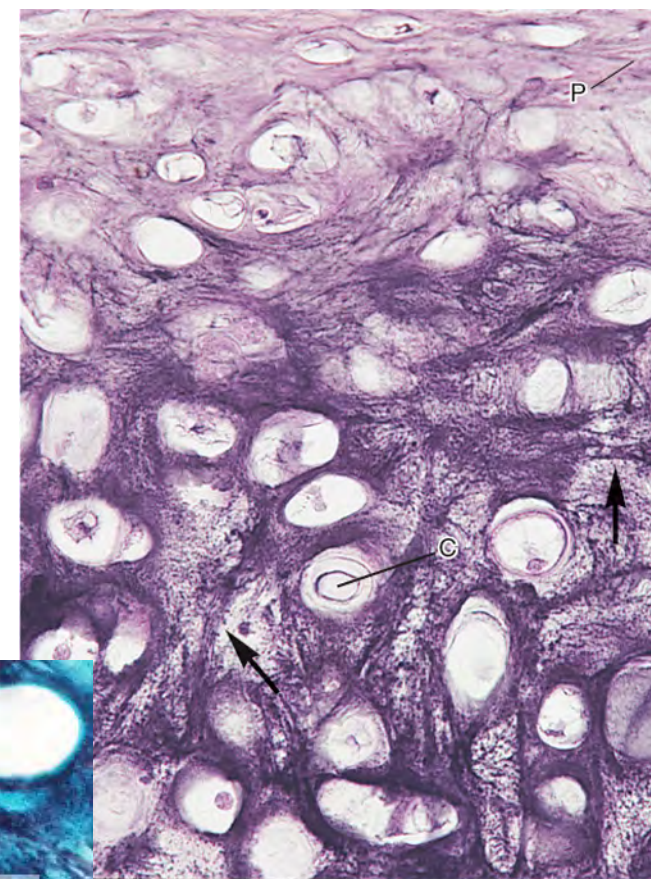
Hyaline cartilage

- Development
 - Proliferation → condensation → chondrification center → differentiation → secretion → matrix formation
- Growth
 - Interstitial growth : at epiphyseal plates, secrete → push → enlarge
 - Appositional growth : at perichondrium



Elastic cartilage

- More flexible than hyaline cartilage
- Elastic fiber
 - Yellowish
 - more opaque than hyaline cartilage
 - More in outer fibrous layer of perichondrium
- More cells and less amount of matrix
- Auricle, external auditory canal, eustachian tube, epiglottis, cuneiform cartilage

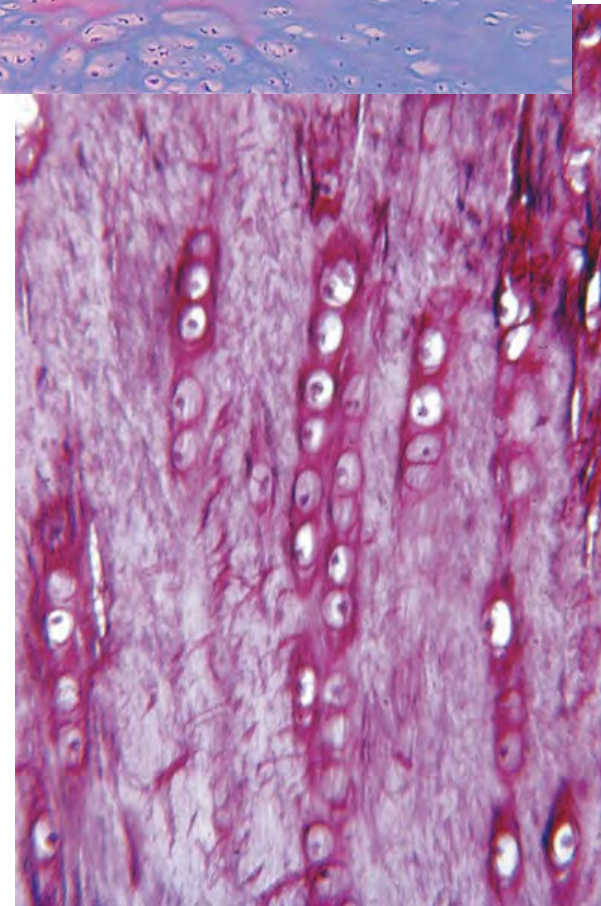
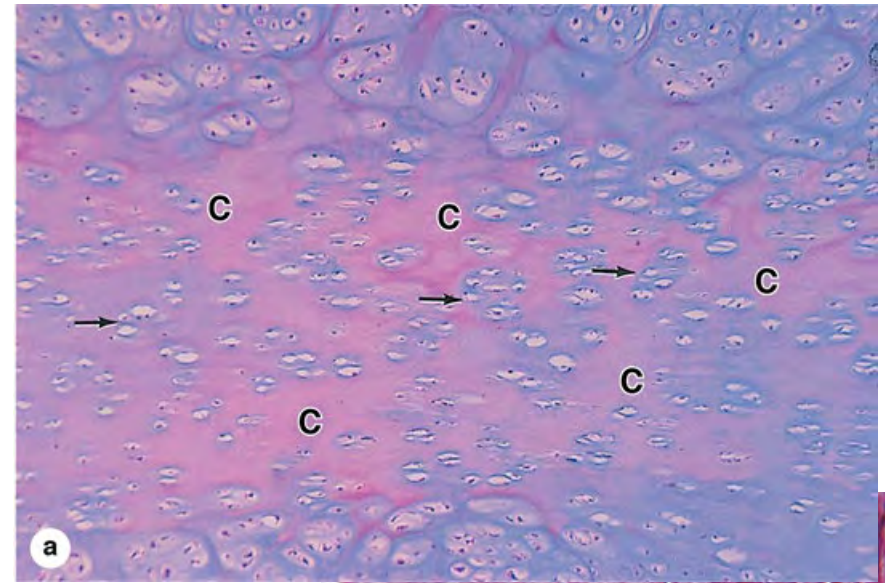


er. Gartner & Hiatt: Color Textbook of Histology 3E - www.studentconsult.com

Fibrocartilage

- Cells
 - Origin : fibroblast → chondrocyte
 - Parallel with collagen fiber bundle
 - One or isogenous cell group
- ECM
 - Type I collagen fiber → acidophilic
 - GAG: Chondrotin sulfate, keratan sulfate
- Associated with dense C-T and hyaline cartilage
- No perichondrium
- Pubic symphysis, intervertebral disk, articular disk, insertion of some tendons

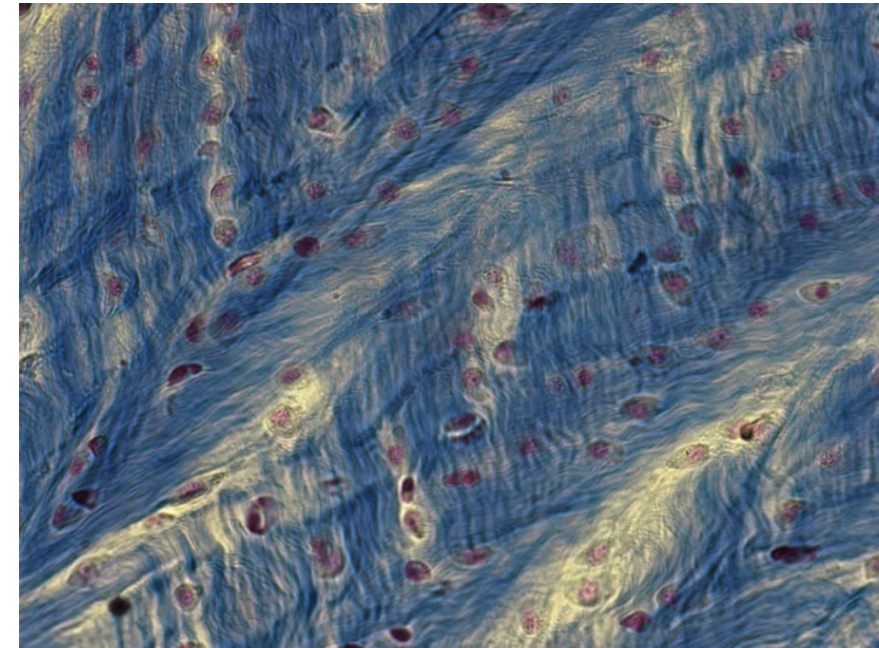
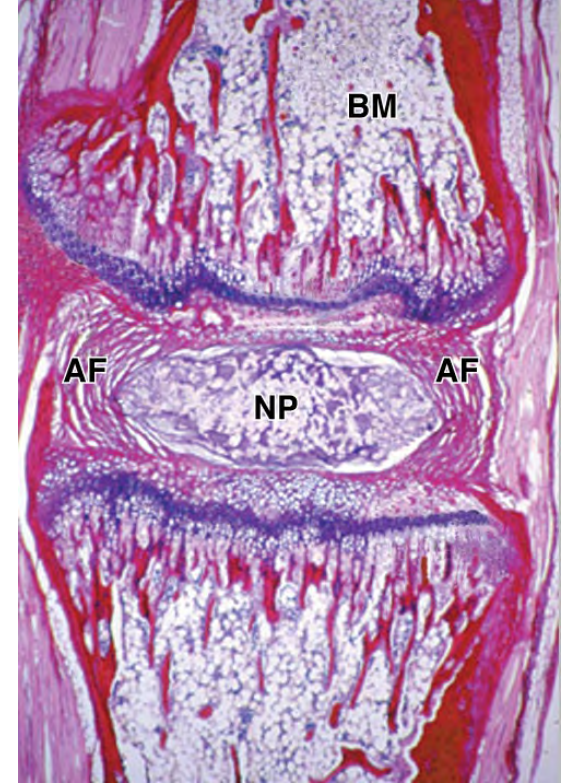
Pubic symphysis

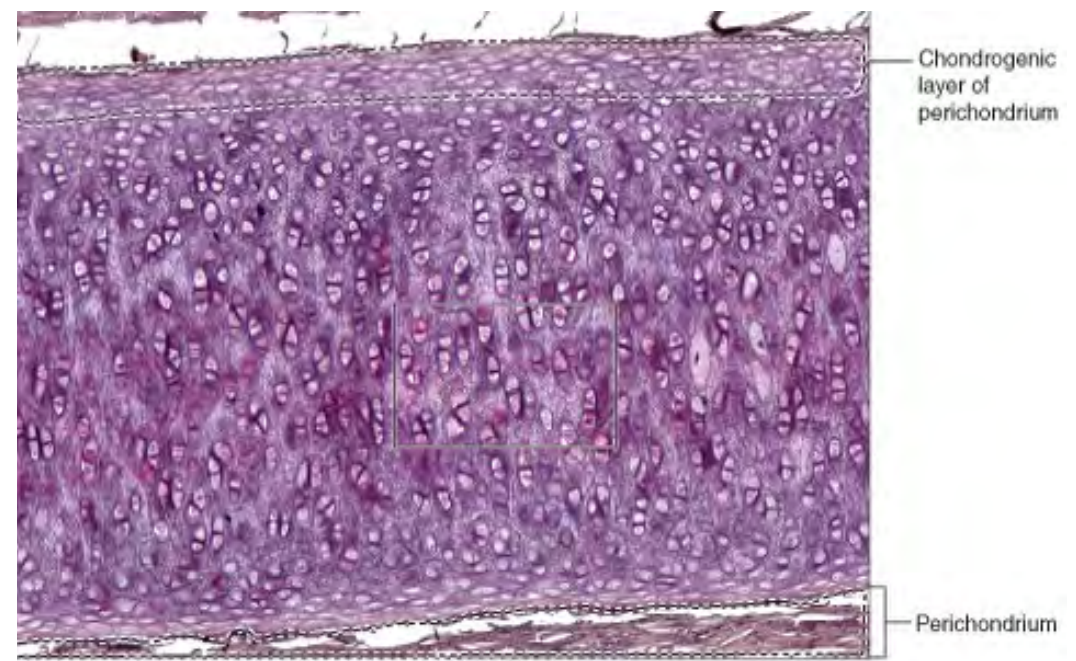
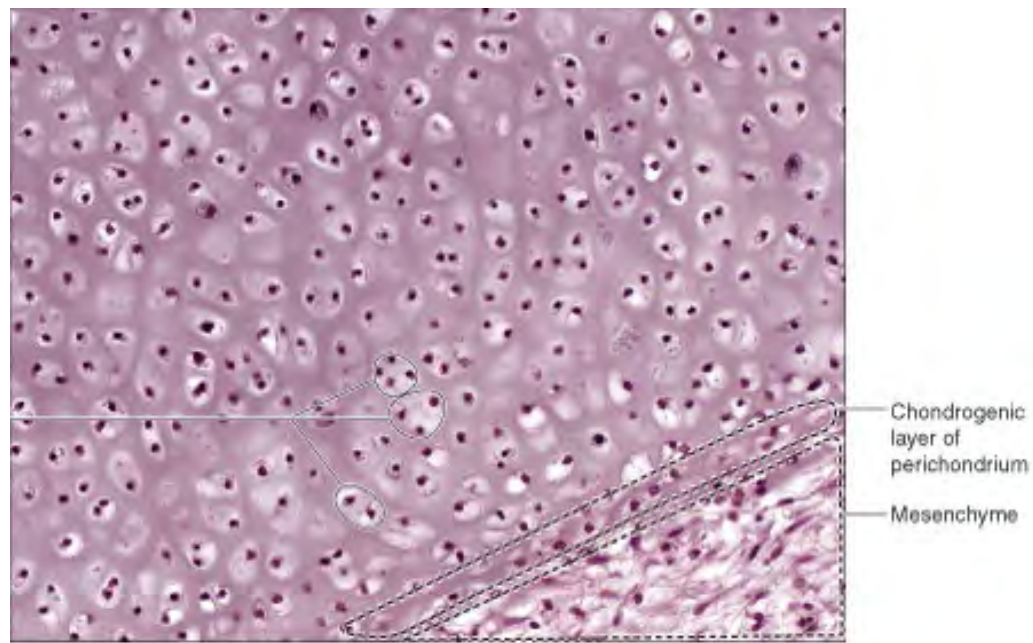


intervertebral disk

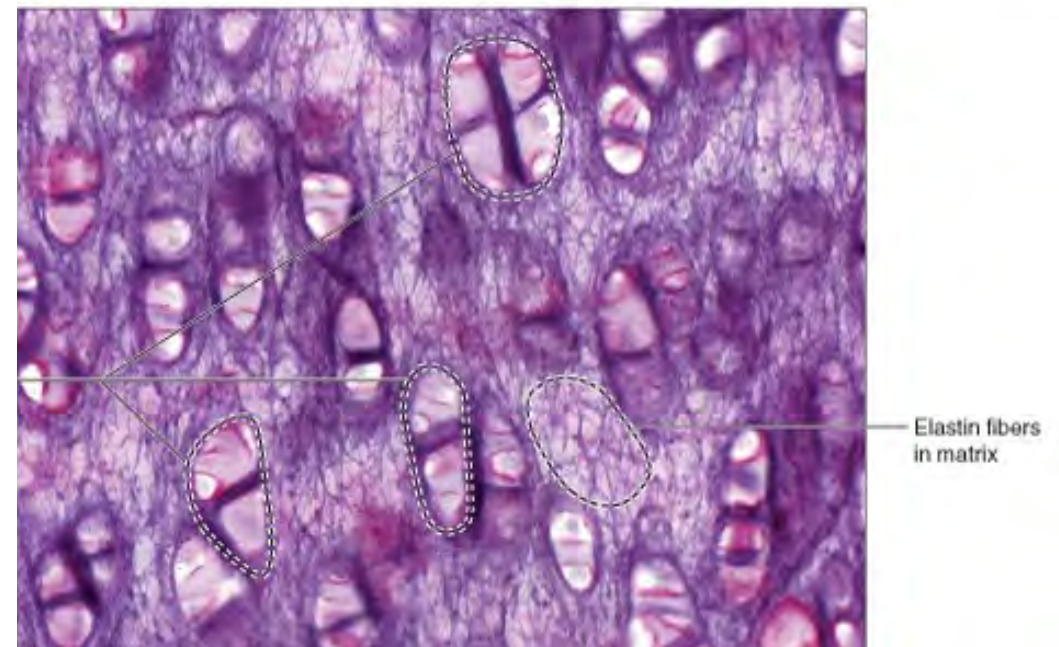
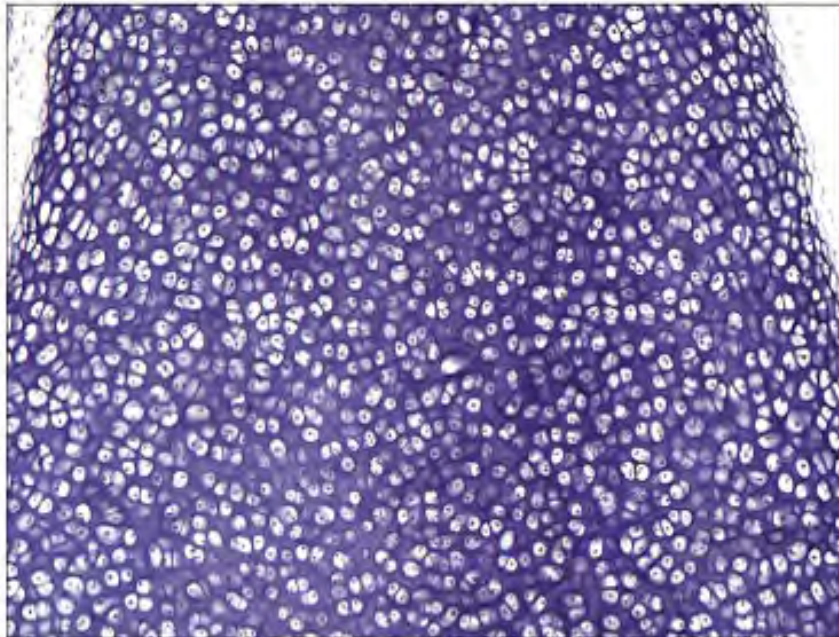
Fibrocartilage

- Intervertebral disk
 - between hyaline cartilage
 - Two parts
 - Annulus fibrosus: fibrous cartilage (tensile forces)
 - Herring-bone appearance:
 - » Multilayer
 - » $45^{\circ} \sim 90^{\circ}$ angulation
 - Nucleus pulposus (compression)
 - From cells of notochord
 - Hyaluronic acid rich
 - Ruptured disk
 - Slipped disk





A



- Classification

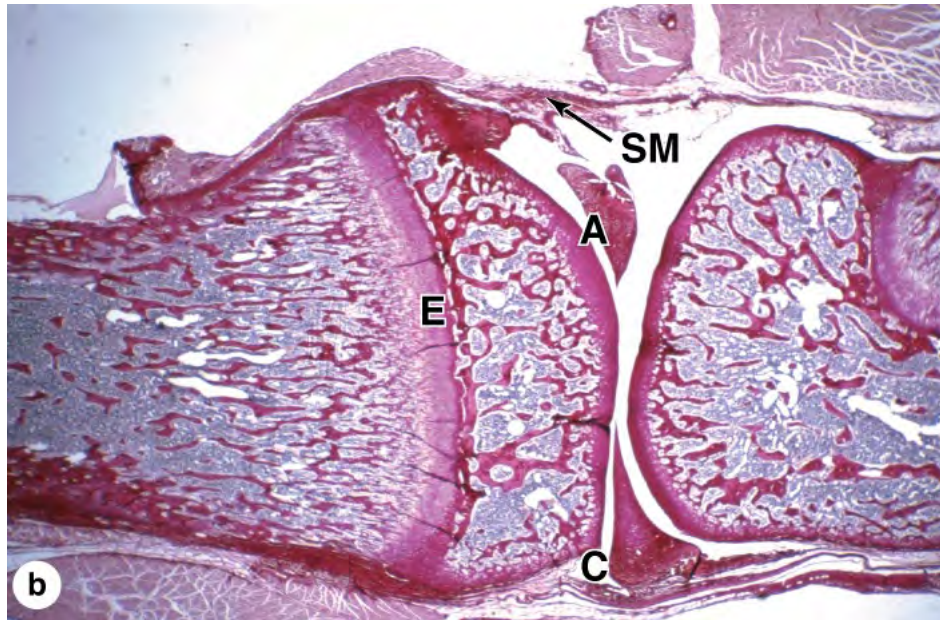
- Synarthrosis

- Synostosis: 뼈 유착 결합 → 머리뼈
 - Synchondrosis: 유리연골성결합 → 성장시(뼈끝판, 두개저), 성인(rib and sternum)
 - Syndesmosis: 인대결합

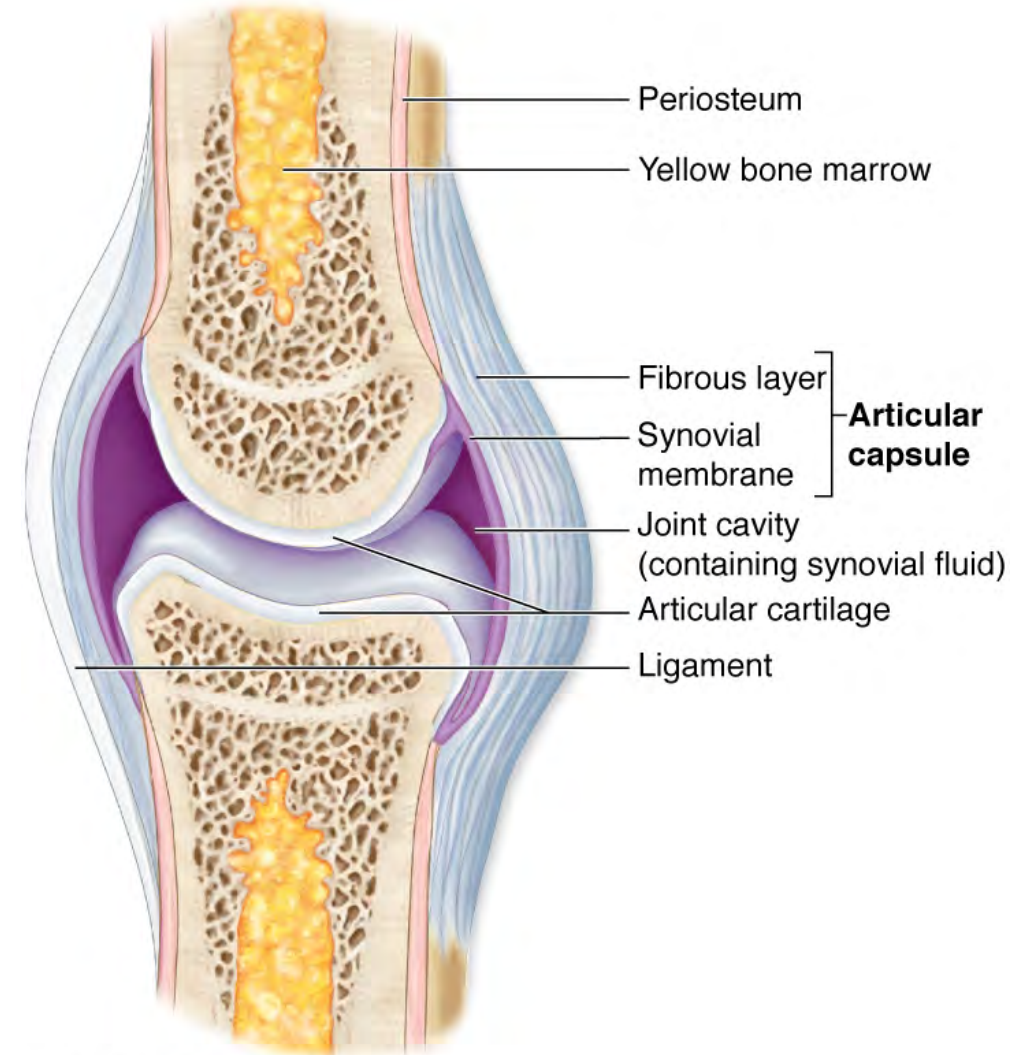
- Diarthrosis

- Structures

- Articular cavity
 - Articular cartilage
 - Joint capsule

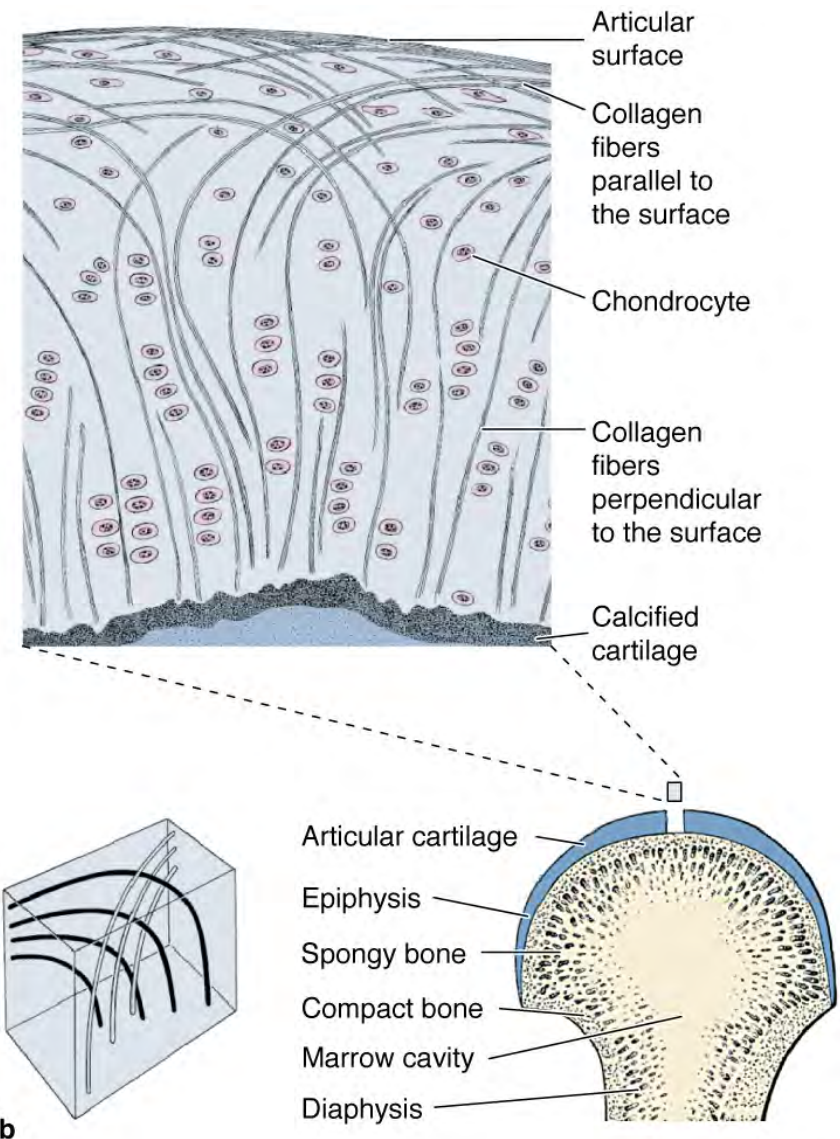


Joint

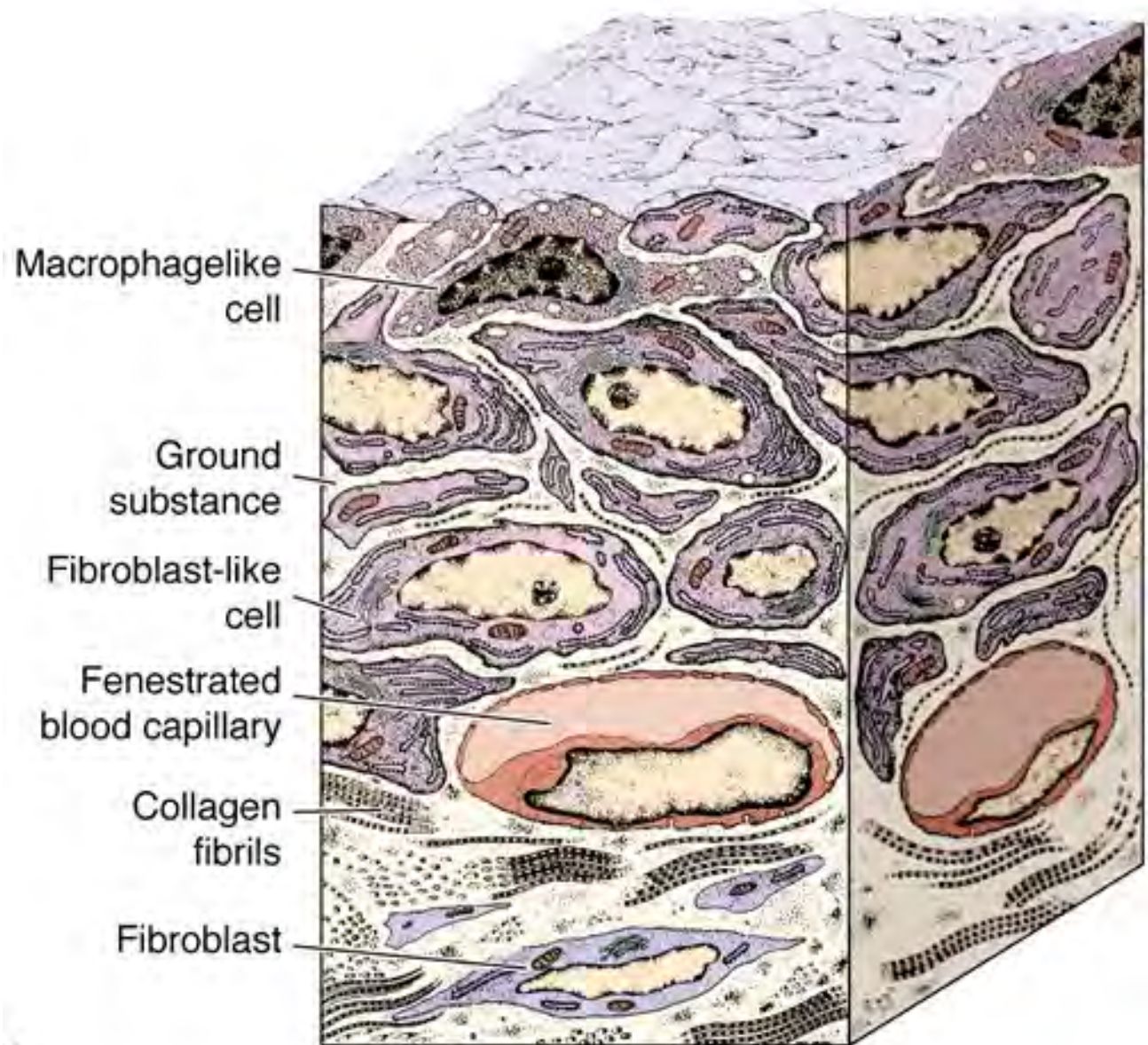
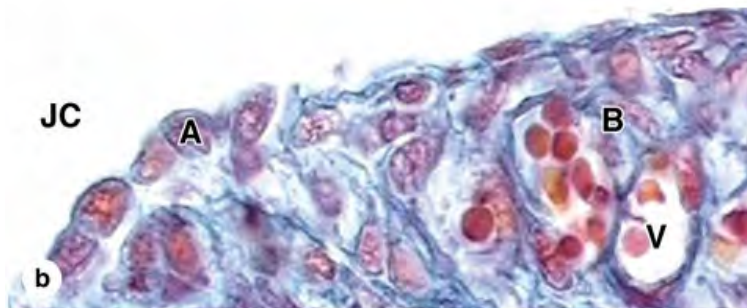
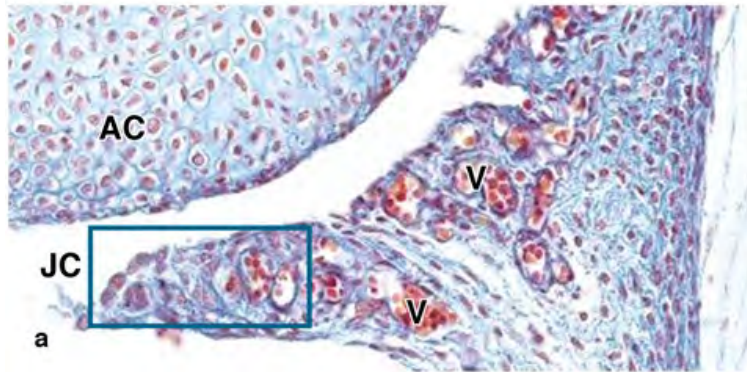
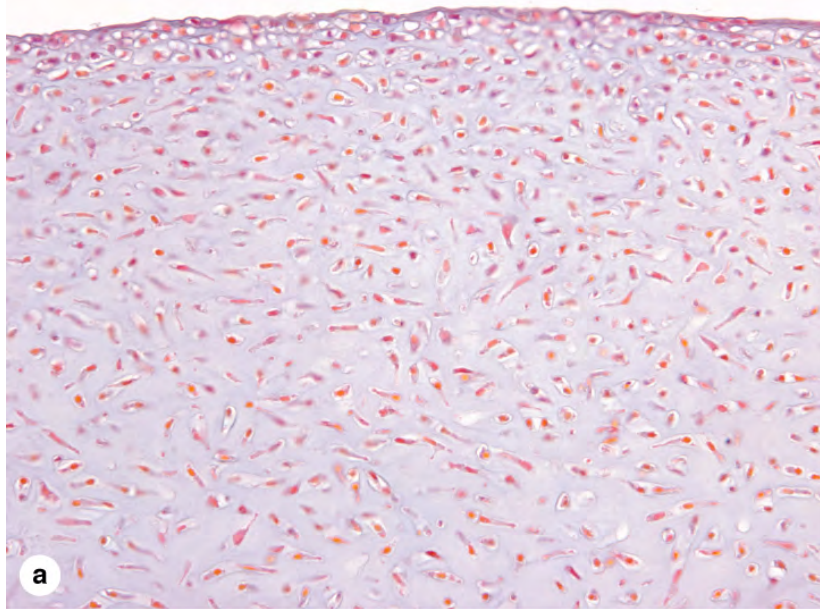


a Typical synovial joint

- Synovial fluid
 - No color, transparent, viscous
 - 성분:
 - Dialyzed plasma component (투석된 혈장성분)
 - Hyaluronic acid
- Articular cartilage
 - Hyaline cartilage, without perichondrium
 - Elasticity
 - Water out from the space in GAGs
 - Electrostatic repulsion between carbonate and sulfate
 - Exchange: gas, nutrient and waste with synovial fluid
- Synovial membrane
 - M-cell: macrophage-like cell, type B
 - F-cell: fibroblast-like cell, type A



Synovial membrane



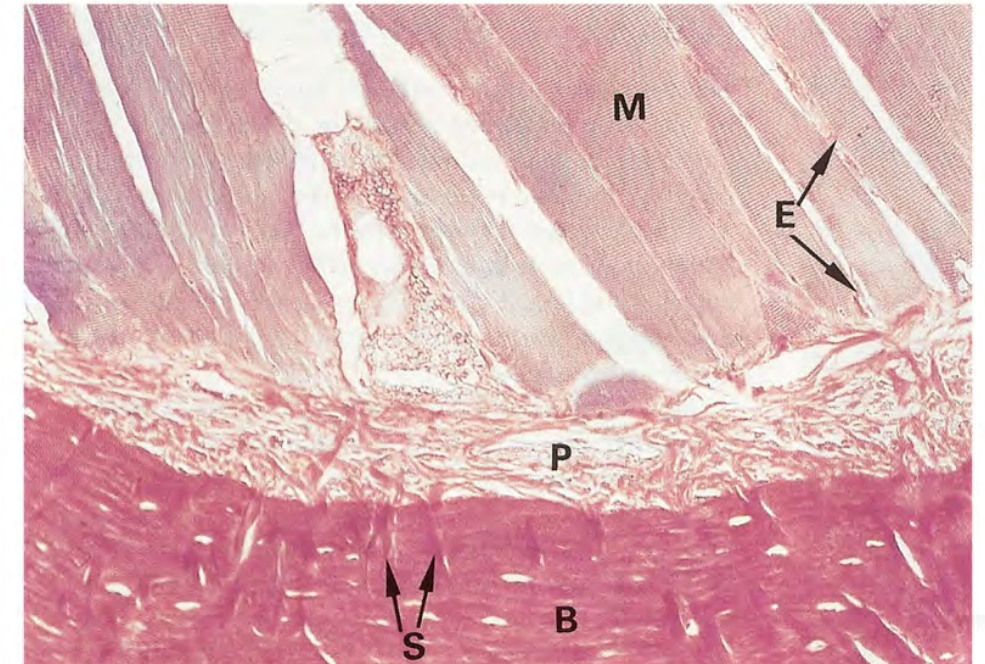
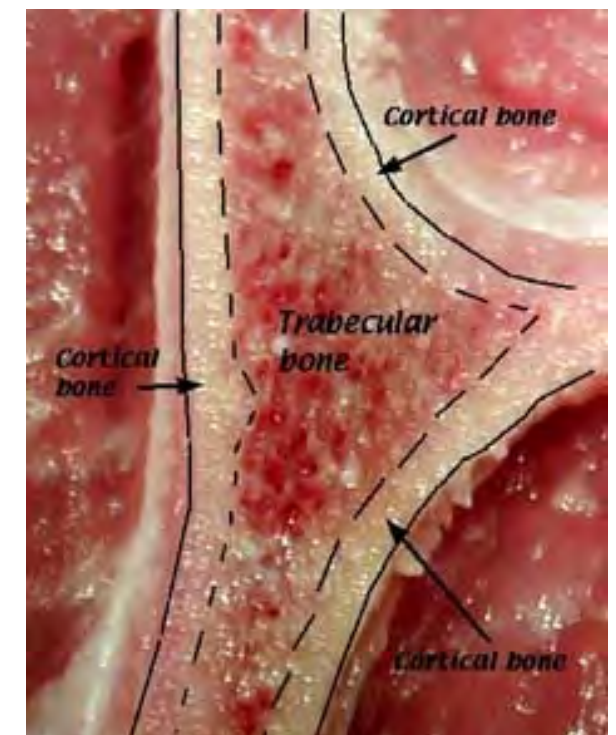
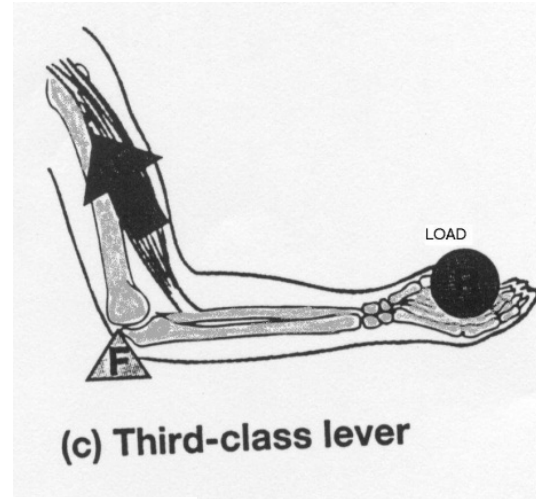
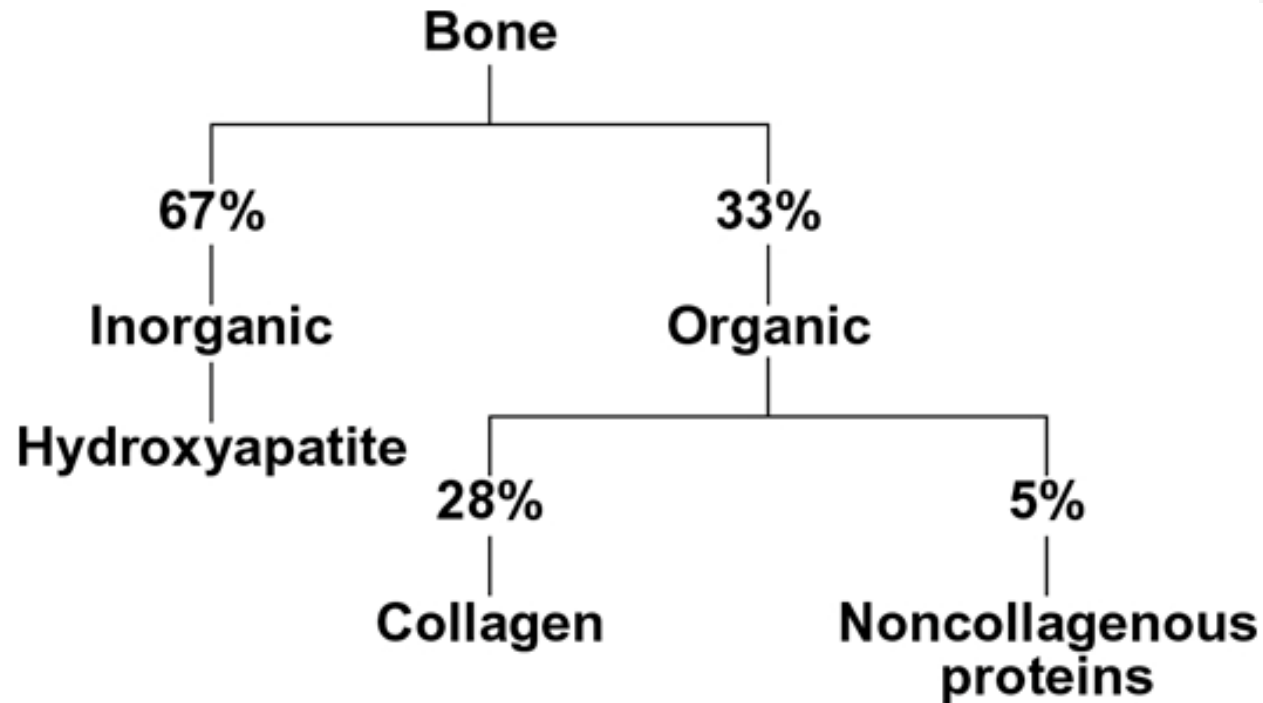
Histophysiology of cartilage

- Turnover : constant change of proteoglycans with age
- Cartilage degeneration
 - With age: ectopic hypertrophy → death → calcification → bone formation → less mobility and pain in joint
- Cartilage regeneration
 - Chondrogenic cells
 - fibroblast

Bone

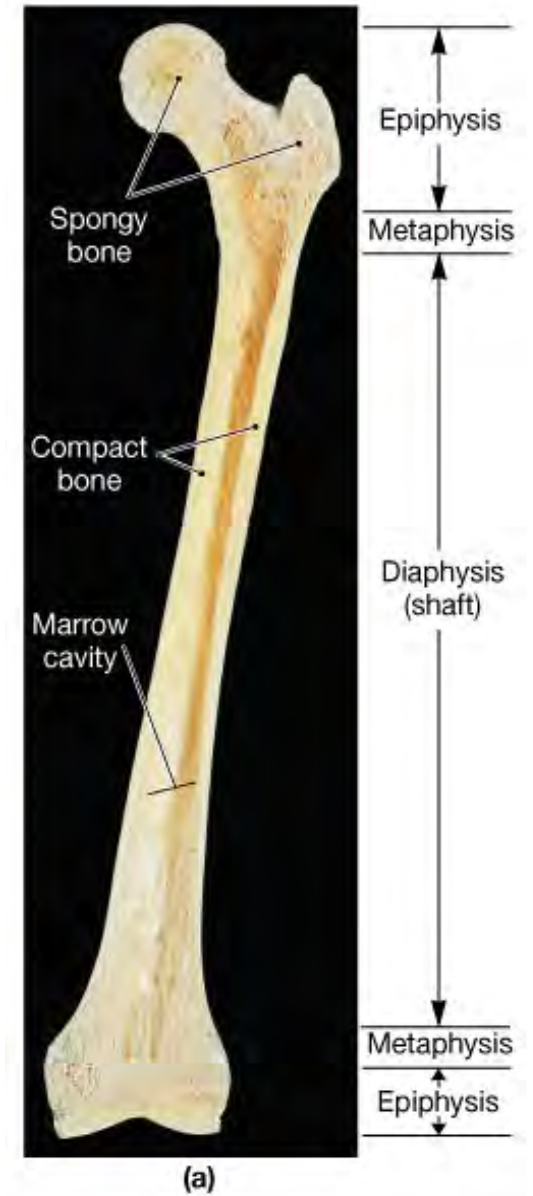
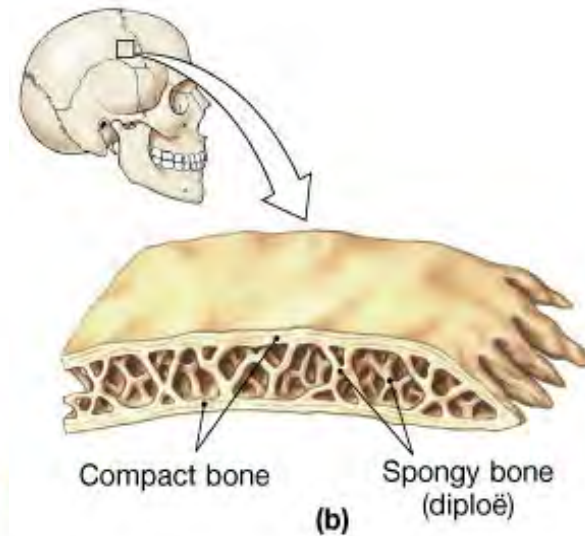
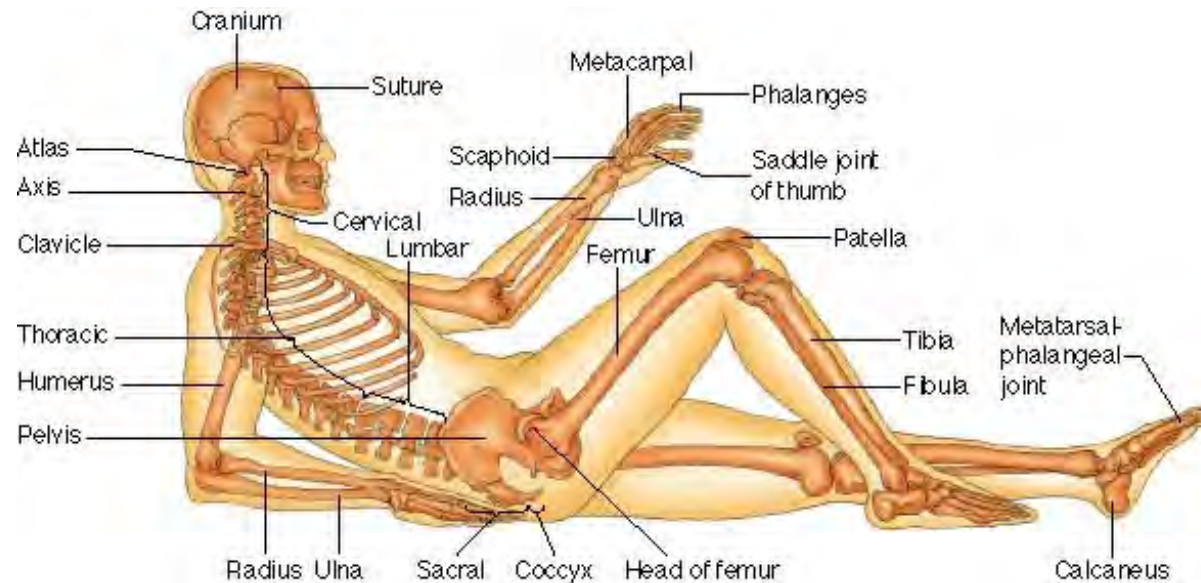
Functions

1. Mineral store *for Ca^{2+} : 99%*
2. Protection
3. Support and motion
4. Blood cell formation



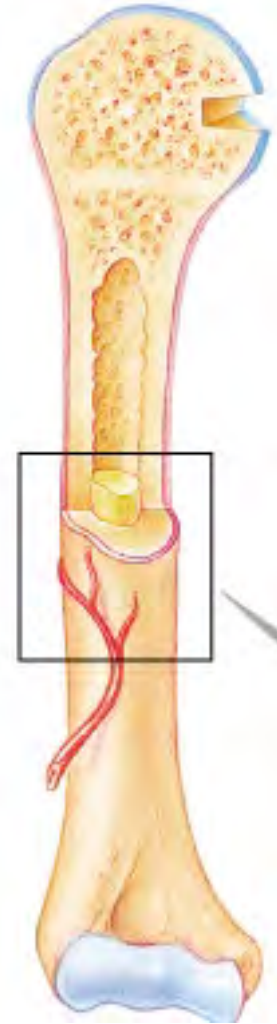
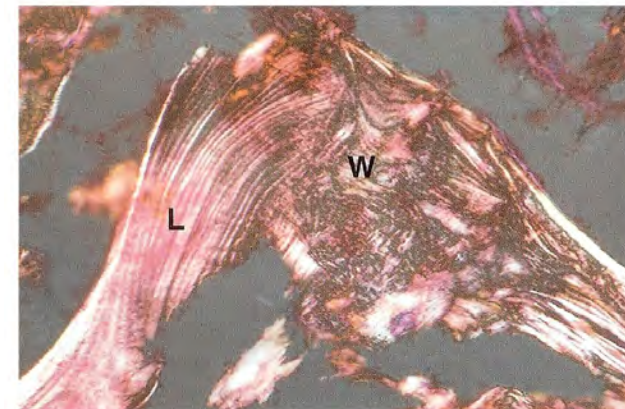
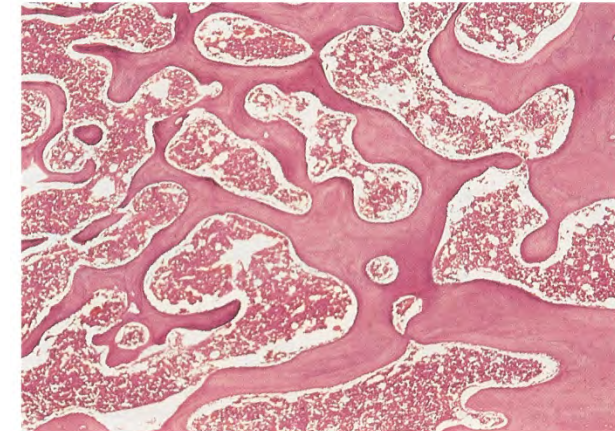
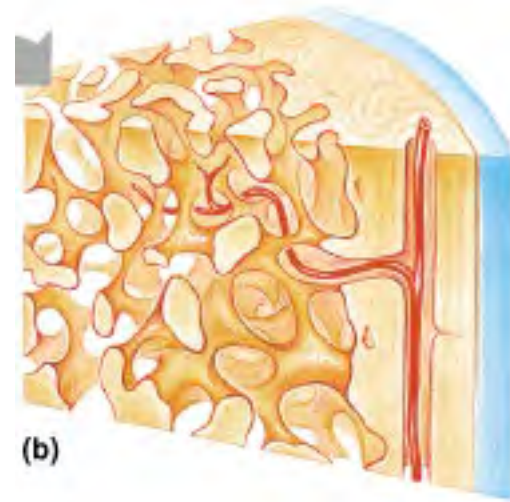
Types of bone

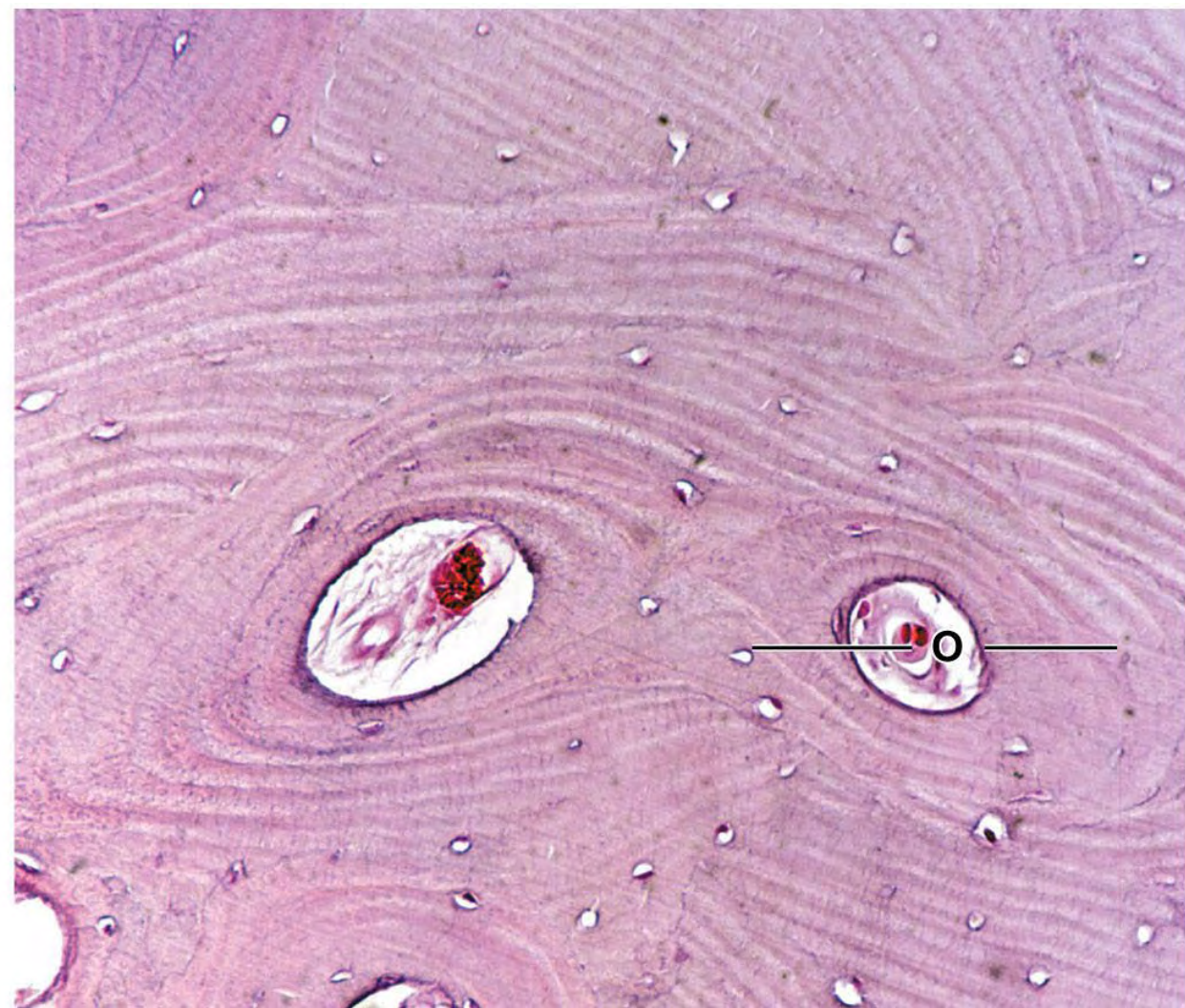
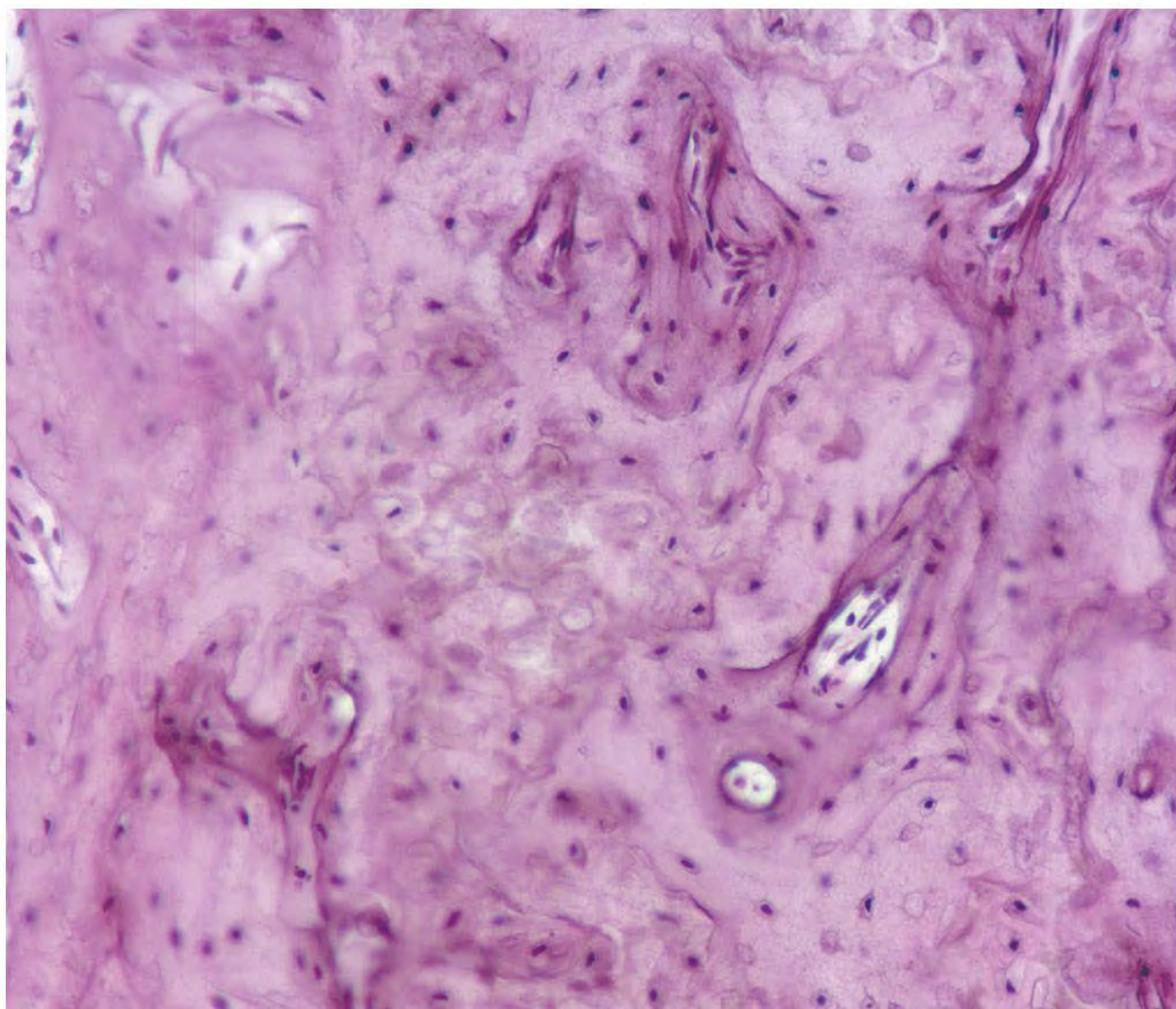
- According to shape
 - Long bones
 - Short bones: carpals and tarsals
 - Flat bones: of skull, diploë (inner and outer tables)
 - Irregular bones: vertebra
 - Sesamoid bones: patella....



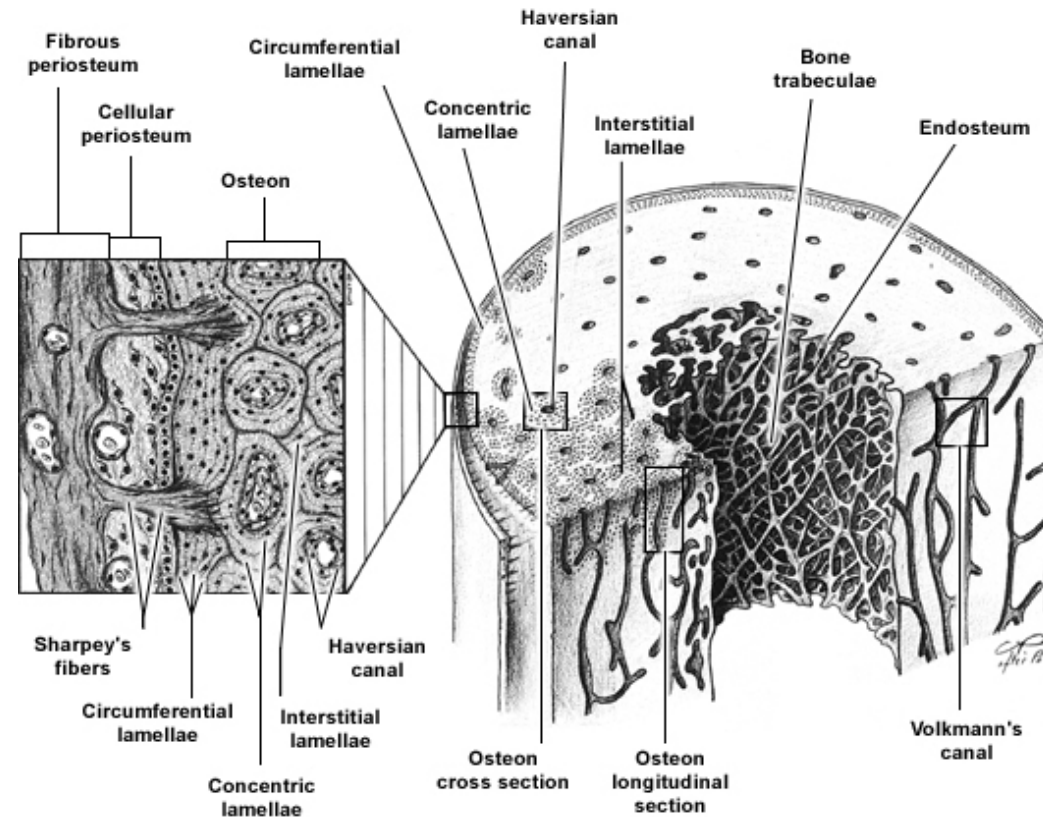
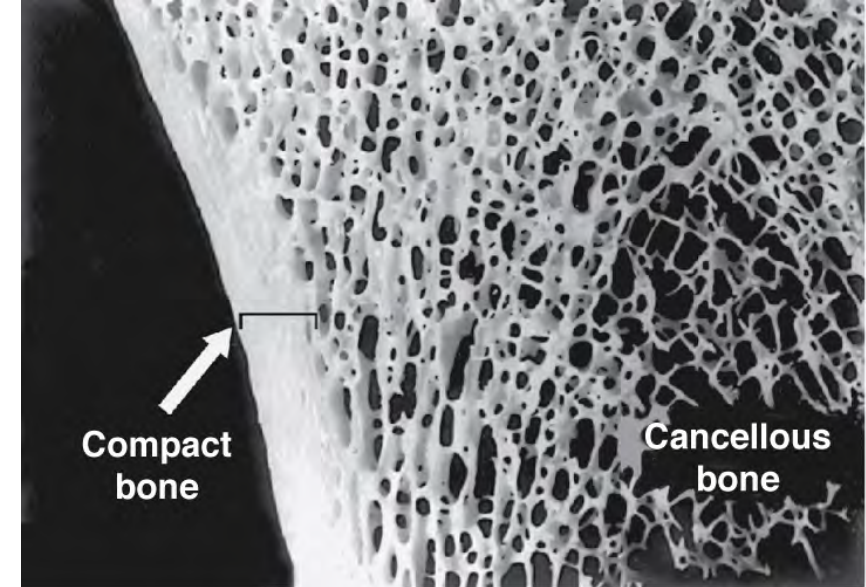
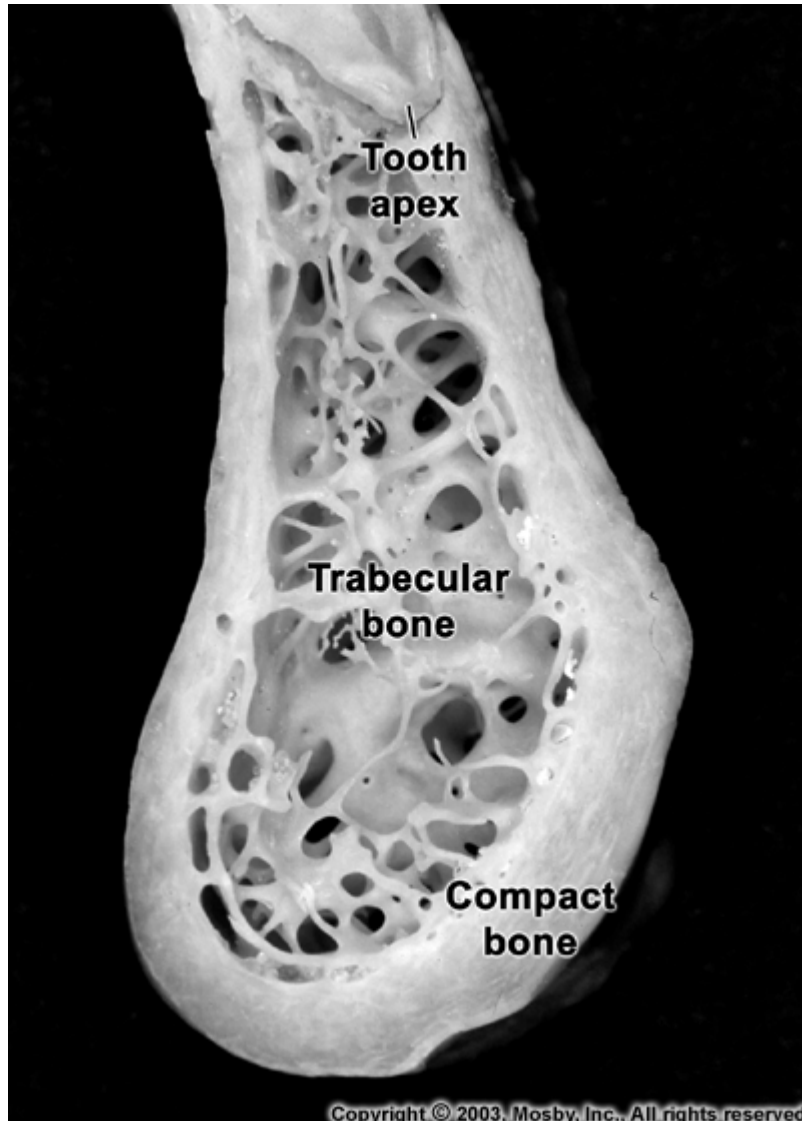
Classification

- Gross observation
 - Compact bone
 - dense, solid, weight-bearing,
 - Spongy(cancellous) bone
 - porous, marrow cavity, trabeculae, spicules, light weight
- Microscopic observation (Maturity)
 - Primary bone: woven bone
 - Immature
 - More cells, irregular fiber bundles
 - Less mineralized
 - During development and bone repair
 - Secondary bone : lamellar bone
 - Mature
 - Lamellae (3~7 μ m thick)
 - Canaliculi
 - More calcified $\rightarrow$ stronger
 - Regular and parallel collagen fiber



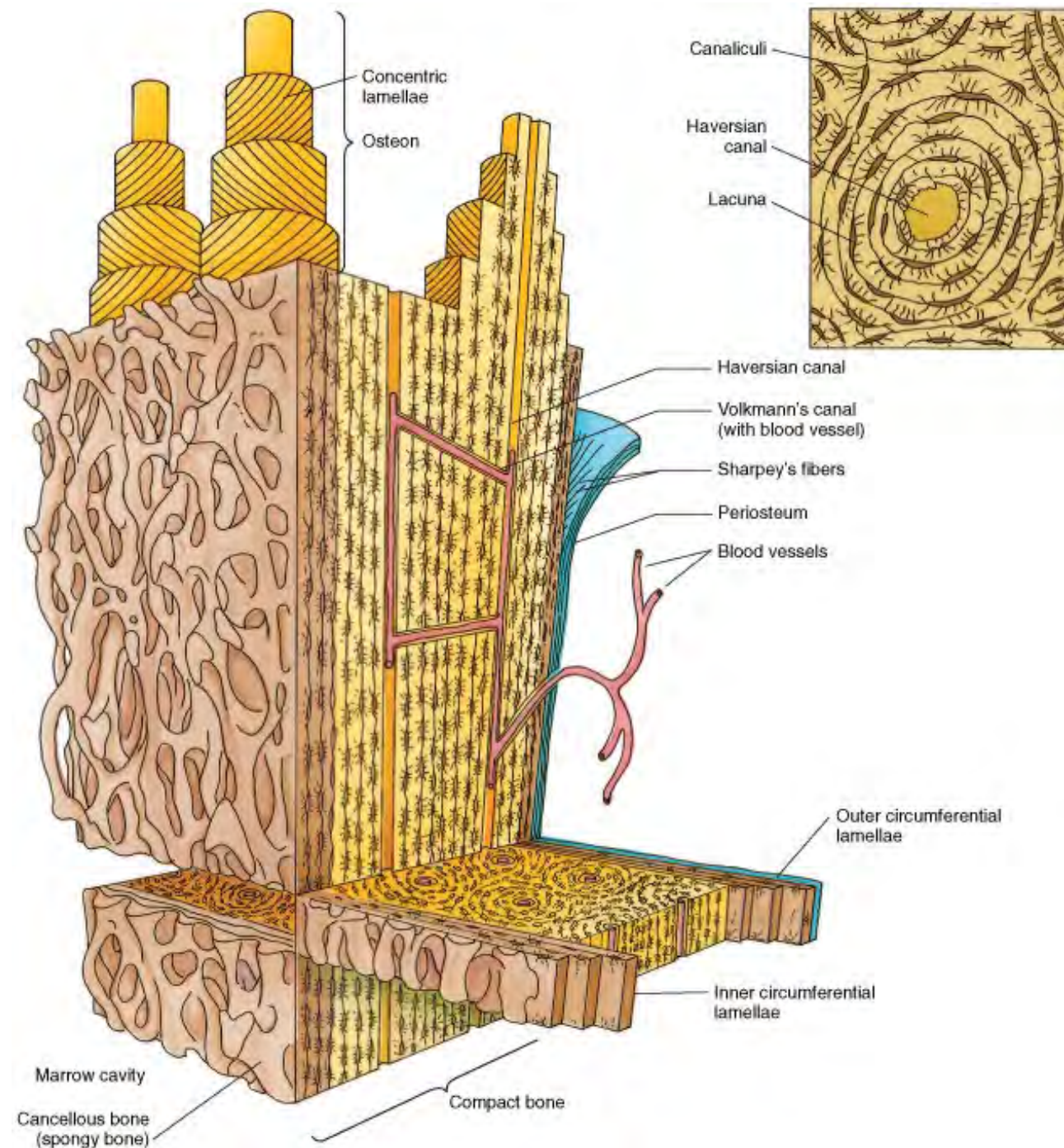


Compact vs. Spongy Bone

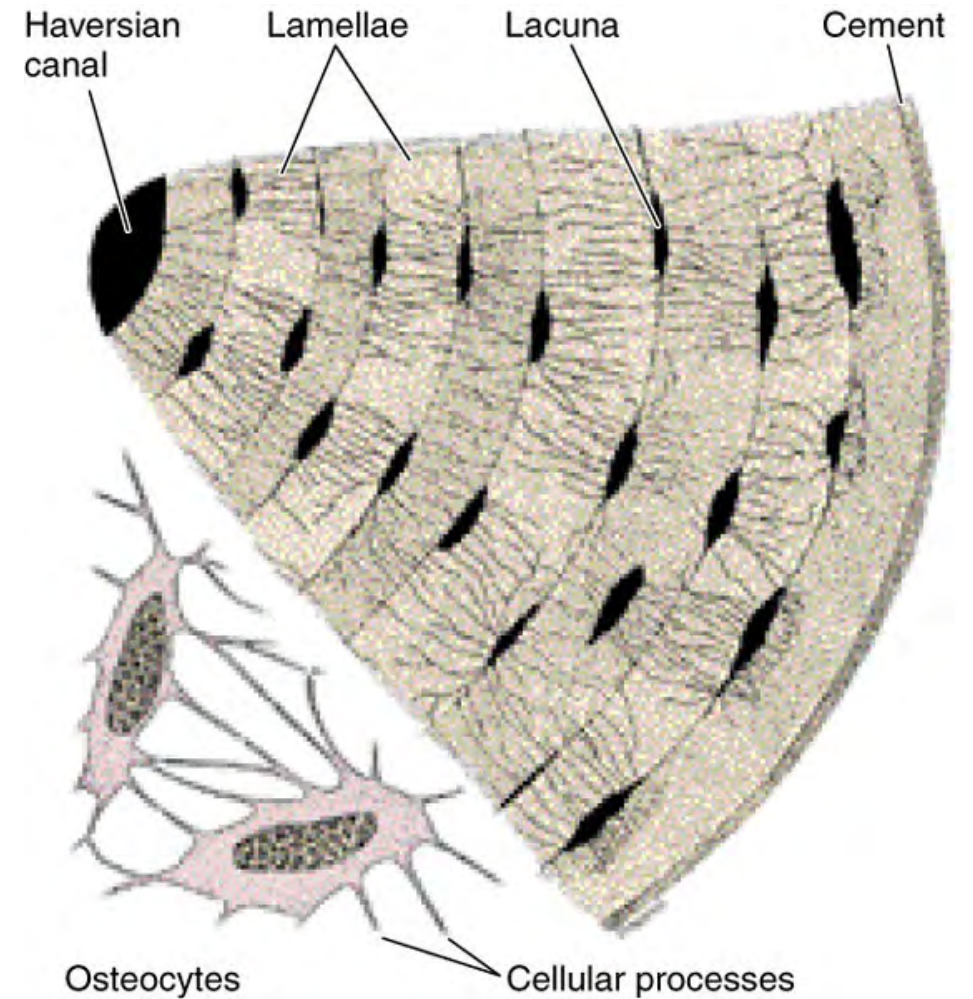
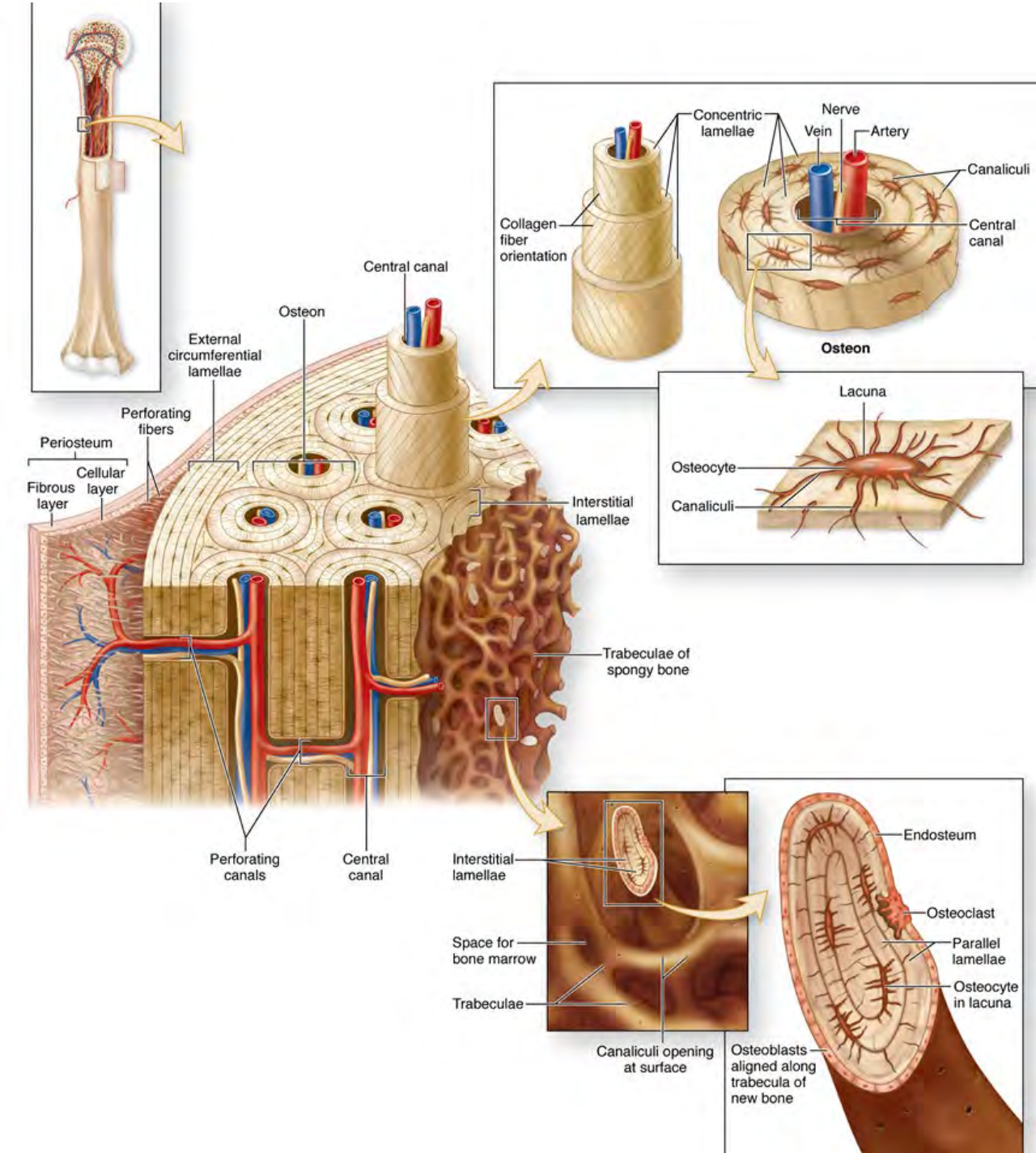


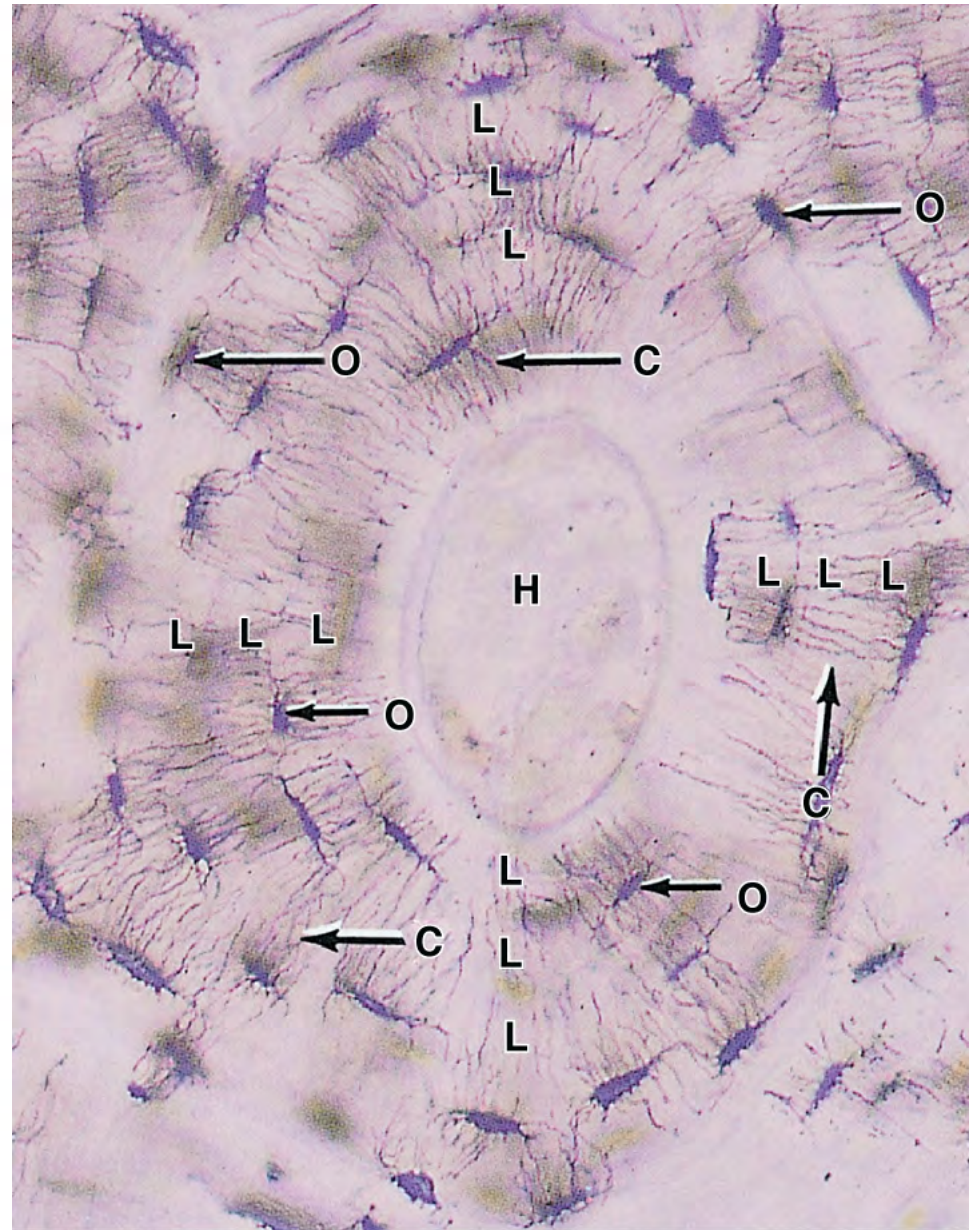
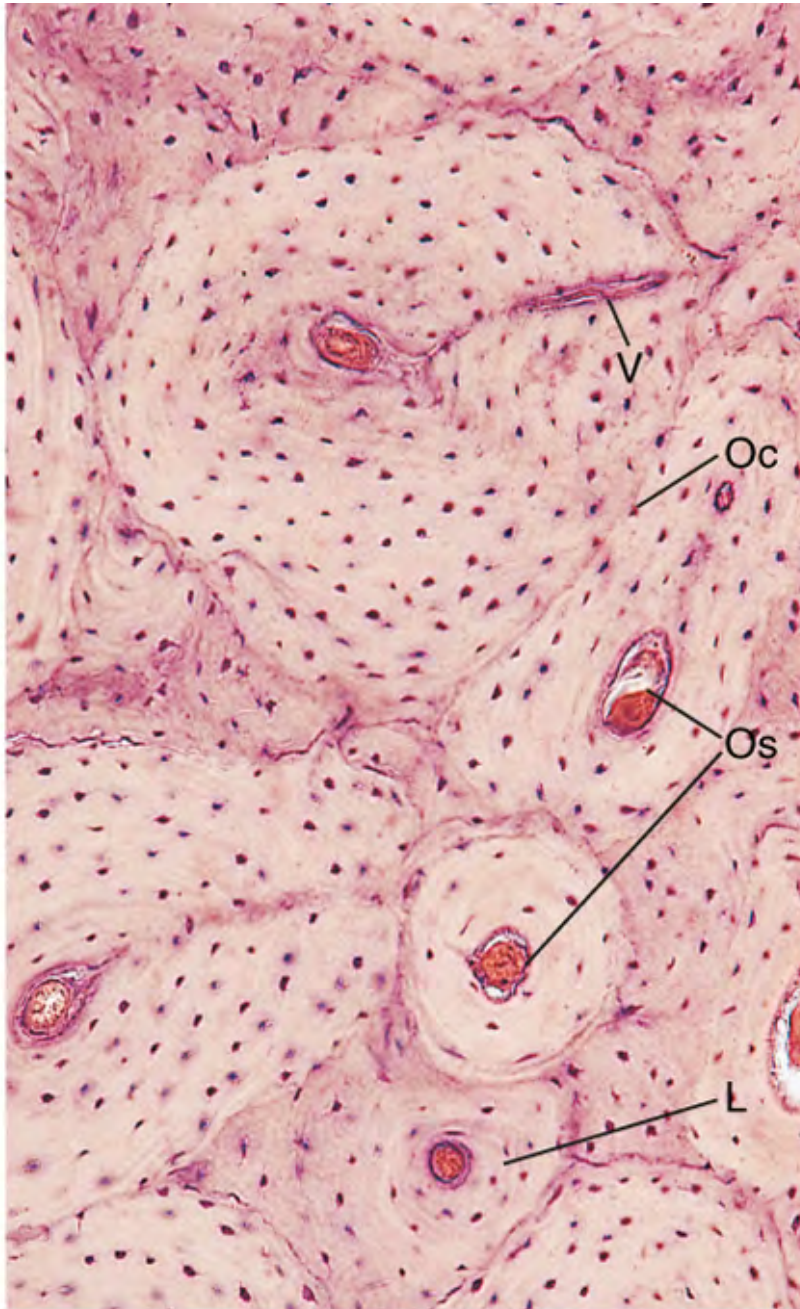
Compact Bone

- Lamellae : increase strength
 - Osteon / Haversian system
 - Structural unit of long bone
 - Cylindrical, concentric
 - Parallel to bone
 - Haversian canal
 - Circumferential lamellae
 - Outer
 - Inner
 - Interstitial lamellae



Bone and blood vessel architecture





Bone

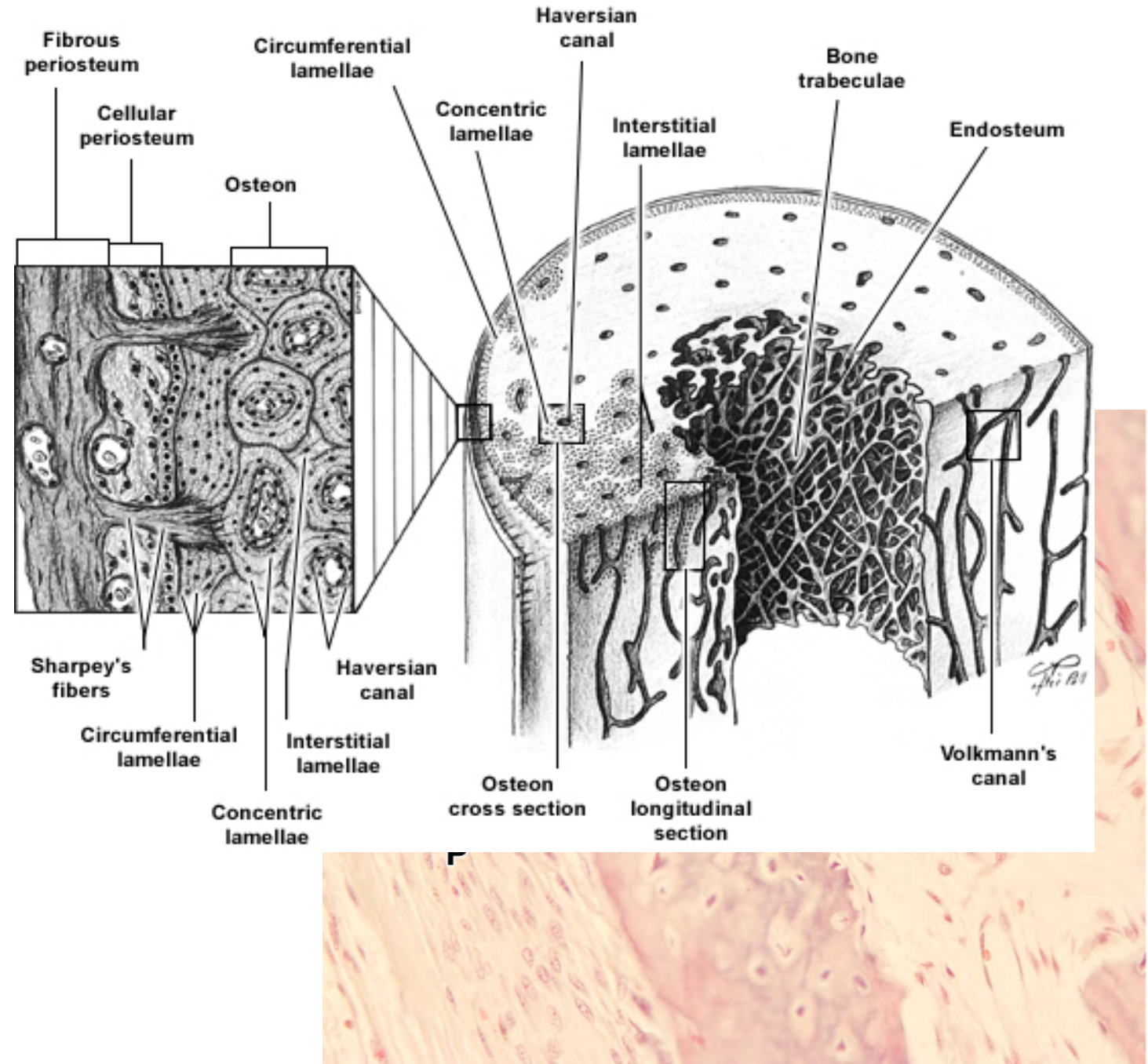
- Periosteum : outer fibrous layer (fibroblast) + inner cellular layer (cells)
- Bone
 - Cells : osteoprogenitor cell, osteoblast, osteoclast, osteocyte\
 - ECM
 - Organic :
 - fiber : type I collagen (80~90%)
 - ground substance
 - Proteoglycan: GAG (chondroitin sulfate)
 - Glycoprotein: osteonectin, osteocalcin, osteopontin, bone sialoprotein
 - Inorganic : 65% of dry weight
 - As a form of Hydroxyapatite crystals $[\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2]$
 - 40nm in length, 25nm in width, 1.5~3nm in thickness
- Endosteum : monolayer of osteoprogenitor cells and osteoblasts, osteoclast
- Bone marrow
 - Red bone marrow
 - Yellow bone marrow

Periosteum

- Non-calcified tissue
- Outer fibrous layer
 - dense irregular CT (fibroblast)
 - Nerves
 - Blood vessels
- Inner cellular layer
 - osteoprogenitor cells
 - osteoblasts,
 - osteoclasts
- Sharpey's fibers

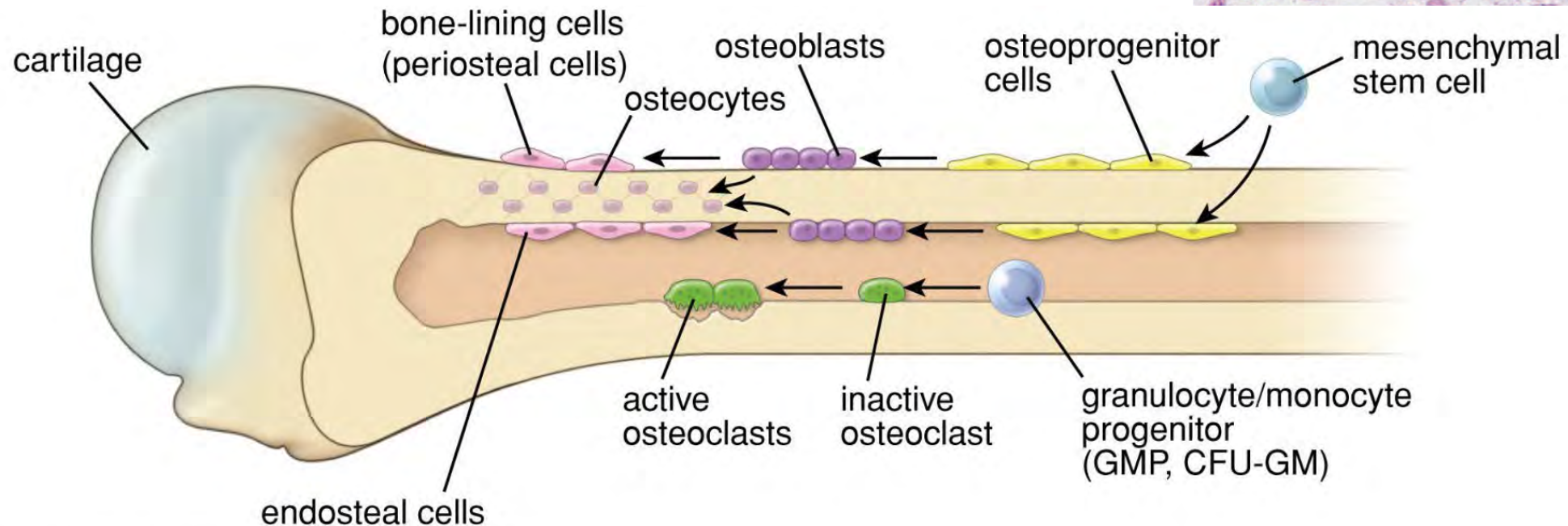
Endosteum

- Inner surface of compact bone
- Covers trabeculae
- Cells
 - osteoprogenitor cells
 - osteoblasts
 - osteoclasts



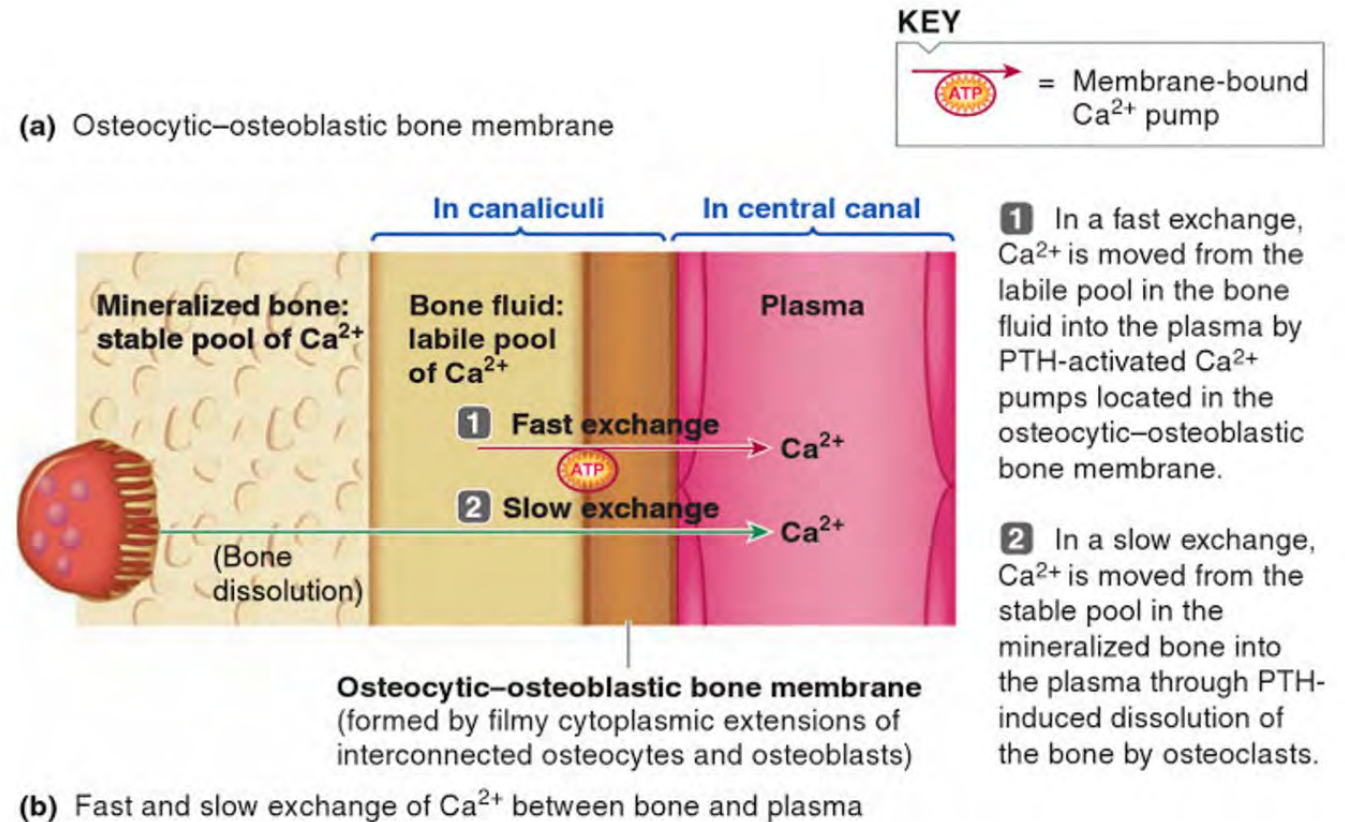
Types of cells in the bone

- Osteocytes
- Osteoblasts – central role in osteogenesis
- Osteoprogenitors- mesenchymal stem cells that produce osteoblasts
- Osteoclasts – bone remodeling



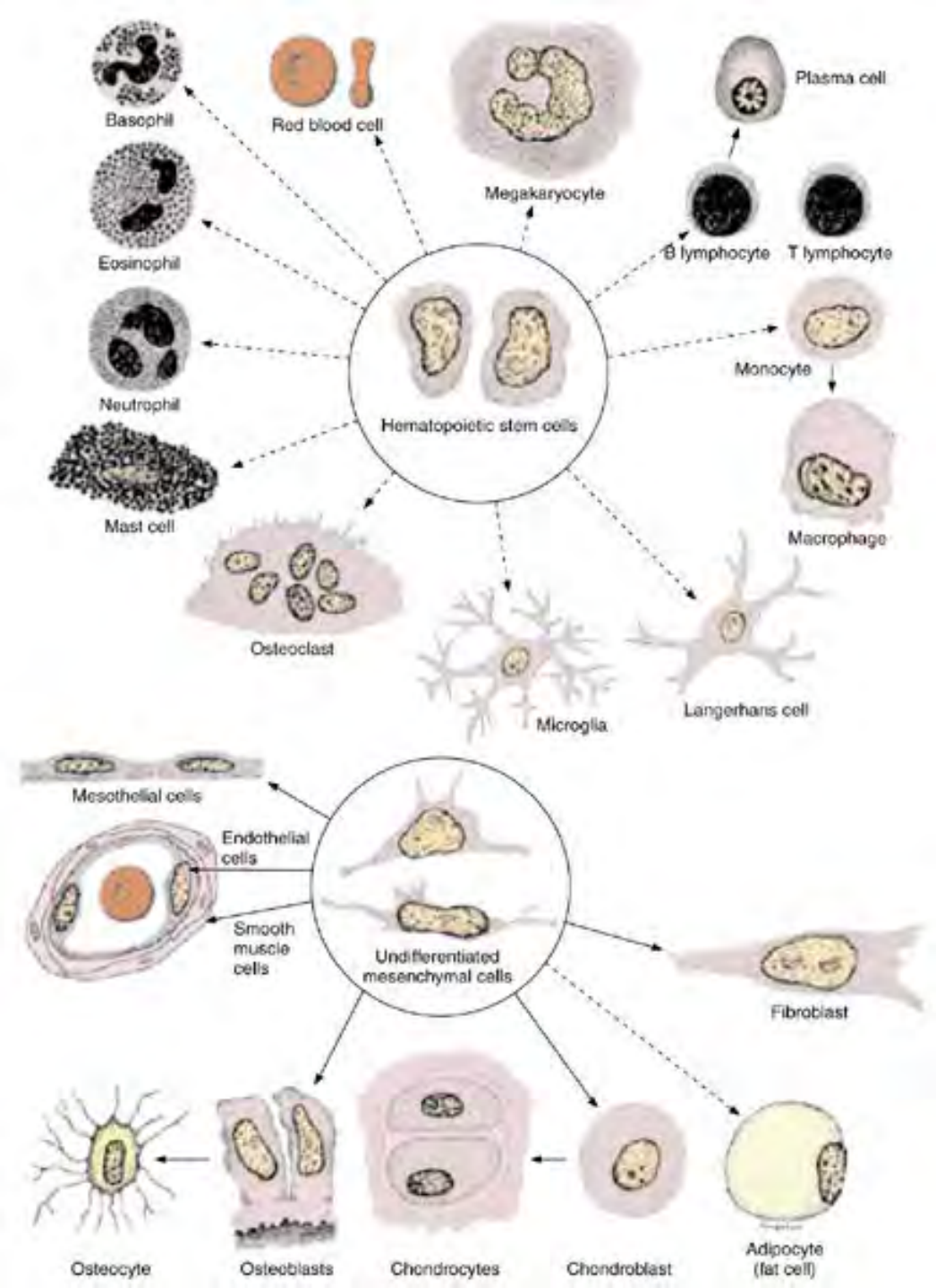
Metabolism in bone tissue

- Mineral store for Ca^{2+}
- Calcium release from bone
 - Simple transfer (fast)
 - in new lamellae
 - bone-Interstitial fluid-blood
 - PTH
 - Hormonal effect (slow)
 - PTH
 - Calcitonin



Origin of bone cells

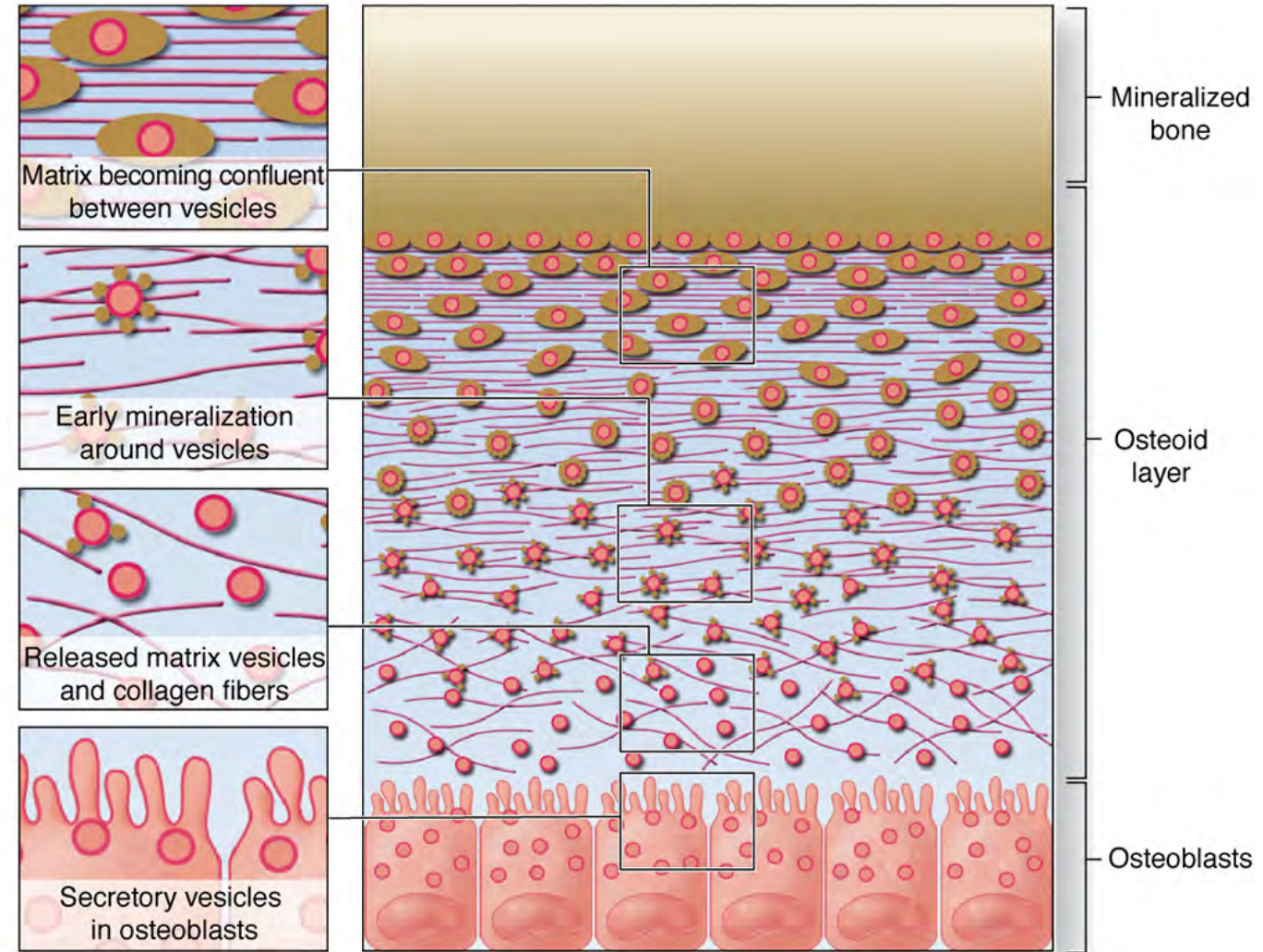
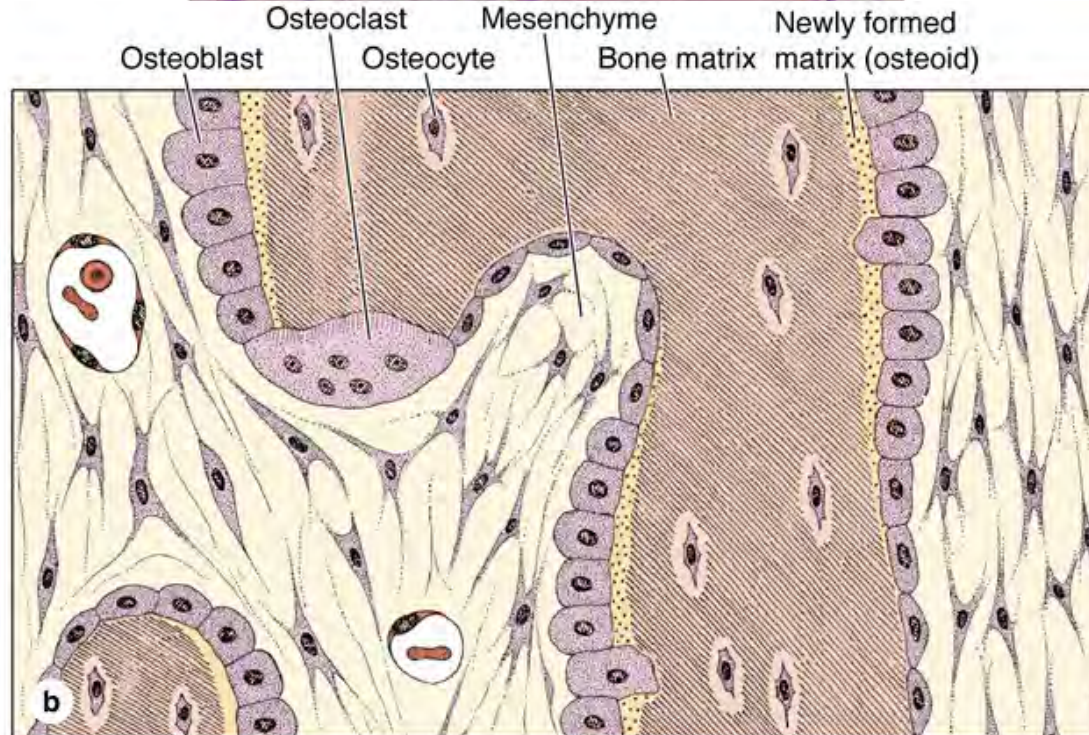
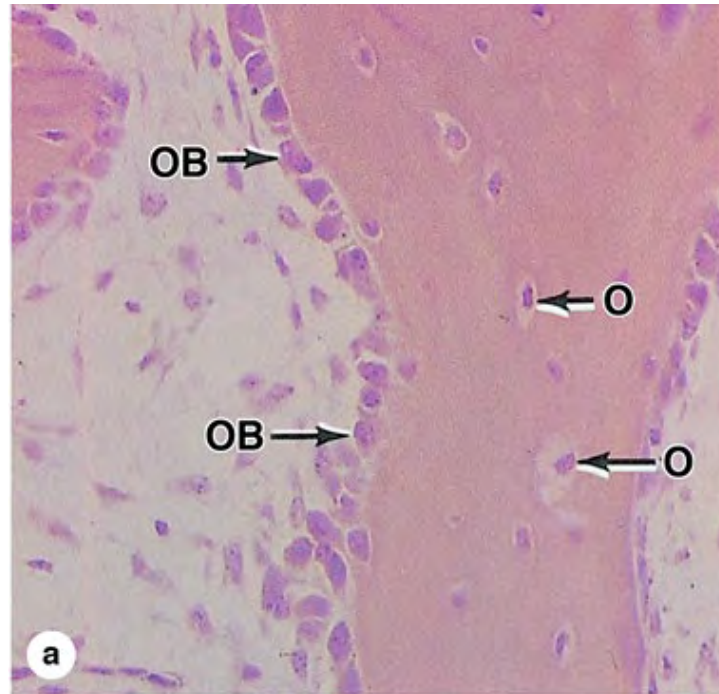
- Osteoprogenitor cells
- Osteoblast
- Osteoclast
- Fibroblast



Osteoblast

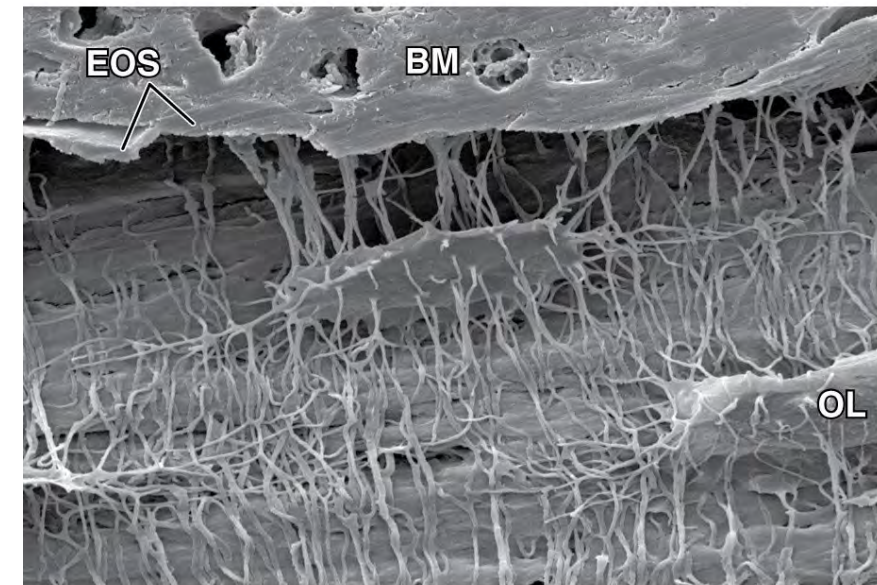
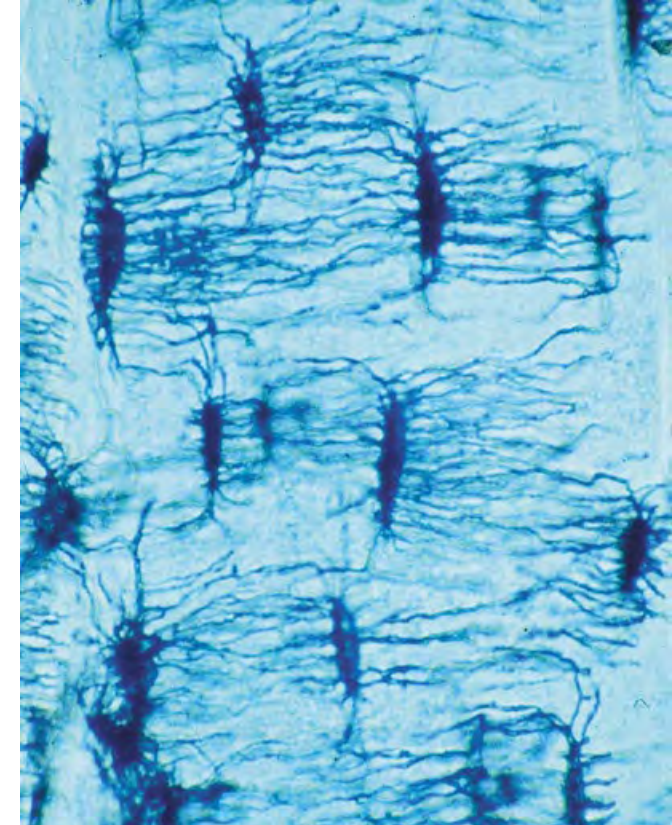
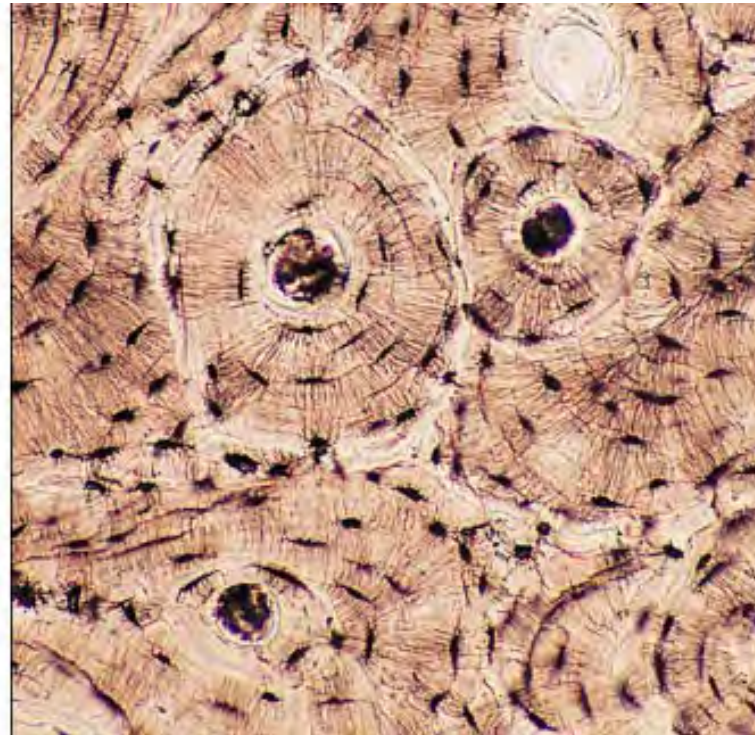
- Origin: mesenchymal cell
- central role in osteogenesis (osteoid), also resorb osteoid
- Sheet-like arrangement
- Cuboidal to columnar shape
- Processes and gap junction
- Secrete: osteoid (a thin uncalcified bone matrix)
 - Type I collagen
 - Proteoglycan (chondroitin sulfate)
 - Glycoprotein
 - RANKL (osteoclast stimulating factor)
- Types
 - Active form: cuboidal or columnar, polarized, basophilic cytoplasm
 - Non-active form: flat, weak basophilic → bone lining cell
- On cell membranes
 - Receptor for PTH (parathyroid hormone)
 - Secrete OPG(osteoprotegerin ligand)
 - Integrin





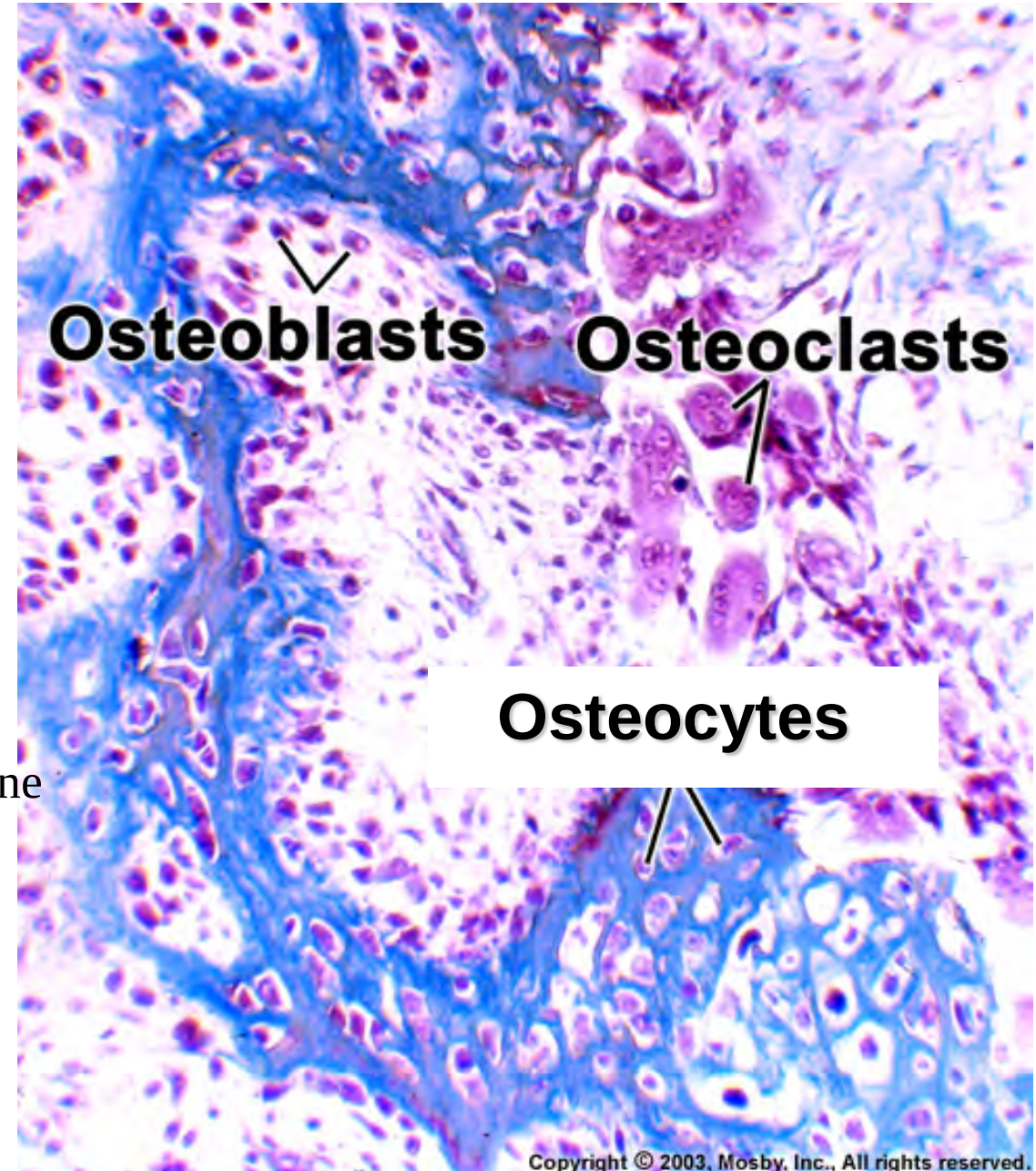
Osteocyte

- 20,000~30,000 osteocytes per mm³
- individually in lacunae
- Flat nucleus, poor cytoplasm
- Process in bony canaliculi
- Cell to cell junction: gap junction
- can not divide
- Inactive but secrete substances
- Function:
 - Maintain and monitor the protein and mineral content of the surrounding matrix.
 - Mechanotransduction: respond to mechanical stress → cAMP, osteocalcin, IGF → recruit preosteoblasts
 - Participate in the repair of the damaged bone

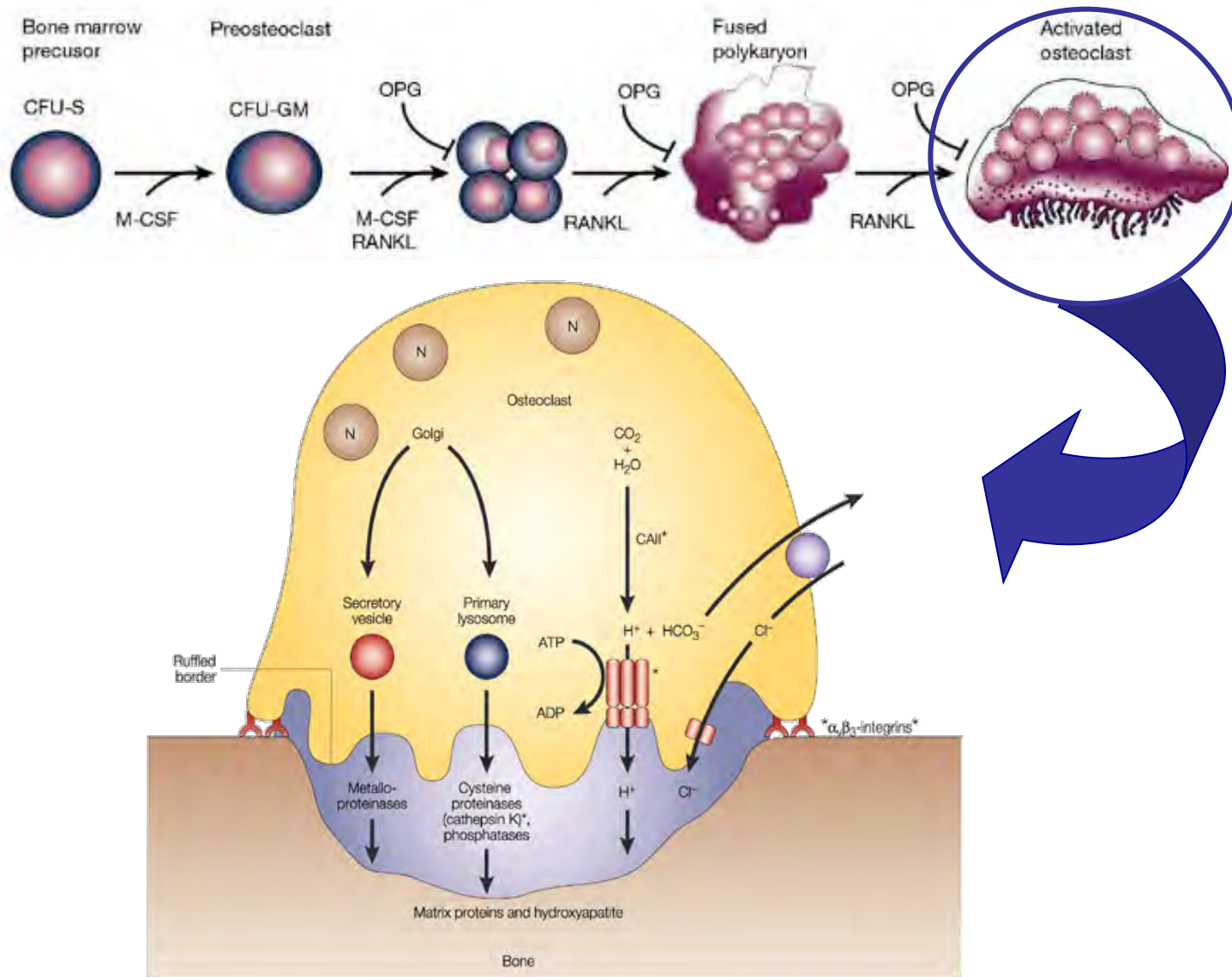


Osteoclast

- Origin: hematopoietic stem cells (rather than monocytes in bone marrow)
- Wandering cells
- appearance:
 - Big (150µm) and Irregular shape
 - Acidophilic cytoplasm
 - many processes
 - Many nuclei (5~50) by cell fusion
 - Ruffled border
 - Clear zone
- In Howship's lacunae (shallow depression)
- Secrete: collagenase, proteolytic enzyme → resorb the bone
- Receptor:
 - for calcitonin (parafollicular cell in thyroid gland)
 - RANK
 - Colony-stimulating factor-1
 - Osteoprotegerin
 - Not for PTH



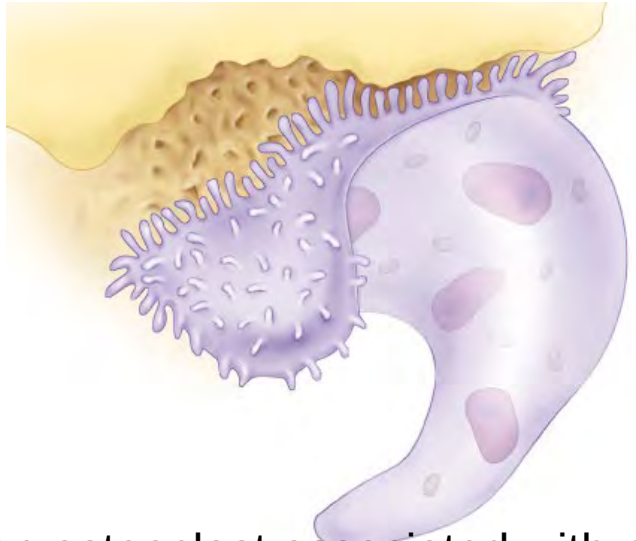
Osteoclastogenesis



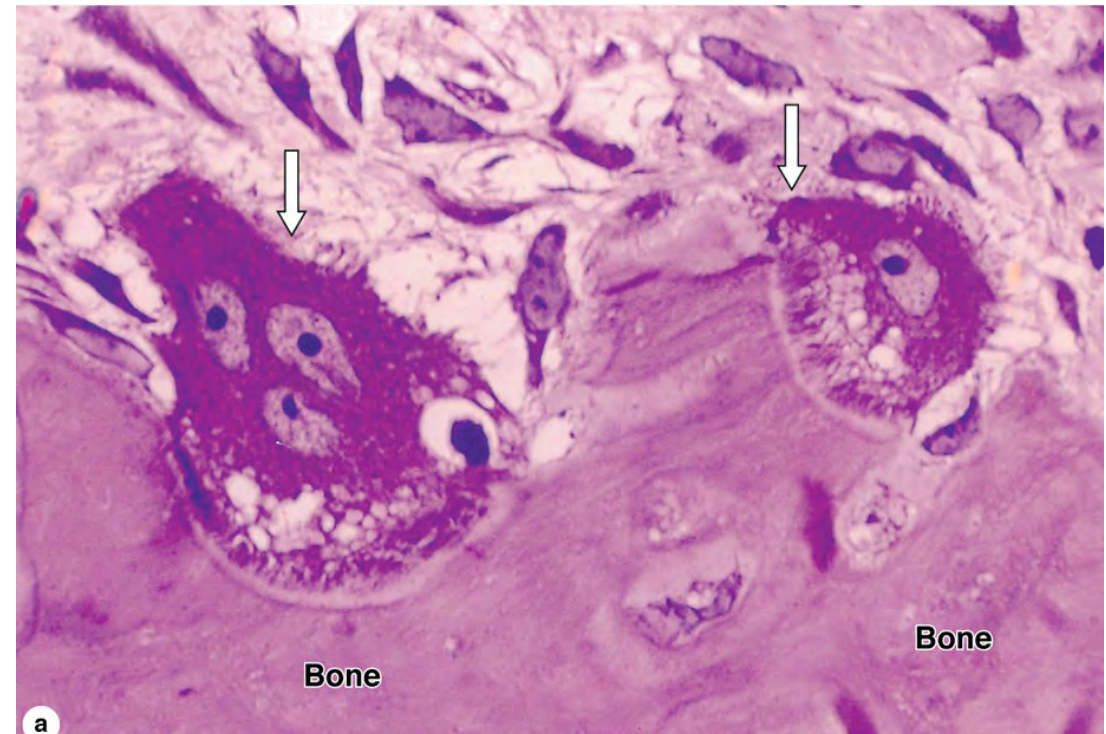
- **M-CSF**
 - macrophage colony-stimulating factor
 - promotes formation of mononucleated osteoclasts
- **RANKL**
 - receptor activator of NF- κ B ligand
 - interacts with RANK expressed by osteoclast
 - promotes formation of multinucleated osteoclasts
- **OPG**
 - osteoprotegerin
 - negative regulates RANKL signaling by binding RANKL
- **Calcitonin:**
 - a hormone of thyroid gland,
 - suppresses the osteoclastic activity.
- **Parathyroid hormone (PTH):**
 - No PTH receptor in osteoclasts.
 - PTH stimulates the osteoblasts to secrete RANKL

Osteoclast

- Bone resorption
 - Decalcification : remove inorganic matrix
 - $\text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{H}^+ + \text{HCO}_3^-$
 - H^+ -ATPase
 - Below pH 4.0
 - Resorption of organic matrix
 - Lysosomal hydrolases, metalloproteinase :
Collagenase and gelatinase



Odontoclast: an osteoclast associated with root resorption of deciduous teeth



Other cells in bone

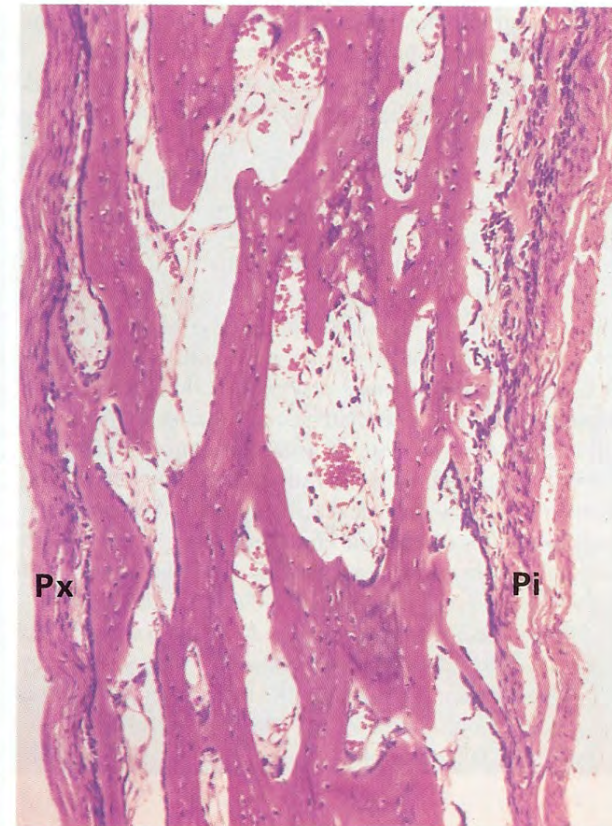
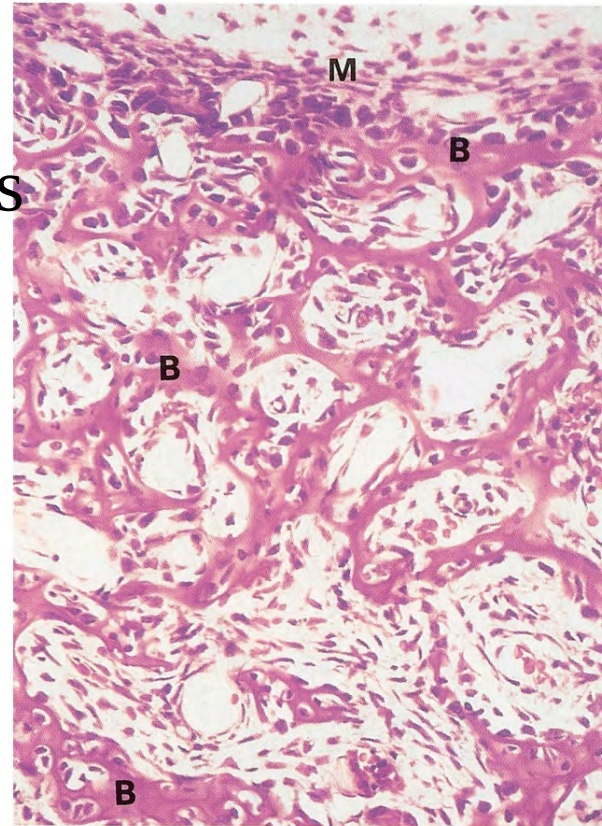
- Osteoprogenitor cells :
 - Lining the endosteum and periosteum
 - Spindle shape
 - Pale-staining oval nucleus
 - Scant cytoplasm
 - Differentiate into
 - Osteoblast
 - Chondroblast: in low oxygen level
- Fibroblasts: Inner and outer layer of periosteum

Bone Formation

- Osteogenesis – development of the skeleton and growth through adolescence (~18 females, ~21 males)
 - Osteoblasts secrete osteoid
 - Osteoblasts become osteocytes
 - Osteoid is mineralized (calcium phosphate precipitates)
 - Periosteum formed
 - Mature lamellar bone formed on surfaces
- Intramembranous ossification: mandible body
- Endochondral ossification: long bones and condyle

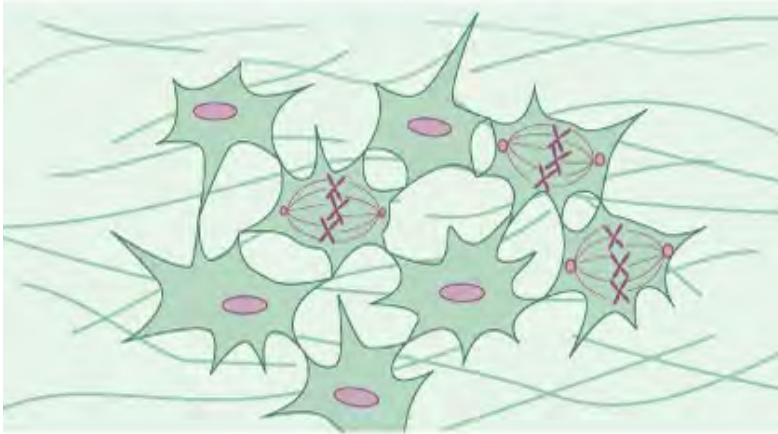
Intramembranous Ossification

- Locations
 - Flat bone develops from mesenchyme (undifferentiated, embryonic CT)
 - Appositional growth in long bone
 - Skull and clavicle
- Mesenchymal cells become osteoblasts
- Osteoblasts form bone
- Processes
 - Primary ossification center
 - Fusion of centers
 - Formation of woven bone
 - Formation of periosteum
 - Formation of lamellar bone

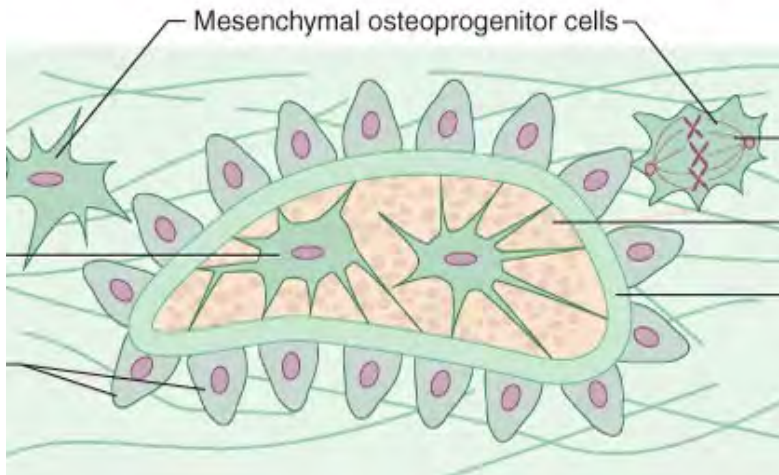


Intramembranous Ossification

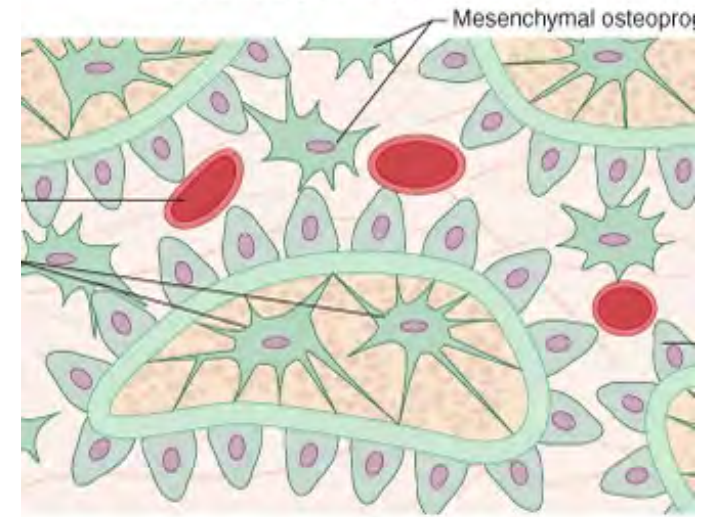
Condensation of mesenchymal osteoprogenitor cells



Further proliferation and differentiation of cells into osteoblasts, and production of osteoid woven bone spicule

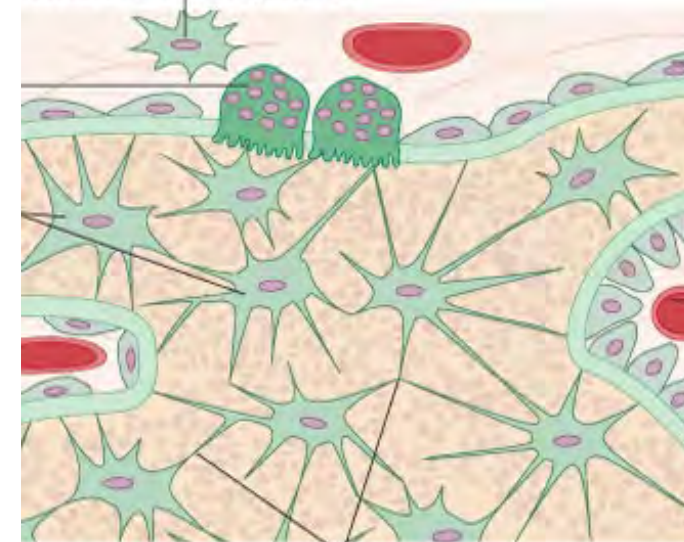


Increase in sizes of bone spicules and invasion of intervening areas by blood vessels

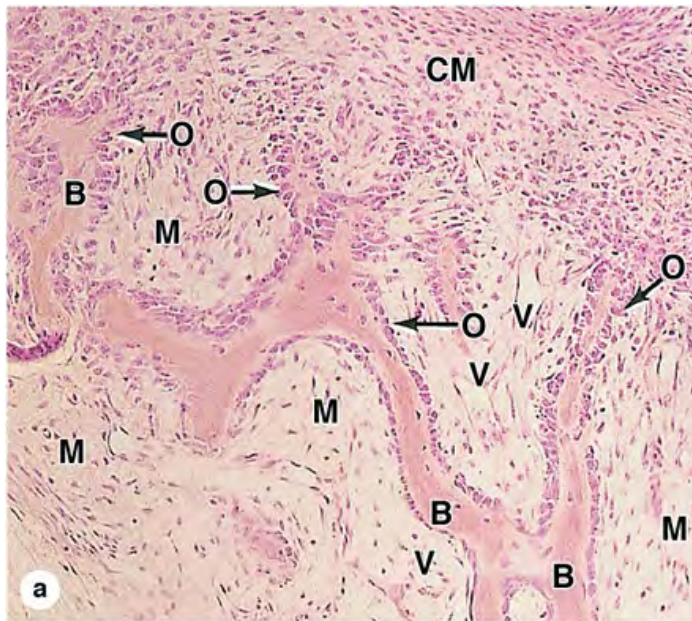


Fusion of bone spicules to form a continuous plate of woven bone and the beginning of remodeling by osteoclasts

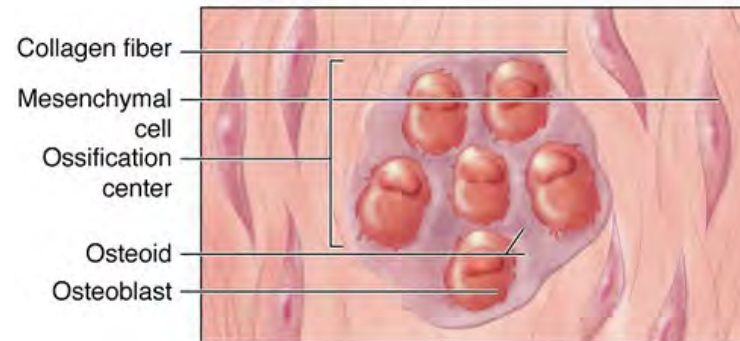
Mesenchymal osteoprogenitor cell



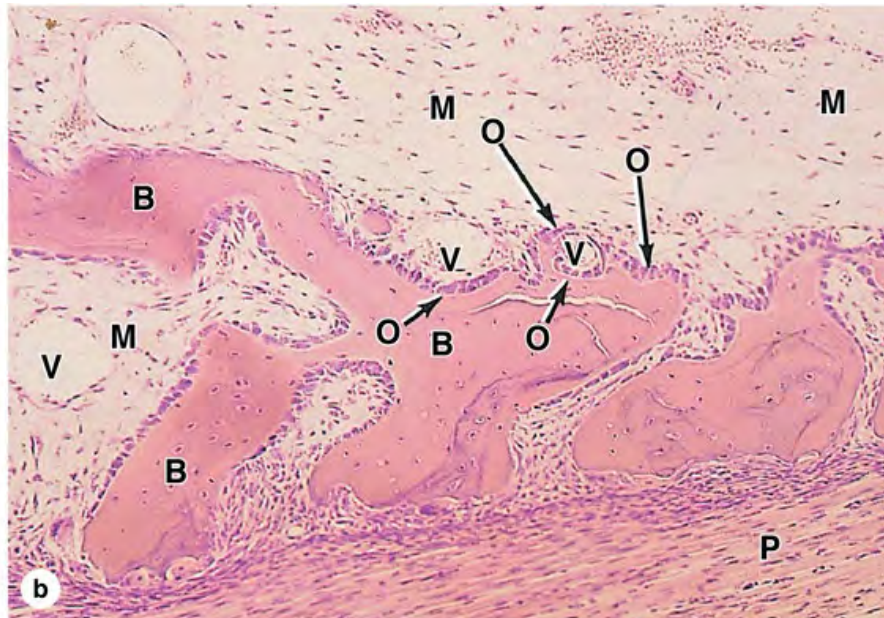
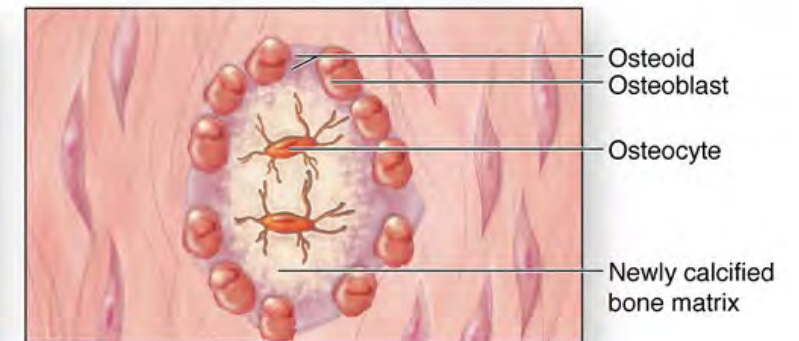
Osteocyte-osteocyte canalicular contact; gap junctions present



(a) Ossification centers form within thickened regions of mesenchyme.



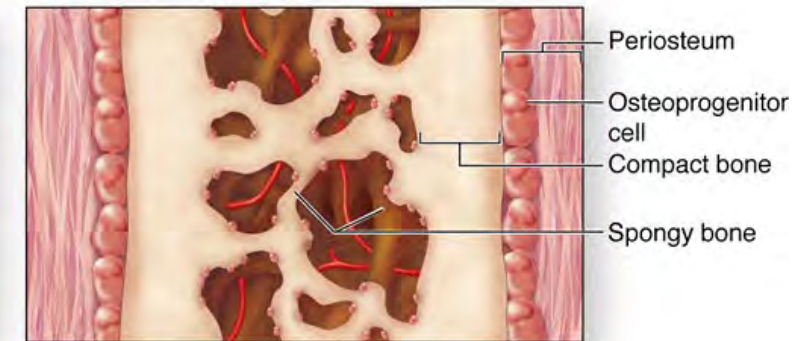
(b) Osteoid undergoes calcification.



(c) Woven bone and surrounding periosteum form.

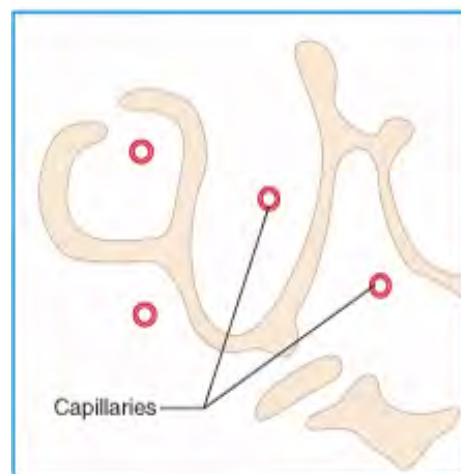


(d) Lamellar bone replaces woven bone, as compact and spongy bone form.

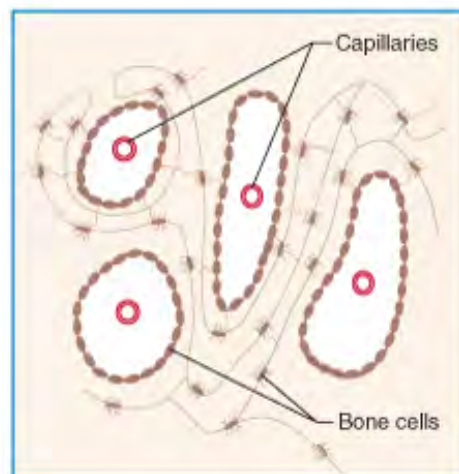




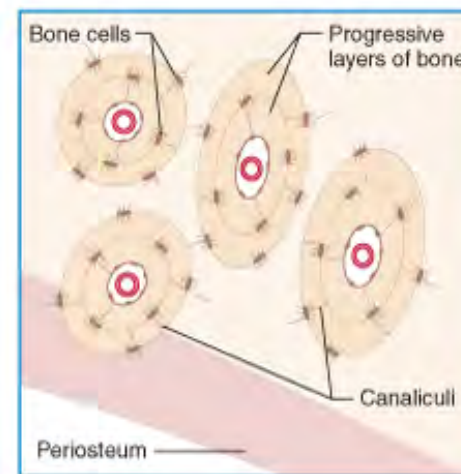
A



B



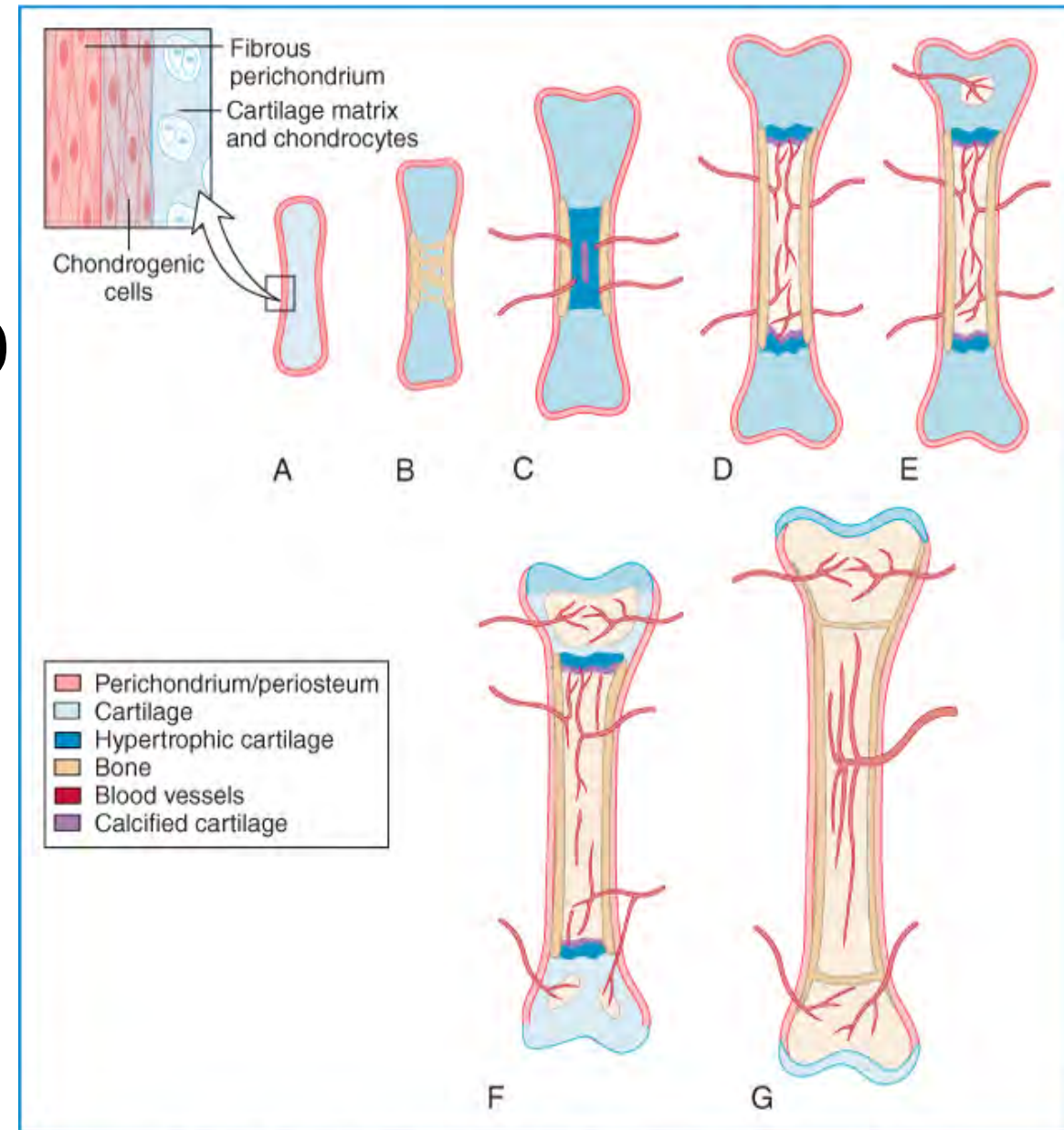
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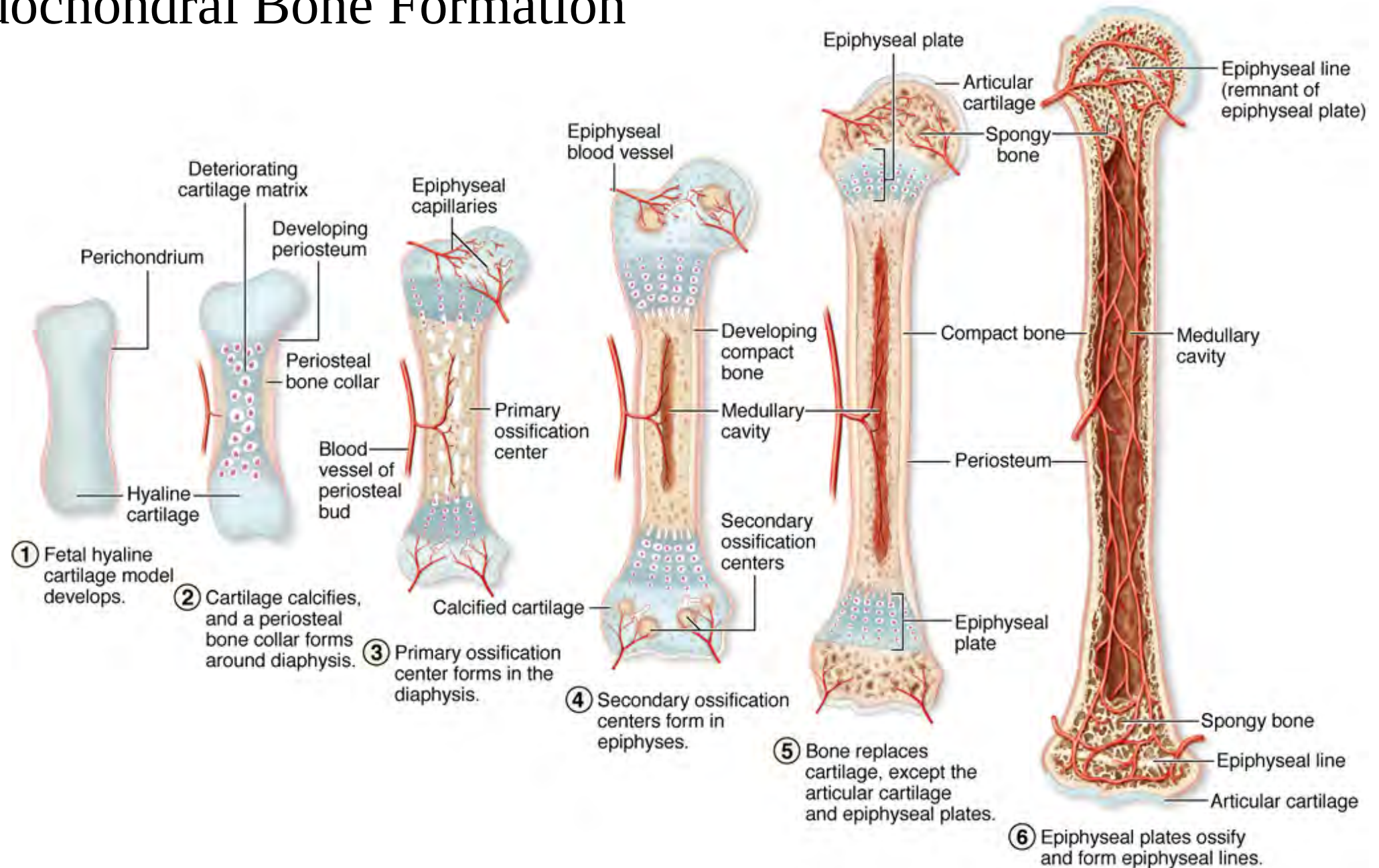
D

Endochondral ossification

- Hyaline cartilage model of bone first
- Cartilage is replaced by bone (long bone)
- Procedures
 - Bone collar
 - Degeneration of chondrocyte
 - Osteoclast
 - Blood vessel invasion
 - Osteoprogenitor cells invasion
 - Osteoclasts, macrophage: remove the cartilage matrix
 - Osteoblast : bone apposition



Endochondral Bone Formation



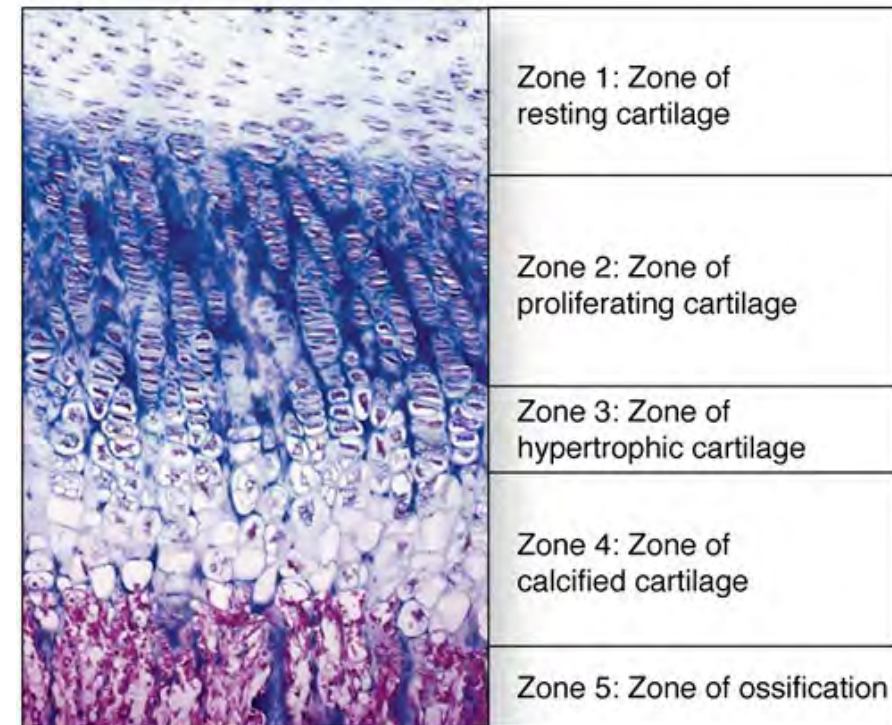
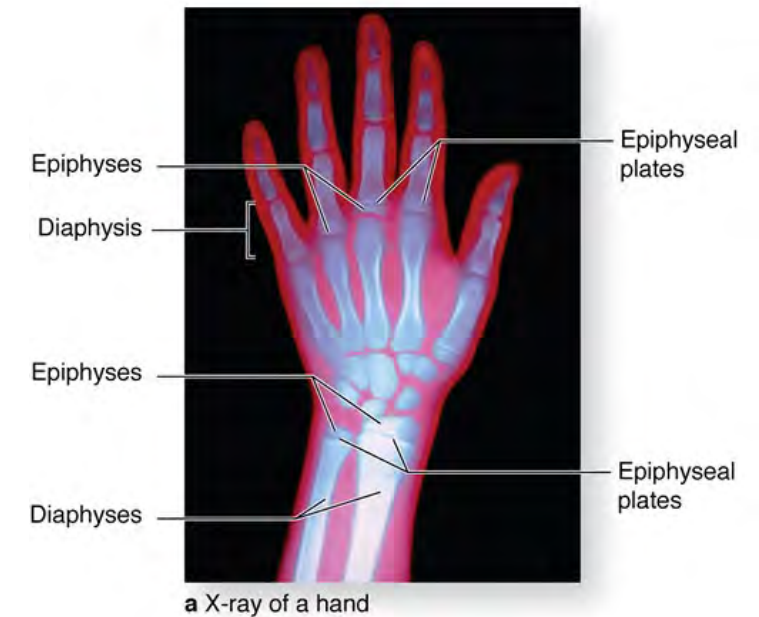
Endochondral Bone Formation

Event	Description
Hyaline cartilage model formed.	Miniature hyaline cartilage model formed in region of developing embryo where bone is to develop; some chondrocytes mature, hypertrophy, and die; cartilage matrix becomes calcified
Primary Center of Ossification	
Perichondrium at the midriff of diaphysis becomes vascularized.	Vascularization of perichondrium changes it to periosteum Chondrogenic cells become osteoprogenitor cells
Osteoblasts secrete matrix, forming subperiosteal bone collar.	The subperiosteal bone collar is formed of primary bone (intramembranous bone formation)
Chondrocytes within the diaphysis core hypertrophy, die, and degenerate.	Presence of periosteum and bone prevents diffusion of nutrients to chondrocytes; their degeneration leaves lacunae, opening large spaces in septa of cartilage
Osteoclasts etch holes in subperiosteal bone collar, permitting entrance of osteogenic bud	Holes permit osteoprogenitor cells and capillaries to invade cartilage model, now calcified, and begin elaborating bone matrix
Calcified cartilage/calcified bone complex is formed.	Bone matrix laid down on septa of calcified cartilage forms this complex (histologically, calcified cartilage stains blue, calcified bone stains red)
Osteoclasts begin resorbing the calcified cartilage/calcified bone complex.	Destruction of the calcified cartilage/calcified bone complex enlarges the marrow cavity
Subperiosteal bone collar thickens, begins growing toward the epiphyses.	This event, over a period of time, completely replaces diaphyseal cartilage with bone
Secondary Center of Ossification	
Ossification begins at epiphysis.	Process begins in same way as at primary center, except that there is no bone collar; osteoblasts lay down bone matrix on calcified cartilage scaffold
Growth of bone occurs at epiphyseal plate.	Cartilaginous articular surface of bone remains; epiphyseal plate persists-growth added at epiphyseal end of plate. Bone is added at diaphyseal end of plate
Epiphysis and diaphysis become continuous.	At the end of bone growth, cartilage of epiphyseal plate ceases proliferation; bone development continues to unite the diaphysis and epiphysis

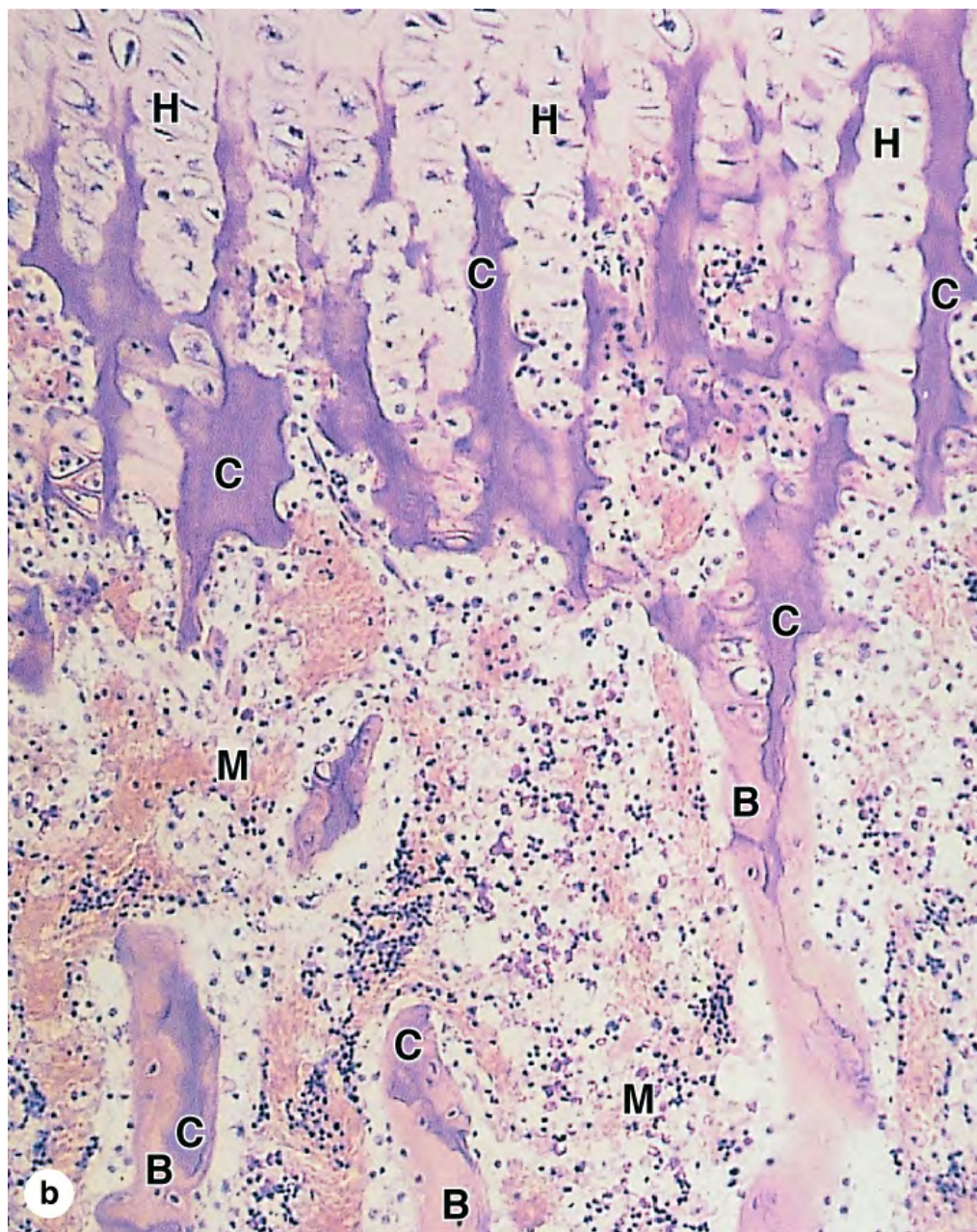
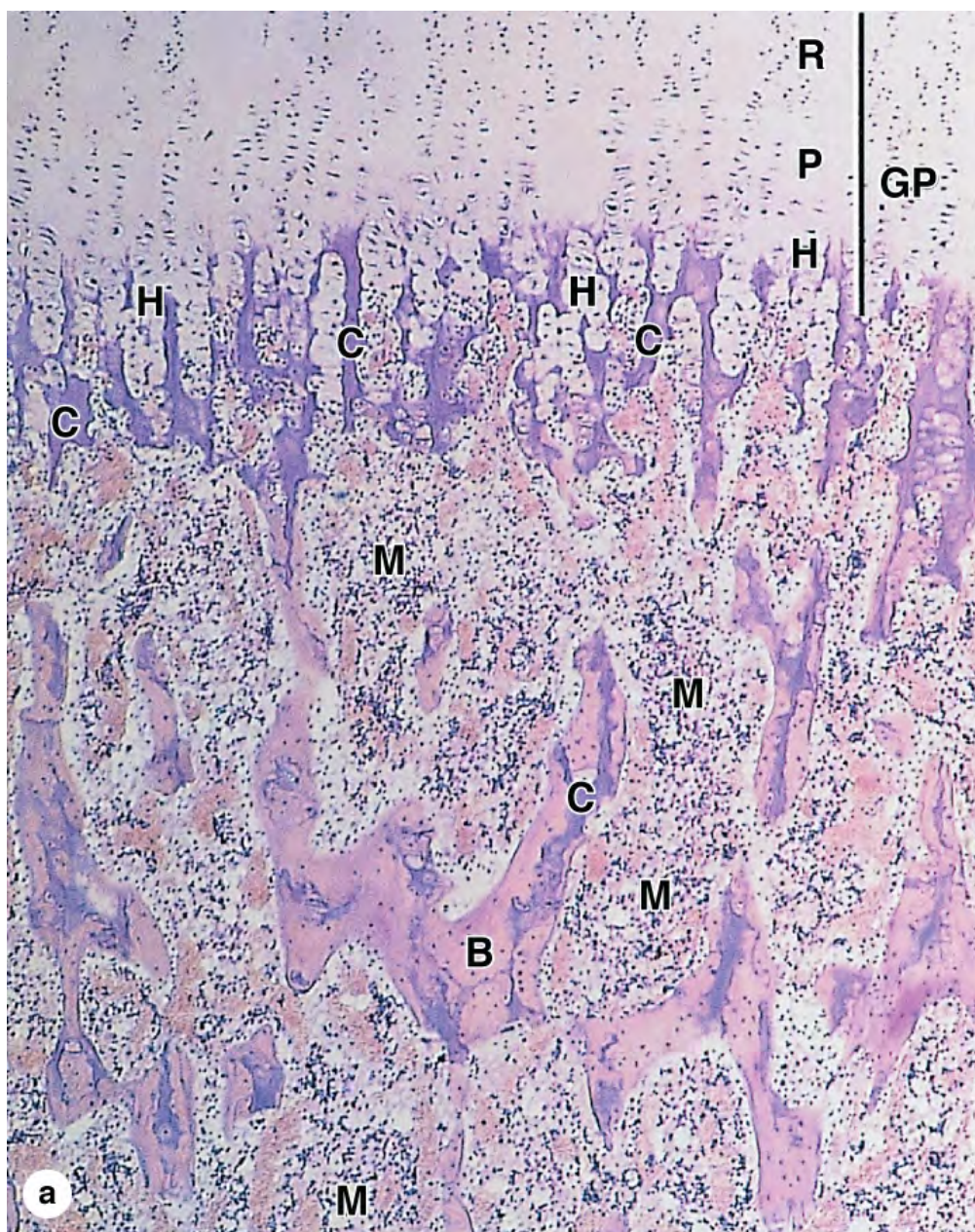
Endochondral ossification;

Five zone in developing epiphyseal plate

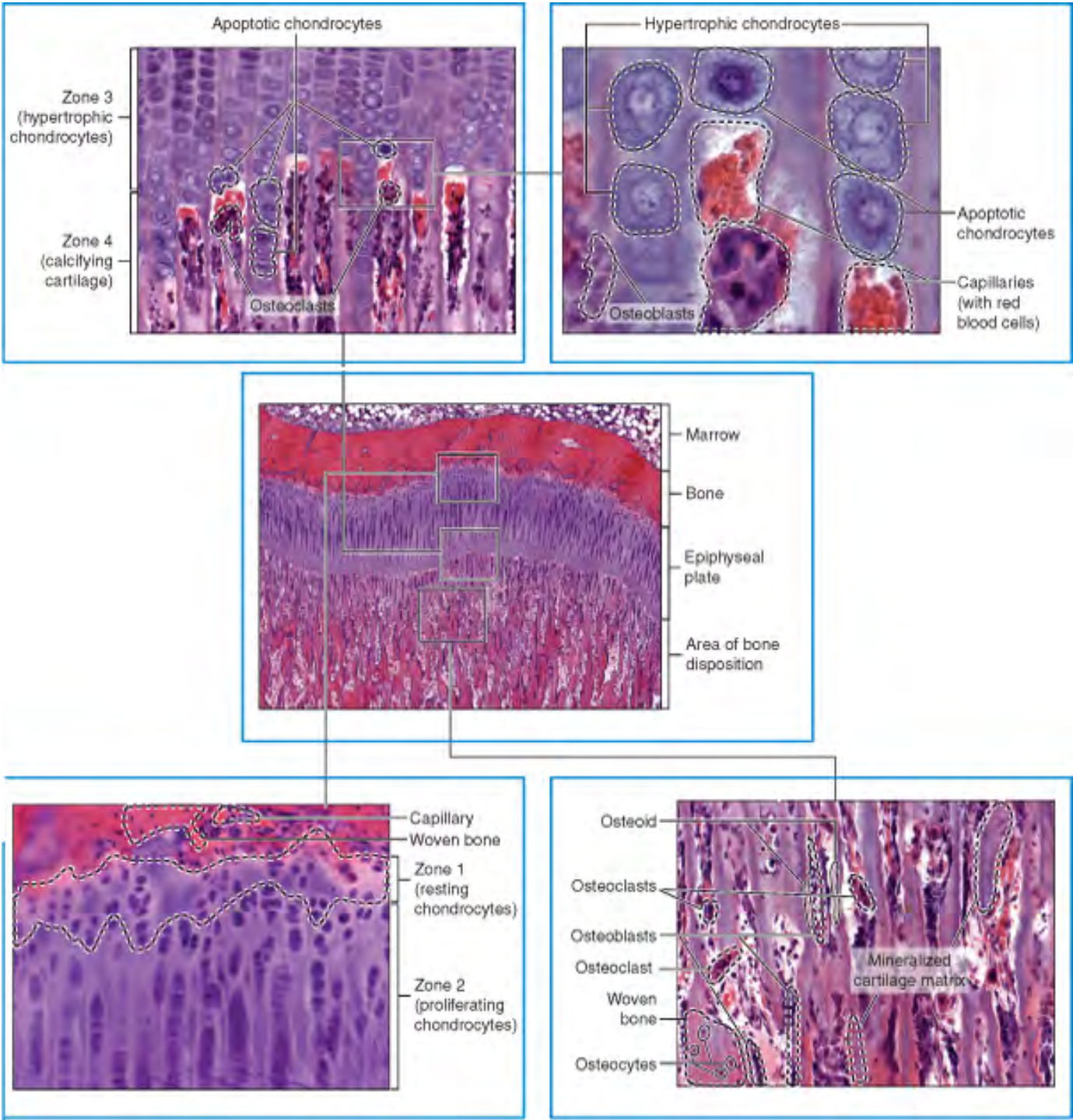
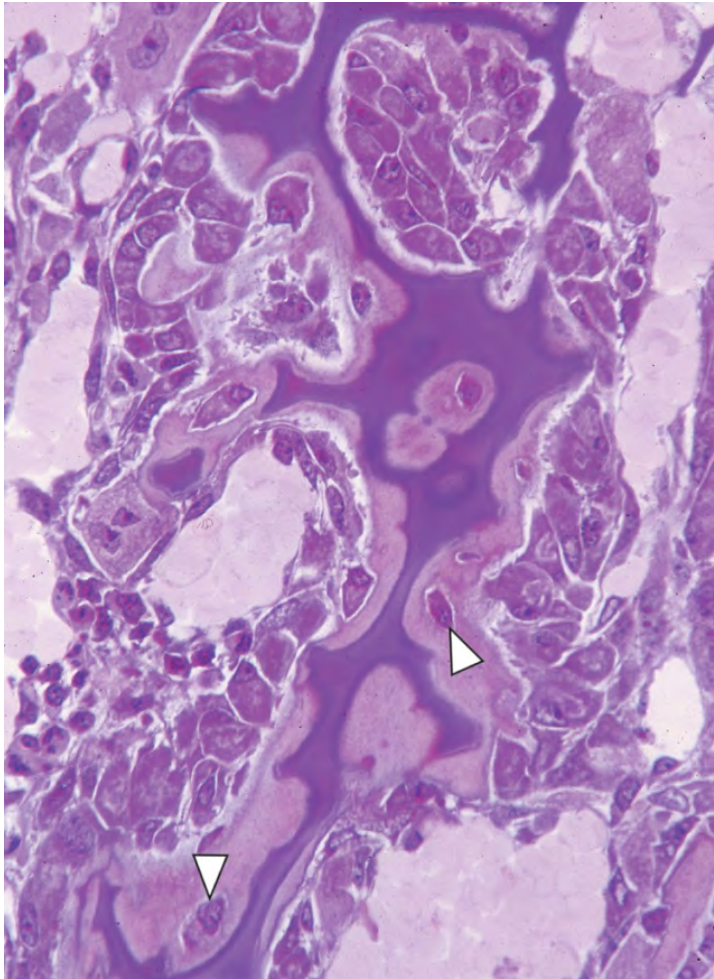
- Reserve zone(=resting zone)
 - Randomly distributed chondrocytes
- Proliferative zone
 - Isogenous cell group
- Hypertrophic zone
 - Mature, hypertrophic
 - Accumulate glycogen
- Calcified cartilage zone
 - Dead chondrocytes
 - calcified matrix
- Ossification zone
 - Bone formation on calcified cartilage



b Epiphyseal plate

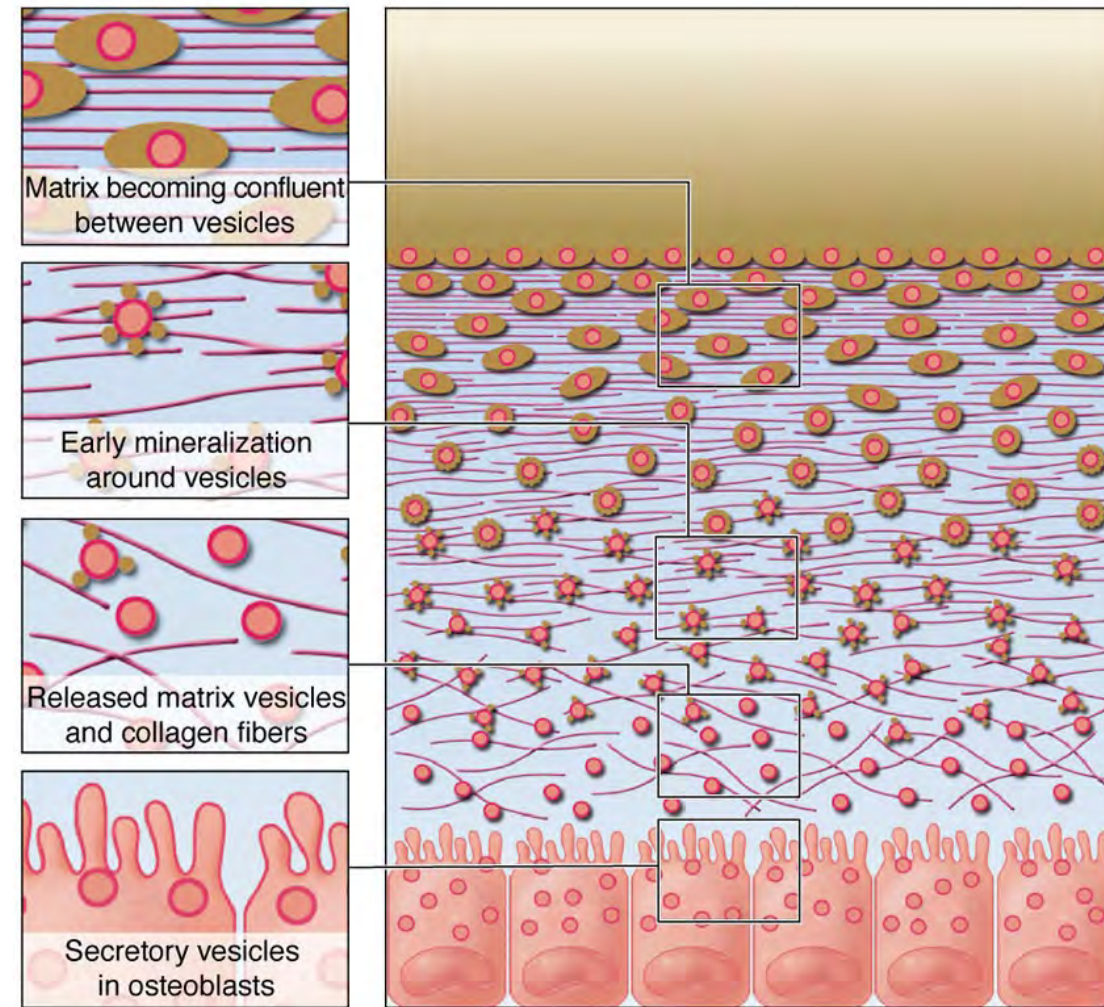


Endochondral ossification



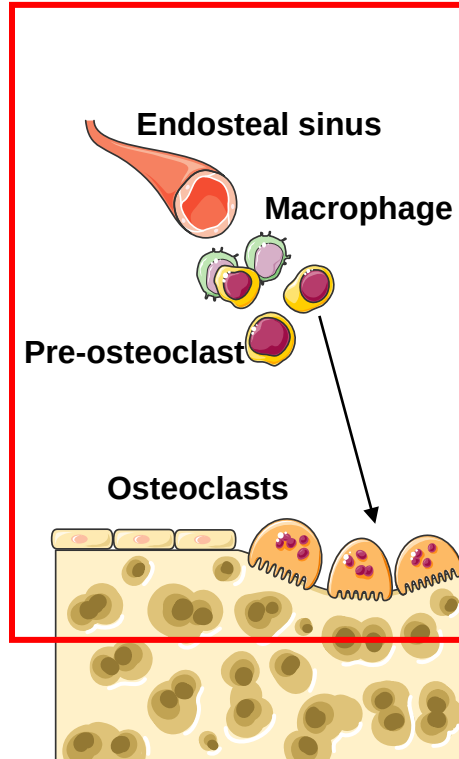
Mechanisms of calcification

- Heterogenous nucleation theory
 - Deposition of calcium phosphate on collagen fiber gap
- Matrix vesicle theory
 - Secreted by osteoblasts
 - Small membrane-bounded vesicle
 - 100~200nm in diameter,
 - Numerous calcium pump
 - High concentration of Ca^{2+} , PO_4^{3-} ions, cAMP, ATP, ATPase, Alkaline phosphatase, pyrophosphatase, calcium-binding proteins, phosphoserine
- Stimulation
 - proteoglycan
 - Calcium-binding glycoprotein
 - osteonectin, bone sialoprotein



Bone remodeling process

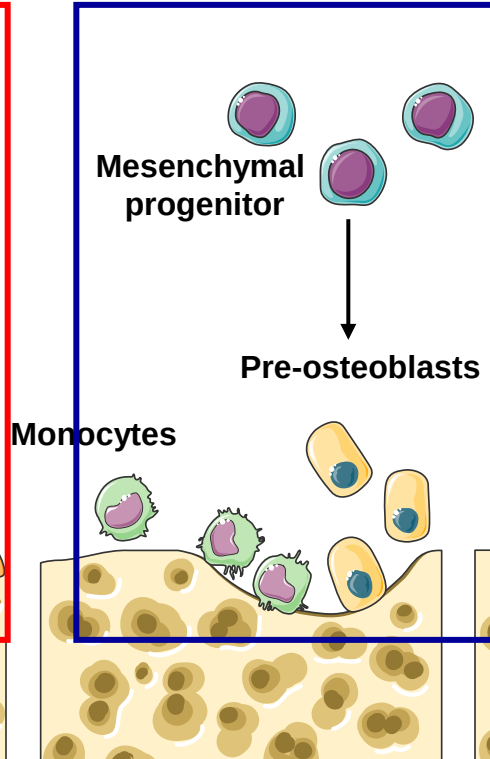
Osteoclastogenesis



Activation & Resorption

3 weeks

Osteoblastogenesis



Reversal

Osteoblasts

Bone Formation

6 months

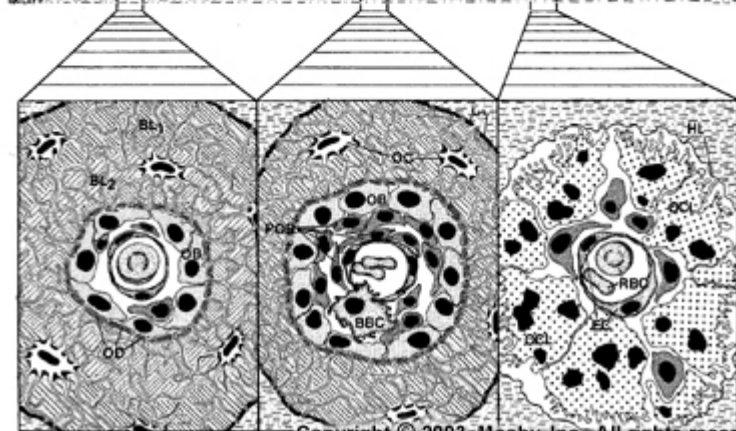
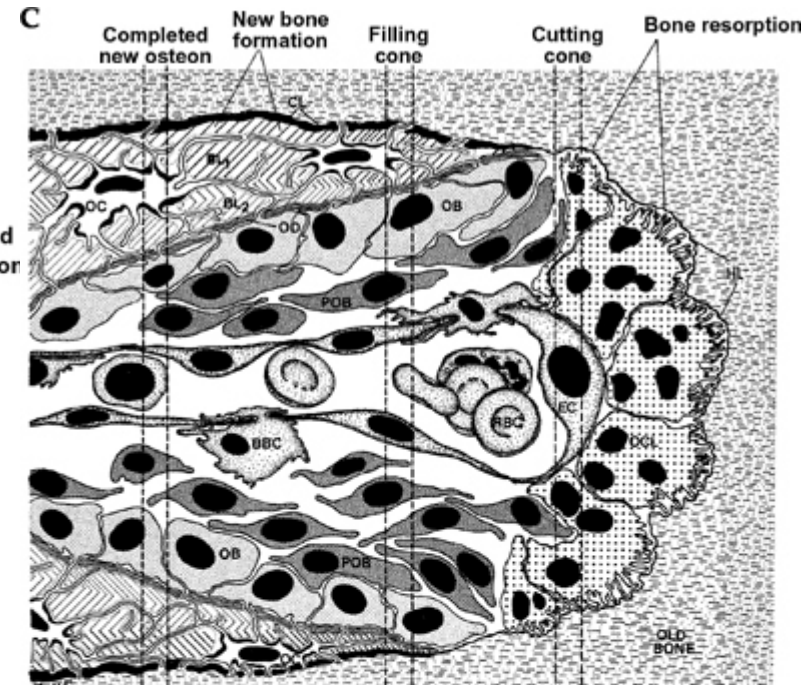
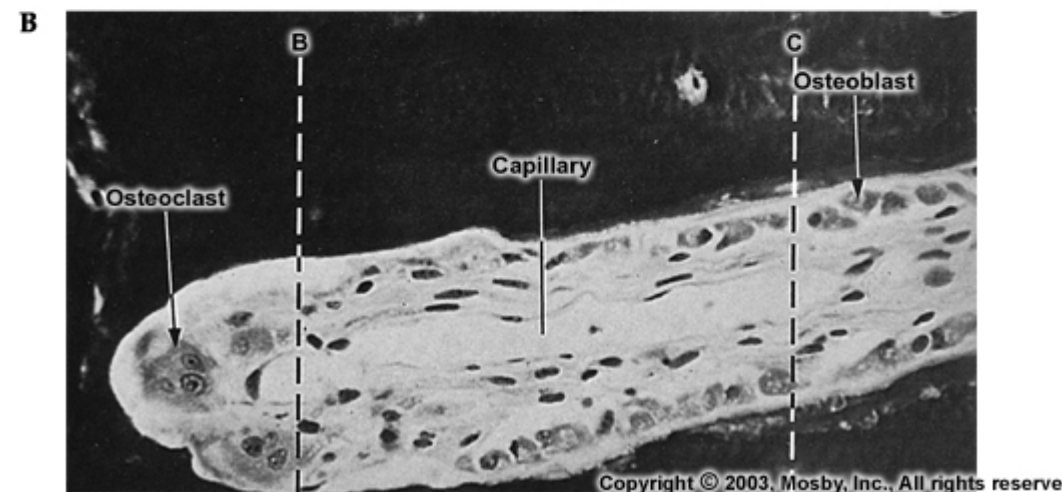
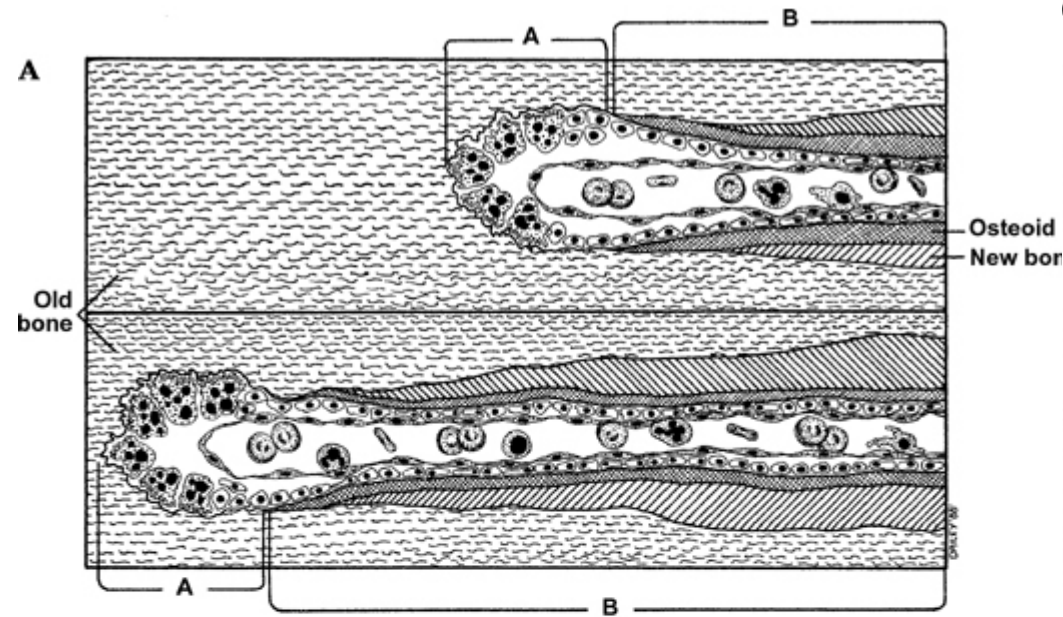
Lining cells

Osteocytes

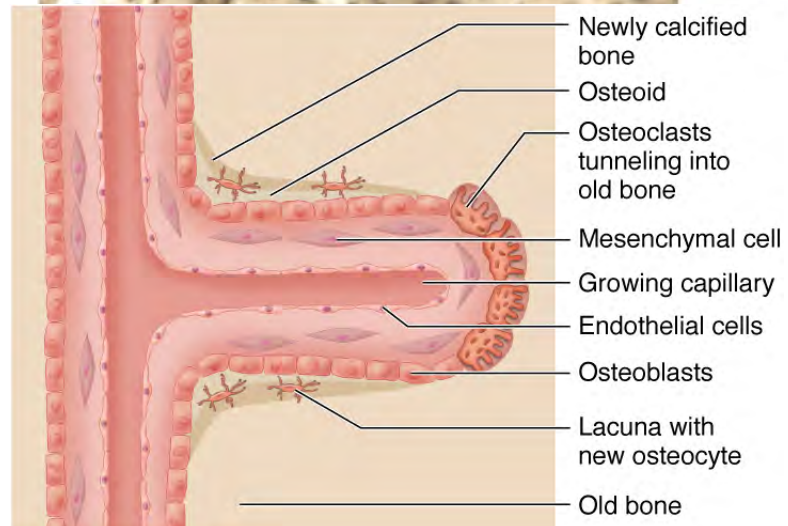
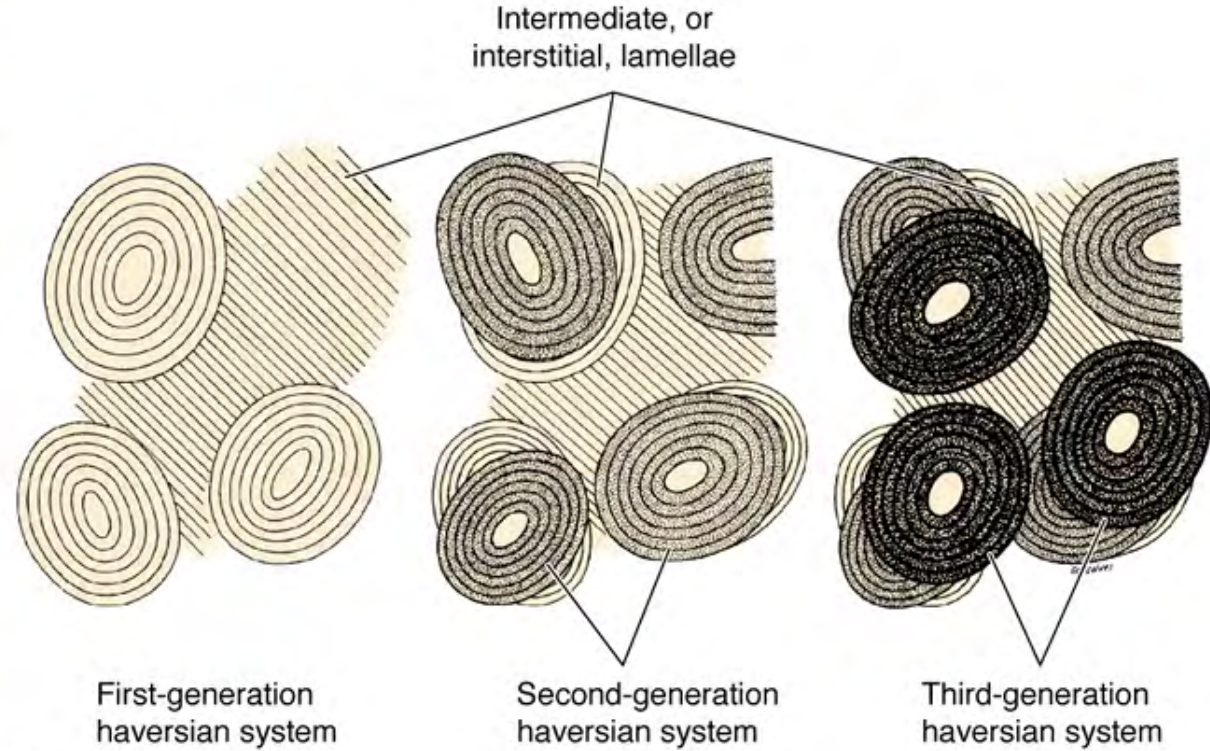
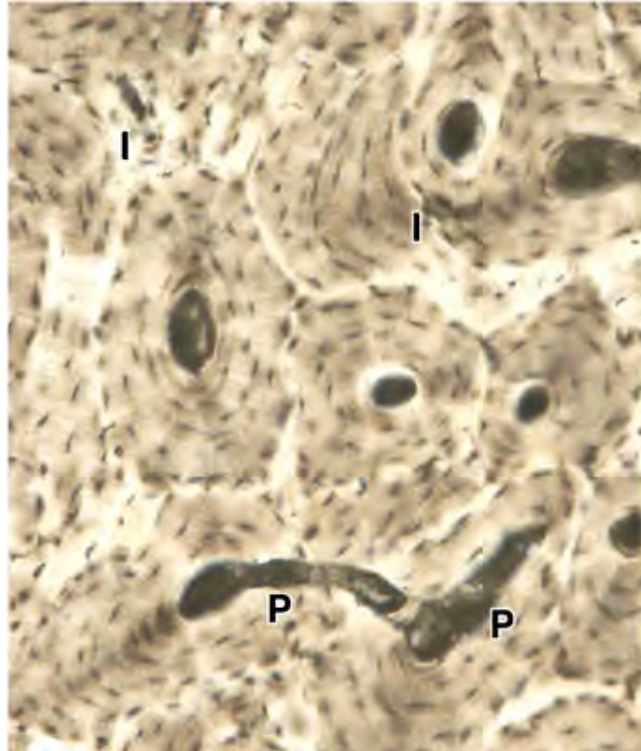
Quiescence

- Dynamic – changing throughout life
 - 5-7% of bone mass turned over each week
- Osteoblasts – form new bone
- Osteoclasts
 - multinucleated, fused cells
 - Lysosomal enzymes digest collagen & matrix
 - Phagocytize debris

Bone remodeling



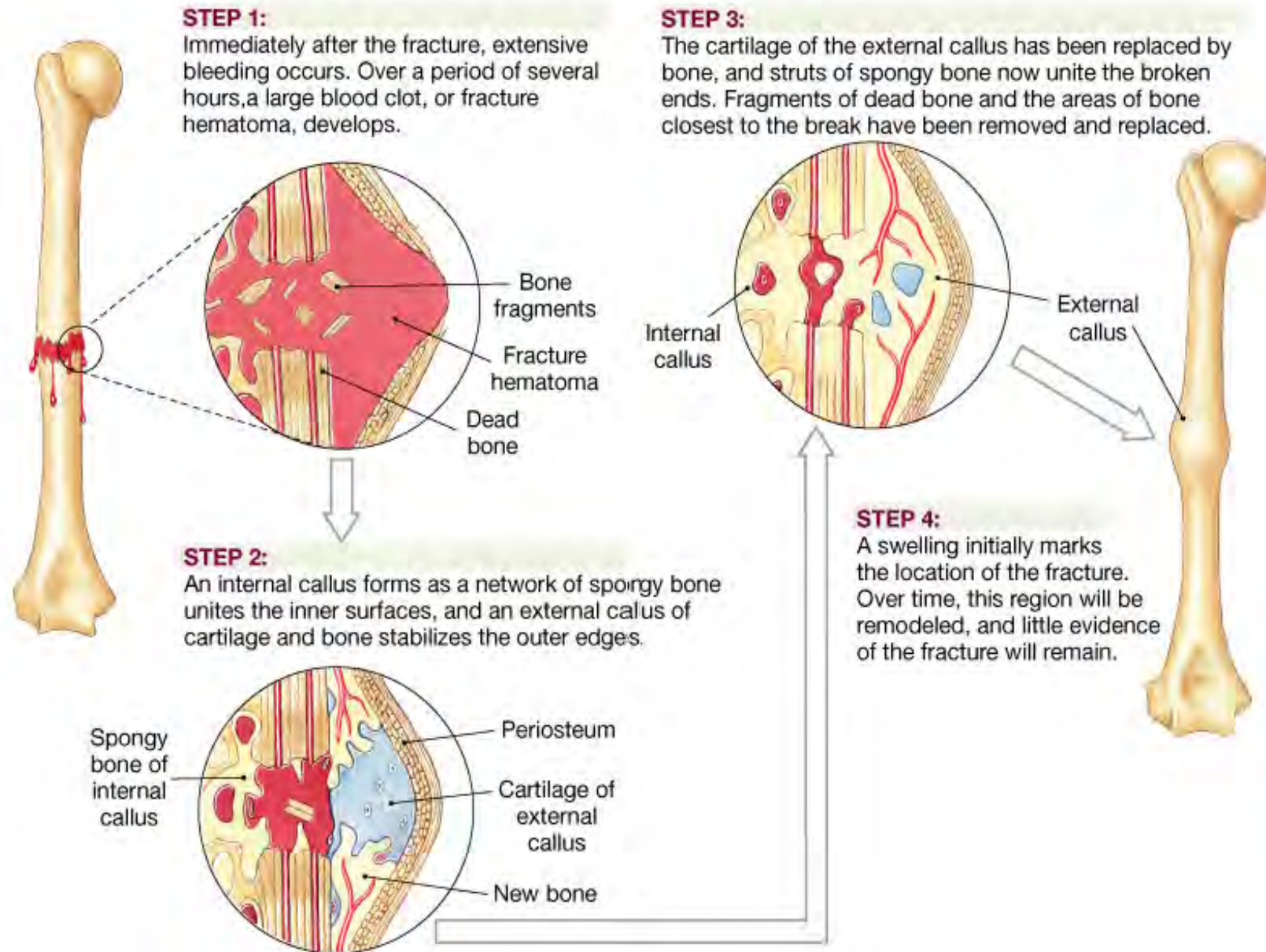
Bone remodeling

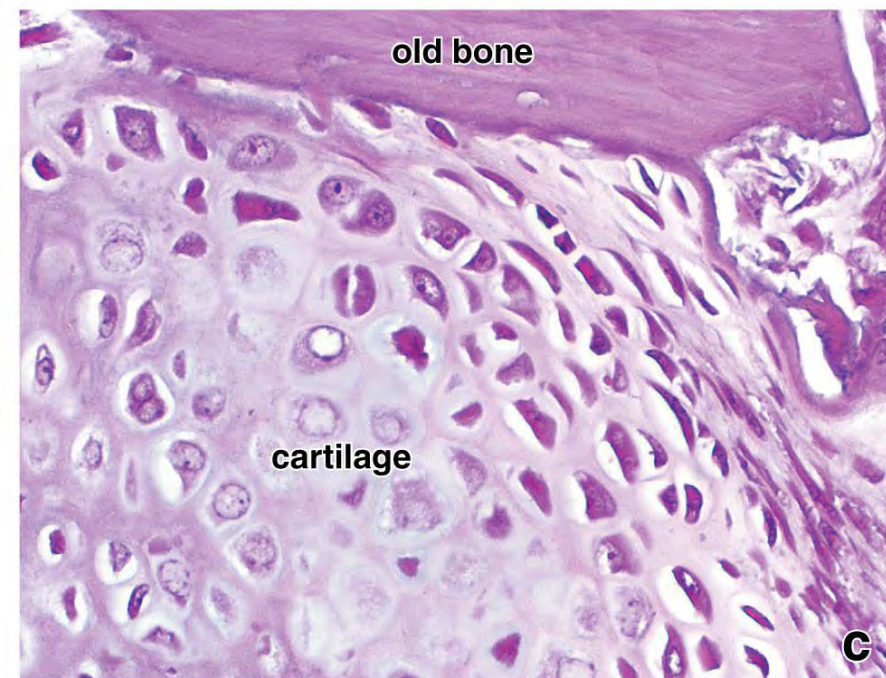
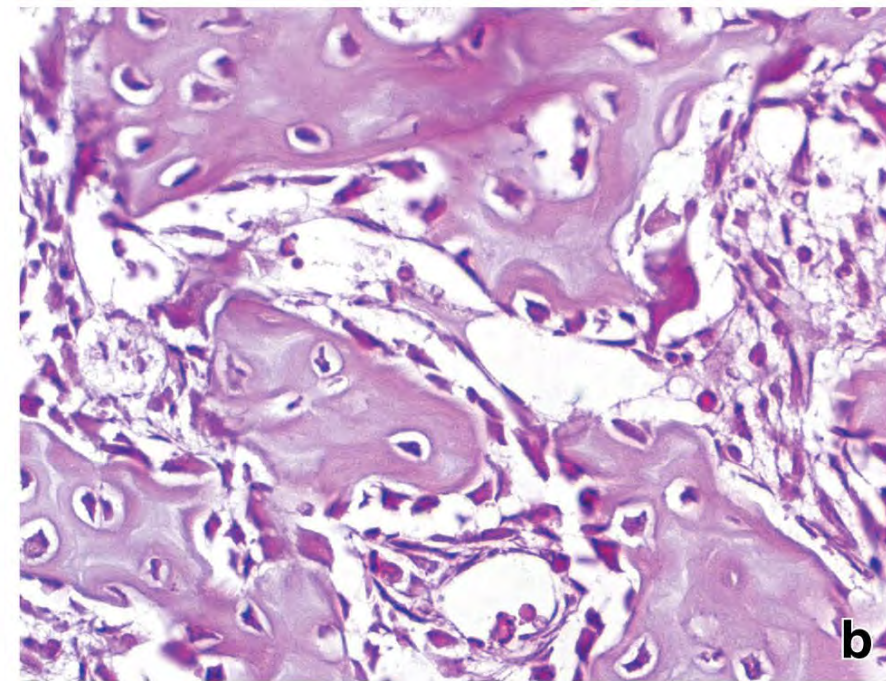
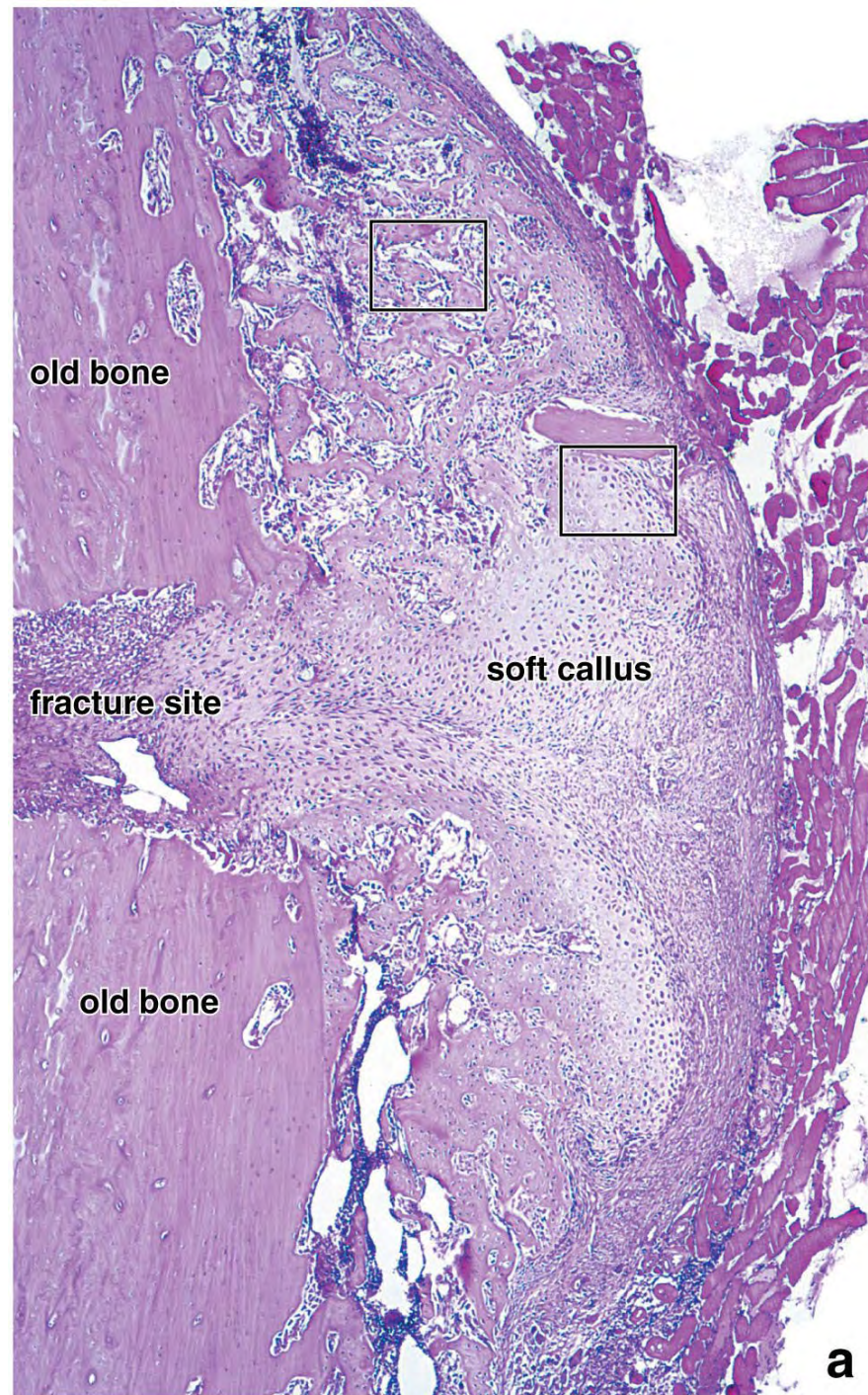


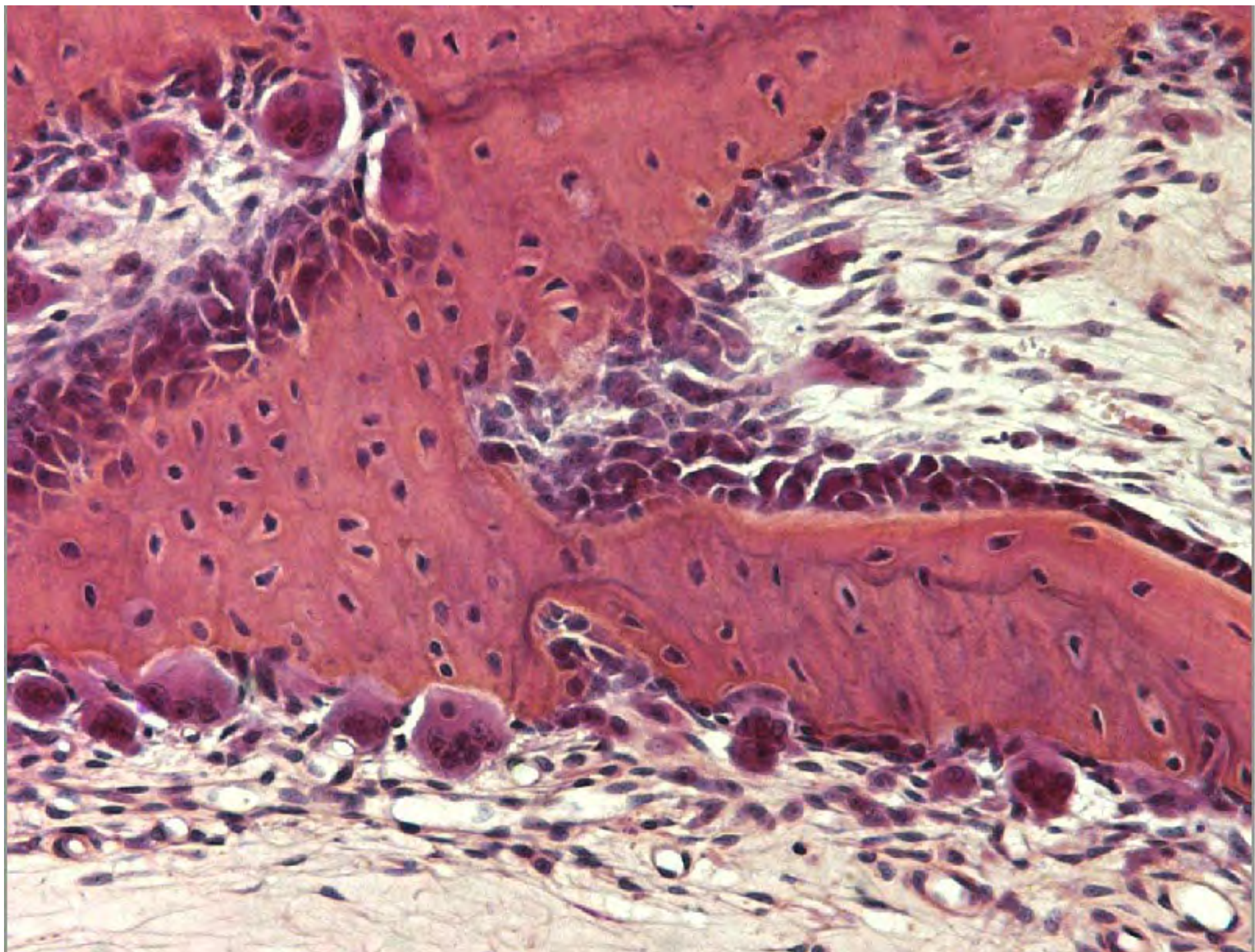
Bone Remodeling

- Fracture repair
 - localized hemorrhage, blood clot
 - Death of osteocyte
 - Macrophage
 - Scavenge the clot, dead osteocyte, bone matrix
 - Proliferation of osteoprogenitor cells
 - Formation of cellular tissue
 - Primary bone formation (Bone callus)
 - By intramembraneous ossification
 - Endochondral ossification
 - bone remodeling to secondary bone

Steps in the fracture repair







		Bone	Cartilage
Cells		Osteoblast, Osteocyte, osteoclast	Chondroblast Chondrocyte
Extracellular matrix (inorganic matrix)		67%	
Extracellular matrix (organic matrix)	Fiber	Collagen fiber	Collagen fiber Elastic fiber
	Ground substance	Collagen fiber	GAG, proteoglycan, glycoprotein, hyaluronic acid, elastin
characteristics	hardness Blood vessel Nerve Lymphatics	hard Yes Yes yes	Gel-like No No no
Function		Mineral store, Protection Support and motion Blood cell formation	Elasticity Shock absorption Sliding area
Growth		Interstitial growth Appositional growth	Interstitial growth Appositional growth
subtypes		Compact bone, Spongy bone, lamellar bone, woven bone	Hyaline cartilage, Elastic cartilage, Fibrous cartilage
coverage		Periosteum endosteum	perichondrium